About the Project

In today's digital era, social media has become an inseparable part of students' daily lives, significantly influencing how they communicate, interact, learn, and utilize their time. Platforms such as Instagram, Facebook, and other emerging applications have transformed the way students engage with the world, providing instant connectivity and access to information. However, this widespread usage has also led to a considerable increase in screen time, raising serious concerns about its potential impact on students' mental health, sleep patterns, academic performance, and overall lifestyle.

The project titled **"Analysis of Social Platforms in Students"** is designed to explore, analyse, and understand the multifaceted effects of social media usage on students' well-being. With the growing dependency on digital platforms, students are increasingly exposed to issues such as sleep deprivation, reduced concentration, heightened stress levels, anxiety, and even addictive behaviour. These challenges can negatively influence both their personal and academic lives, making it essential to study and address these patterns in a systematic and data-driven manner.

The primary objective of this project is to examine the relationship between social media usage and critical factors such as mental health scores, addiction levels, sleep duration, and academic performance. By collecting and analysing relevant data, the project seeks to identify how different levels of social media engagement affect students' daily routines and long-term outcomes. This includes understanding whether excessive usage leads to poor academic results, how reduced sleep impacts mental stability, and how addiction to digital platforms develops over time.

To achieve these objectives, the project employs data analytics and visualization techniques. Raw data collected from students is first processed and cleaned to ensure accuracy and consistency. This includes handling missing values, removing inconsistencies, and structuring the data in a usable format. Once prepared, various analytical methods are applied to uncover meaningful patterns, trends, and correlations. For example, the system can highlight whether students who spend more hours on social media tend to have lower academic performance or higher stress levels.

A key strength of this project lies in its ability to transform complex data into easily understandable insights through interactive dashboards and graphical representations. Visual tools such as bar charts, line graphs, and key performance indicators (KPIs) are used to present findings in a clear and engaging manner. These visualizations not only simplify data interpretation but also enable users—such as students, educators, and researchers—to quickly grasp important insights without requiring advanced technical knowledge.

The scope of this project extends to identifying behavioural patterns among students with varying levels of social media usage. It examines different categories of users, such as low, moderate, and heavy users, and compares their mental health, sleep quality, addiction scores, and academic outcomes. By doing so, the project provides a comprehensive understanding of how digital habits influence different aspects of student life. Additionally, the use of statistical evaluation techniques ensures that the findings are reliable and backed by data.

Furthermore, the project adopts a proactive and data-driven approach to highlight potential risks associated with excessive social media usage. By identifying early warning signs—such as high addiction scores or reduced sleep hours—it becomes possible to suggest preventive measures and promote healthier digital habits. The insights generated can be valuable for students in managing their time effectively, for educators in understanding student behaviour, and for researchers in further exploring this domain.

In conclusion, the **"Analysis of Social Platforms in Students"** project serves as a comprehensive analytical framework that bridges the gap between raw data and actionable insights. It not only examines the growing influence of social media on students' lives but also provides a structured approach to understanding and addressing its impacts. By leveraging modern data analytics and visualization techniques, the project contributes to promoting responsible social media usage, improving academic outcomes, and enhancing overall student well-being. Ultimately, it supports informed decision-making and encourages a balanced and mindful approach to digital engagement.

Existing System

Before the development of this project, there was no dedicated or structured system available within the college to analyse the impact of social media usage on students. Despite the widespread use of platforms such as Instagram, Facebook, and others among students, there existed a significant gap in understanding how these platforms influenced various aspects of student life, including academic performance, mental health, sleep patterns, and daily routines. Social media had become deeply embedded in students' lifestyles, yet there was no formal mechanism to study or evaluate its effects in a systematic and scientific manner.

Traditionally, any insights into student behaviour related to social media usage were derived from general assumptions, informal discussions, or limited surveys conducted occasionally. These approaches were often unorganized and lacked consistency, as they did not follow a standardized methodology for data collection or analysis. As a result, the information gathered was often incomplete, biased, or unreliable. Without proper tools or frameworks, it was difficult to draw accurate conclusions or identify meaningful trends from such scattered data.

Moreover, there was no centralized platform within the college to collect, store, process, and analyse data related to students' social media usage. Information, if collected, remained fragmented and was not utilized effectively for decision-making or awareness initiatives. The absence of such a system prevented educators and administrators from gaining a comprehensive understanding of how students' digital habits were evolving and what impact they were having on their overall well-being.

Another major limitation of the existing system was the inability to detect patterns and correlations. For instance, there was no structured way to examine whether students who spent more time on social media experienced lower academic performance, or whether reduced sleep due to excessive screen time contributed to increased stress and anxiety levels. These important relationships remained unexplored due to the lack of analytical tools and techniques. Consequently, potential risks associated with excessive social media usage went largely unnoticed and unaddressed.

In addition, the existing scenario lacked the integration of modern data analytics technologies. Techniques such as data preprocessing, statistical analysis, and visualization were not employed, making it challenging to interpret raw data effectively. Without the use of charts, graphs, or dashboards, even the available data could not be presented in a meaningful or user-friendly manner. This limited the ability of stakeholders to understand complex information and act upon it. Furthermore, advanced approaches such as predictive analysis or machine learning were completely absent, which could have otherwise helped in forecasting trends and identifying at-risk students.

Faculty members and college administrators also faced challenges due to the absence of a data-driven system. Without access to reliable insights or analytical dashboards, they had to rely on intuition or anecdotal evidence when addressing issues related to student performance or well-being. This often led to delayed or ineffective interventions, as decisions were not supported by concrete data. Similarly, students themselves lacked awareness about how their social media habits were affecting their academic and personal lives.

To overcome these limitations, there was a clear need for a comprehensive and structured system that could systematically study the impact of social media on students. The absence of such a platform highlighted the importance of developing a solution that integrates data collection, analysis, and visualization into a single framework. This gap served as the foundation for the development of the **Analysis of Social Platforms in Students System**, which aims to provide accurate, reliable, and actionable insights. In conclusion, the existing system can be described as unstructured, informal, and lacking analytical depth. It relied heavily on assumptions rather than data, offered no centralized mechanism for analysis, and failed to utilize modern technological tools. These shortcomings emphasized the need for a robust and data-driven approach to understanding student behaviour, which this project successfully addresses by introducing a well-organized and efficient analytical system.

Proposed System

The proposed system, titled **"Analysis of Social Platforms in Students"**, is designed to offer a comprehensive, structured, and data-driven approach to analysing the impact of social media usage on students' academic performance, mental health, sleep patterns, and overall behaviour. In contrast to the previously existing unstructured and informal methods, this system integrates modern data analytics and visualization techniques to generate meaningful, accurate, and actionable insights. The primary goal of the system is to assist students, educators, and administrators in understanding the consequences of social media usage and to support informed decision-making within the academic environment.

The system begins with the systematic collection of relevant data from students. This data includes various parameters such as average daily time spent on social media, most frequently used platforms, sleep duration, mental health scores, addiction levels, and academic performance indicators like grades or productivity levels. The data collection process is designed to be organized and consistent, ensuring that the information gathered is reliable and suitable for further analysis. By capturing multiple dimensions of student behaviour, the system provides a holistic view of how social media influences different aspects of student life.

Once the data is collected, it undergoes a preprocessing stage, which is essential for ensuring data quality and accuracy. This stage involves handling missing or incomplete values, removing inconsistencies, correcting errors, and transforming the data into a structured format suitable for analysis. Data preprocessing plays a crucial role in improving the reliability of the results, as inaccurate or unclean data can lead to misleading conclusions. Techniques such as normalization, encoding of categorical variables, and filtering are applied to prepare the dataset effectively.

Following preprocessing, the system applies various analytical techniques to uncover patterns, trends, and correlations within the data. Statistical methods and data analysis tools are used to examine relationships between different variables. For example, the system can analyse whether increased social media usage is associated with lower academic performance, or how reduced sleep duration may contribute to higher stress or poorer mental health. These analyses help in identifying key factors that influence student well-being and performance. By relying on data-driven insights rather than assumptions, the system ensures that the conclusions drawn are both accurate and meaningful.

One of the most significant features of the proposed system is its use of interactive dashboards and visualizations. Complex datasets are transformed into intuitive visual representations such as bar charts, line graphs, pie charts, and key performance indicators (KPIs). These visual tools make it easier for users to interpret data and quickly grasp important insights. The dashboards are designed to be interactive, allowing users to filter data, explore different variables, and view customized results based on their needs. This makes the system accessible not only to technical users but also to non-technical stakeholders such as students and faculty members.

In addition to visualization, the system also categorizes students based on their social media usage patterns. Students can be grouped into categories such as low, moderate, and high users, and their corresponding mental health, sleep quality, and academic performance can be compared. The system also highlights potential risk factors, such as high addiction scores, excessive screen time, or insufficient sleep. This enables early identification of students who may be at risk and allows for timely intervention. Such insights can be valuable for counselling, awareness programs, and academic support initiatives.

Furthermore, the proposed system has the capability to be extended with advanced features such as predictive analysis. By using historical data, the system can forecast future trends and predict potential outcomes. For instance, it can estimate the likelihood of a decline in academic performance based on increasing social media usage patterns. This predictive capability enhances the usefulness of the system by enabling proactive decision-making rather than reactive responses.

Another important aspect of the system is its user-friendly design and accessibility. The system is developed as a web-based platform, making it easily accessible within the college environment from various devices. It ensures that data is securely stored and managed, with provisions for regular updates and maintenance. The interface is designed to be simple and intuitive, allowing users to navigate through

dashboards and reports with ease.

In conclusion, the **"Analysis of Social Platforms in Students"** represents a significant advancement over traditional methods by providing a structured, efficient, and scalable solution for analysing student behaviour. It transforms raw data into meaningful insights through systematic processing, analysis, and visualization. By addressing the limitations of the existing system, it not only enhances understanding of the impact of social media on students but also promotes better academic performance, improved mental well-being, and more responsible digital habits. Ultimately, the system serves as a valuable tool for fostering a data-driven approach to student development and institutional decision-making.

Requirements

The successful implementation of the Student Social Media Analytics project requires a combination of data collection methods, analytical tools, visualization platforms, and web development technologies. The following points describe the complete set of requirements in a structured and detailed manner:

A. Data Collection Requirements

- **Manual Data Collection using Google Sheets:**
  The first step in the project involves collecting primary data directly from students. This is achieved using Google Sheets or Google Forms, where a structured questionnaire is created.
  - The form includes fields such as:
    - Average daily social media usage (in hours)
    - Most used social media platform
    - Sleep duration per night
    - Mental health score (based on self-assessment)
    - Addiction level
    - Academic performance indicators (grades or performance rating)
  - Google Sheets is used to store responses in real-time, ensuring easy access and organization of data.
  - This method ensures that the data is relevant, customized to the project, and reflects actual student behaviour within the college.
- **Secondary Data Collection from Kaggle:**
  In addition to primary data, secondary datasets are obtained from Kaggle to enhance the scope and reliability of the analysis.
  - Kaggle provides pre-existing datasets related to social media usage, student behaviour, and mental health.
  - These datasets help in:
    - Increasing the volume of data
    - Comparing trends across different populations
    - Validating findings from the primary dataset
    - The integration of Kaggle data ensures that the analysis is not limited to a small sample size and improves the robustness of the project.

B. Data Processing and Analysis Requirements:

- **Python Programming Language**:
  Python is required as the primary programming language for data analysis due to its simplicity and powerful libraries.
  It is used for:
    - Data cleaning and preprocessing
    - Statistical analysis
    - Identifying patterns and correlations
    - Python provides flexibility and efficiency in handling large datasets.
- **Jupyter Notebook Environment:**
  Jupyter Notebook is used as the development environment for writing and executing Python code.
  It allows step-by-step execution of code, making it easier to test and debug.
  It supports the integration of code, output, and explanations in a single document.
  This makes it ideal for data analysis and documentation of the process.
- **Python Libraries**
  Several libraries are required for efficient data handling and analysis:

- Pandas: For data manipulation, cleaning, and transformation
- NumPy: For numerical computations
- Matplotlib / Seaborn: For basic data visualization during analysis
- Scikit-learn: For data analysis

These libraries help in converting raw data into structured and meaningful formats.

C. Development Environment Requirements

- **Visual Studio Code (VS Code)**:
  VS Code is required as the primary code editor for the project.
  It provides support for multiple programming languages including Python, HTML, CSS, and JavaScript.
  Features include:
  - Syntax highlighting
  - Code completion
  - Extensions for Python and web development

It acts as a central platform for managing all parts of the project.

D. Data Visualization Requirements

- **Tableau Software**:
  Tableau is used for creating interactive and visually appealing dashboards.
  It enables the transformation of analysed data into charts, graphs, and KPIs.
  Features include:
  - Drag-and-drop interface for easy dashboard creation
  - Interactive filters and drill-down capabilities
  - Real-time data visualization
  - Dashboards created in Tableau help in presenting insights such as:
    - Relationship between social media usage and academic performance
    - Impact of sleep on mental health
    - Distribution of addiction levels among students
    - Tableau enhances the clarity and effectiveness of the findings.

E. Web Development Requirements

- **Frontend Technologies (HTML, CSS, JavaScript)**:
  These technologies are required to develop the user interface of the project website.
  - HTML: Structures the content of the website
  - CSS: Styles the website for better visual appearance
  - JavaScript: Adds interactivity and dynamic behaviour
  - The website includes sections such as:
    - Login page
    - Home page
    - About the project
    - Dashboards and insights
    - Key findings
    - About the developer
    - Contact us page
- Integration of Dashboards
- Tableau dashboards are embedded into the website using exported pictures of the dashboards in <img> tag.
- This allows users to interact with visualizations directly on the website.
- Responsive Design: The website is designed to be responsive, ensuring compatibility across devices such as desktops, tablets, and mobile phones.

F. System and Hardware Requirements

- A computer or laptop with sufficient processing power
- Internet connection for:
  - Accessing Google Sheets and Kaggle
  - Downloading datasets
  - Publishing the website
  - Basic storage capacity for datasets and project files

G. Firebase Requirements:

Firebase was used as the backend platform to support authentication, data storage, and hosting functionalities for the Student Social Media Analytics website. The following requirements were identified for successful implementation:

- **Firebase Project Setup:**

A Firebase project must be created using a Google account through the Firebase Console. This project serves as the central environment for managing all backend services such as authentication, database, and hosting.

- **Firebase Authentication:**

Firebase Authentication is required to enable secure user login and logout functionality. It supports email and password-based authentication, ensuring that only authorized users can access the dashboard. Session persistence is configured to maintain user sessions during active use and automatically log out users when required.

- **Fire store Database:**

Cloud Fire store is used to store and manage data in a structured format. It is required for:

- Storing contact form submissions (name, email, phone number, message, timestamp)
- Managing user-related data if needed in future enhancements
  The database operates in real-time and provides scalable and secure data storage.

- **Firebase Hosting:**

Firebase Hosting is used to deploy and host the website. It enables fast and secure delivery of static web content (HTML, CSS, JavaScript). The hosting setup includes:

- Configuring a public directory
- Deploying files using Firebase CLI
- Providing a live URL for accessing the website

- **Firebase SDK Integration:**

Firebase SDK must be integrated into the project using JavaScript modules. This allows the website to interact with Firebase services such as authentication and Fire store. Proper configuration using API keys and project credentials is necessary.

- **Security Rules:**

Basic security rules are required to control access to Fire store data. These rules ensure that only authenticated users can read or write data, thereby maintaining data privacy and integrity.

- **Internet Connectivity:**

Since Firebase is a cloud-based service, a stable internet connection is required for authentication, data storage, and hosting operations.

Firebase provides a scalable, secure, and real-time backend solution, making it suitable for modern web applications without requiring dedicated server management.

Conclusion

The Student Social Media Analytics project requires a well-integrated combination of data collection tools, analytical software, visualization platforms, and web technologies. Starting from manual data collection using Google Sheets and external datasets from Kaggle, the project progresses through data analysis using Python and Jupyter Notebook, followed by visualization in Tableau. Finally, the insights are presented through a responsive website developed using modern web technologies. Together, these requirements ensure the successful development of a comprehensive, data-driven analytical system.

Objective of the Project

The Student Social Media Analytics project is designed with the primary aim of understanding and analysing the impact of social media usage on students' academic performance, mental health, sleep patterns, and overall lifestyle. In today's digital age, where students spend a significant portion of their time on online platforms, it becomes essential to evaluate how these habits influence their well-being and productivity. The objectives of this project are structured to provide a clear, data-driven understanding of these effects and to support informed decision-making.

1. To Analyse Social Media Usage Patterns Among Students:

One of the key objectives of this project is to study how students use social media in their daily lives. This includes analysing:

- The average number of hours spent on social media
- The most frequently used platforms
- The time of usage (day/night patterns)

By examining these patterns, the project aims to identify different categories of users, such as low, moderate, and heavy users. Understanding these usage trends is crucial for establishing a foundation for further analysis and determining how varying levels of engagement affect other aspects of student life.

2. To Evaluate the Impact on Academic Performance:

Another important objective is to investigate the relationship between social media usage and students' academic outcomes.

- The project seeks to determine whether increased screen time leads to reduced concentration and lower academic performance.
- It also aims to identify whether moderate usage has any neutral or positive effects, such as access to educational content.

By analysing academic indicators such as grades or performance ratings alongside usage data, the project provides insights into how digital habits influence learning and productivity.

3. To Assess the Effect on Mental Health:

The project aims to explore how social media usage affects students' mental health.

- Factors such as stress, anxiety, mood fluctuations, and emotional well-being are considered.
- The project examines whether excessive use of social media contributes to negative mental health outcomes.

This objective is particularly important as mental health has become a growing concern among students. By identifying patterns and correlations, the project helps in understanding the psychological impact of digital engagement.

4. To Study the Relationship Between Social Media and Sleep Patterns:

Sleep is a critical factor in maintaining both physical and mental health. This project aims to analyse how social media usage affects students' sleep duration and quality.

- It examines whether students who spend more time online tend to sleep less.
- It also evaluates how irregular sleep patterns impact overall well-being and academic performance.

This objective helps in highlighting the indirect effects of social media usage on health and productivity.

5. To Measure Addiction Levels and Behavioural Changes:

Another key objective is to identify signs of social media addiction among students.

- The project evaluates addiction levels based on usage frequency and dependency patterns.
- It also studies behavioural changes such as reduced focus, increased procrastination, and decreased physical activity.

By measuring addiction levels, the project aims to raise awareness about the risks associated with excessive usage and promote healthier digital habits.

6. To Identify Correlations and Trends Using Data Analytics:

A major objective of this project is to apply data analytics techniques to uncover relationships between different variables.

- For example, the project analyses correlations between:
  - Social media usage and academic performance
  - Sleep duration and mental health
  - Addiction levels and productivity
- Statistical methods are used to ensure that the findings are accurate and reliable.

This objective emphasizes the importance of a data-driven approach in understanding complex behaviours and patterns.

7. To Develop Interactive Dashboards for Data Visualization:

The project aims to present analysed data in a clear and visually appealing manner using dashboards.

- Interactive charts, graphs, and KPIs are created to simplify complex data.
- These dashboards allow users to explore insights easily and make comparisons.

This objective ensures that the results of the analysis are accessible to a wide range of users, including those without technical expertise.

8. To Promote Awareness and Responsible Social Media Usage:

One of the broader objectives of the project is to create awareness among students about the impact of their digital habits.

- By presenting clear insights, the project encourages students to manage their time effectively.
- It highlights the importance of maintaining a balance between online activities and real-life responsibilities.

This objective contributes to the overall well-being of students by promoting responsible usage of social media.

9. To Support Educators and Institutions in Decision-Making:

The project also aims to assist faculty members and administrators in understanding student behaviour.

- Insights from the analysis can help in designing policies, awareness programs, and support systems.
- It enables institutions to take data-driven decisions to improve student performance and well-being.

This objective extends the impact of the project beyond individual students to the institutional level.

10. To Build a Scalable and Extendable Analytical System:

Finally, the project aims to develop a system that can be expanded in the future.

- The system can be updated with new data for continuous monitoring.
- Advanced features such as predictive analytics can be added later.

This ensures that the project remains relevant and useful over time, adapting to changing trends and requirements.

Conclusion:

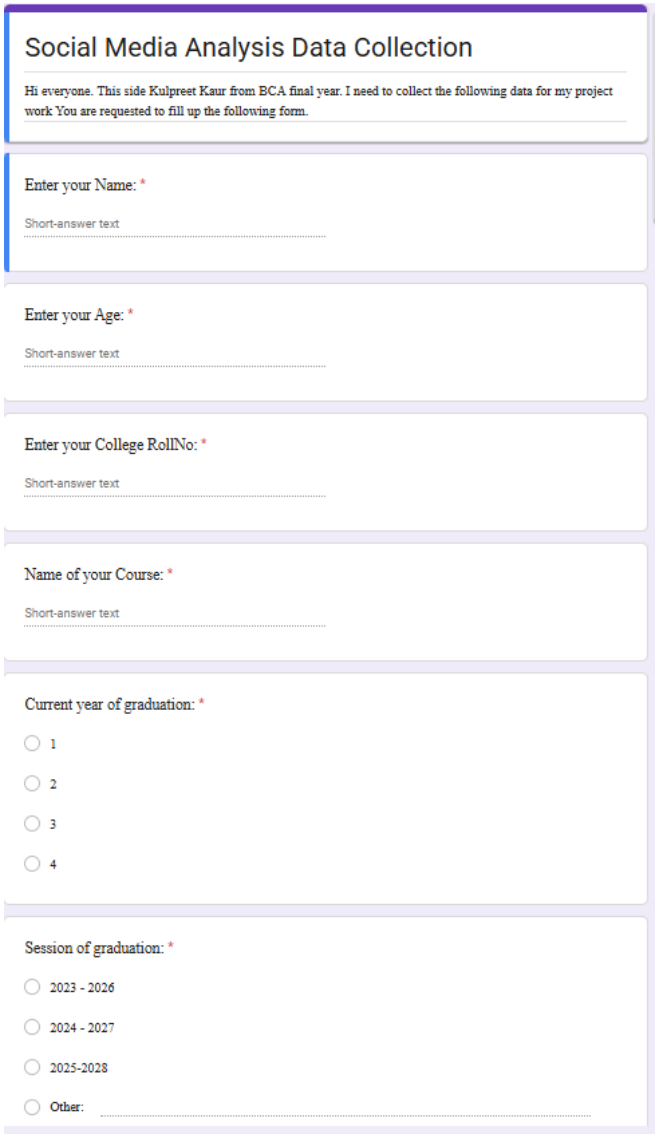
In conclusion, the objectives of the Student Social Media Analytics project are centred around understanding, analysing, and improving the relationship between social media usage and student well-being. By combining data collection, analysis, and visualization, the project provides a comprehensive framework for identifying patterns, assessing impacts, and promoting healthier digital habits. These objectives not only contribute to academic research but also provide practical benefits for students, educators, and institutions, making the project both meaningful and impactful.

Project Development

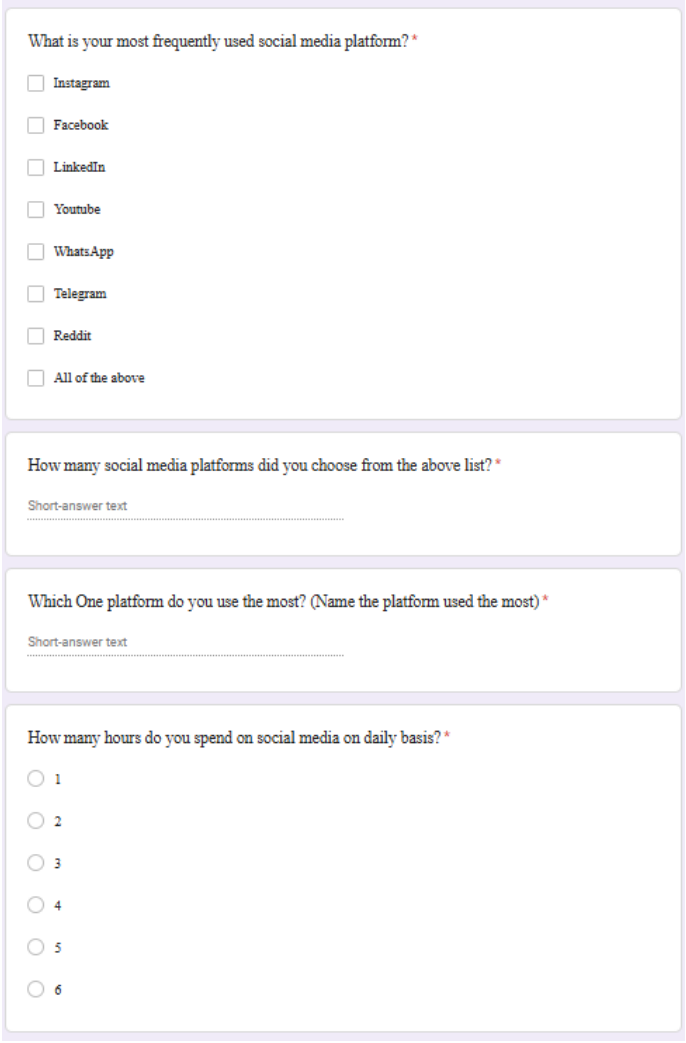
I. Data Collection:

Data collection is a fundamental step in the development of the **Student Social Media Analytics** project, as the accuracy and reliability of the analysis largely depend on the quality of data gathered. In order to ensure a comprehensive and well-balanced dataset, this project utilizes a combination of primary and secondary data collection methods. By integrating both approaches, the project is able to capture real-world student behavior while also enhancing the dataset with larger and more diverse samples.

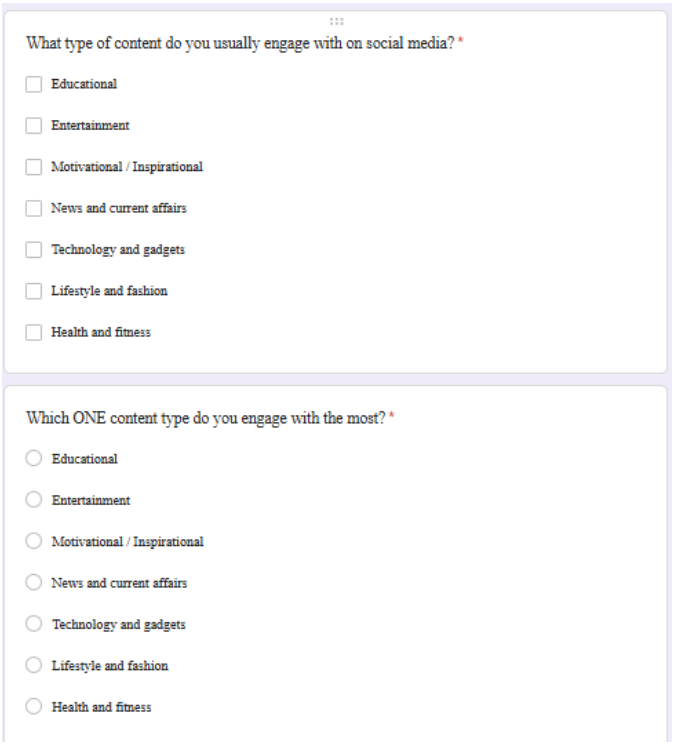
a) Manual Data Collection: The first method involves the collection of primary data directly from students using Google Forms. This approach allows for the creation of a structured questionnaire tailored specifically to the objectives of the project. It ensures that relevant information - such as social media usage patterns, sleep duration, mental health indicators, and academic performance - is collected in a consistent and organized manner. The responses are automatically stored and managed using Google Sheets, making the data easily accessible for further processing and analysis. Following is a showcase of the surveyed form used for collecting data manually using Google Forms which was able to collect data of almost 100 students of my department:



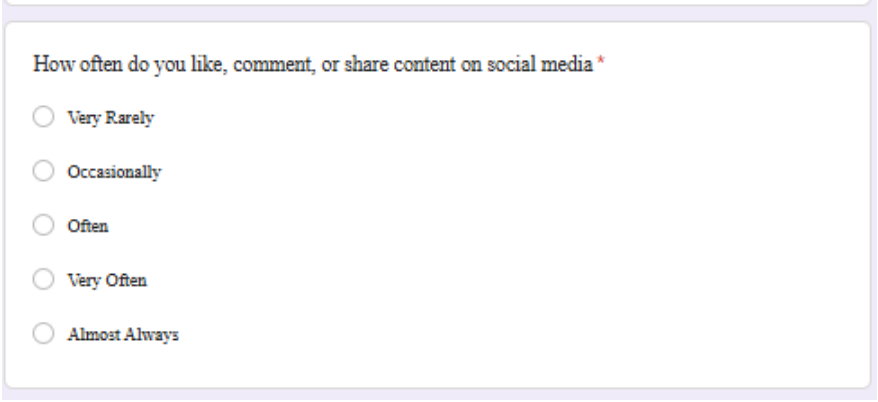
Form 1: Sample of Google form used for manual data collection



Form 1: Sample of Google form used for manual data collection

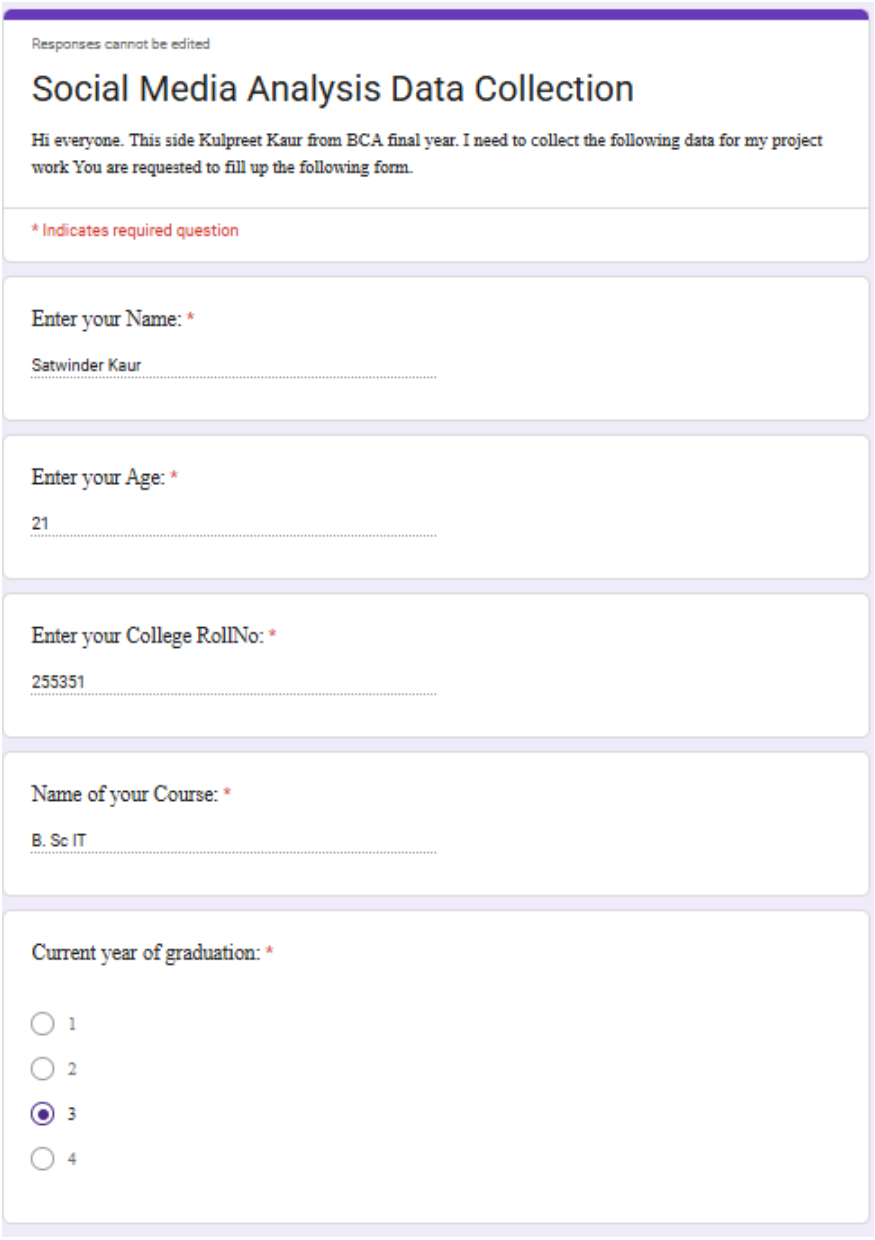


Form 1: Sample of Google form used for manual data collection

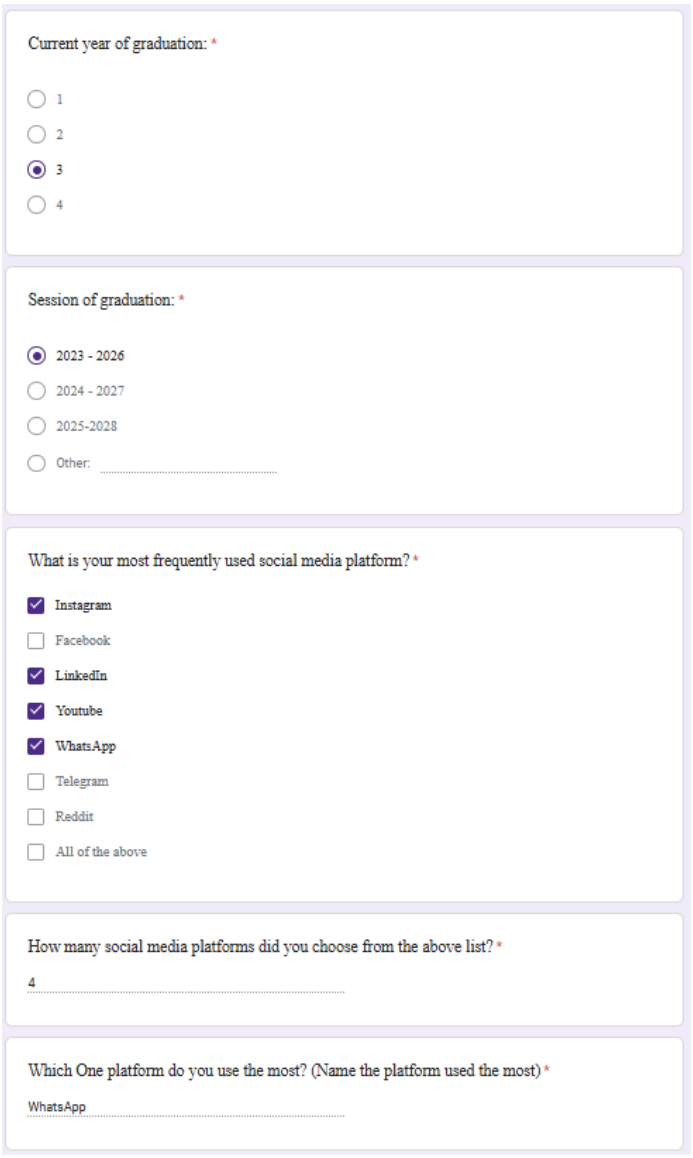


Form 1: Sample of Google form used for manual data collection

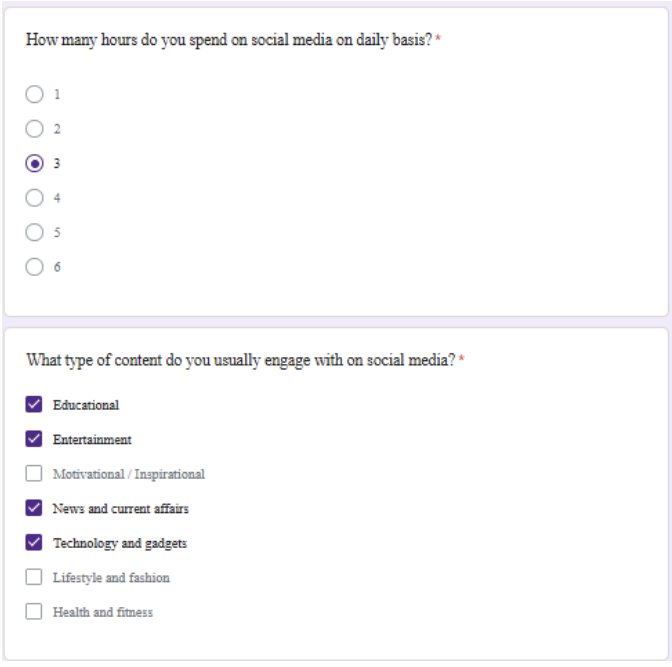
The Above was the sample questionnaire that was surveyed among the students, and the statistics of the responses gathered. The following is a sample of response given by one of the students as a depiction of how the questionnaire was filled:



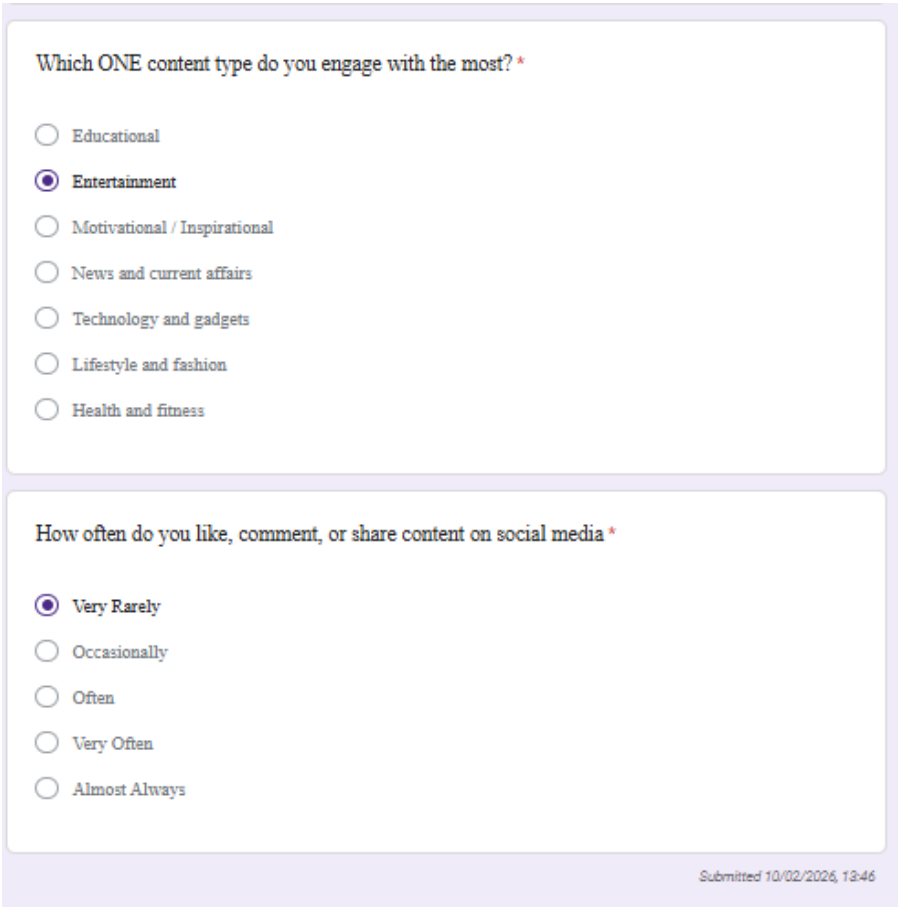
Form 2: Sample of response to the above survey by students



Form 2: Sample of response to the above survey by students



Form 2: Sample of response to the above survey by students



Form 2: Sample of response to the above survey by students

This survey resulted in 100 rows of collected data shown as follows:

Timestamp	Name	Age	RollNo	Course	Cur_yea	Session_of_gra	platforms_used_multiple	platforms_	platform_used	hours_spent	content_preference_mul	content_preference_	engagemen
10/02/2026 13:46:18	Satwinder Kaur	21	255351	BSc IT	3	2023 - 2026	Instagram, LinkedIn, Youtube, WhatsApp	4	WhatsApp	3	Educational, Entertainment, I	Entertainment	Very Rarely
10/02/2026 13:47:44	Arpita	20	255220	BCA	3	2023 - 2026	Instagram	1	Instagram	1	Entertainment	Entertainment	Very Rarely
10/02/2026 13:57:32	Kirti	21	255361	Bsc IT	3	2023 - 2026	Youtube, WhatsApp	2	You tube	3	Educational, Entertainment, I	Entertainment	Very Rarely
10/02/2026 14:07:55	Sharanjeet	18	255205	BCA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	instagram	1	Entertainment	Entertainment	Very Rarely
10/02/2026 14:28:09	Vriddhi	20	255232	BCA	3	2023 - 2026	Instagram, WhatsApp	2	Instagram	2	Educational, Entertainment, `	Lifestyle and fashion	Occasionally
10/02/2026 14:48:31	Jasmeet Kaur	20	..	BCA	3	2023 - 2026	Instagram, LinkedIn, Youtube, WhatsApp	4	Instagram	6	Educational, Entertainment, I	Entertainment	Often
10/02/2026 15:00:10	Prabhjot kaur	20	255202	BCA	3	2023 - 2026	Youtube, WhatsApp	2	Instagram	2	Educational, Entertainment	Educational	Very Rarely
10/02/2026 15:08:08	Niharika	20	255206	BCA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	Instagram	3	Educational, Entertainment, I	Entertainment	Often
10/02/2026 15:16:13	Amritjot kaur	21	2511563	BA	3	2023 - 2026	Youtube	4	YouTube	2	Educational, Entertainment, I	Motivational / Inspiratior	Occasionally
10/02/2026 15:16:21	Paramveer	26	255362	BSc IT	3	2023 - 2026	Instagram, Youtube	5	Instagram	5	Educational, Entertainment	Entertainment	Often
10/02/2026 15:18:29	Yashika	20	2511595	BA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	Instagram	3	Entertainment	Entertainment	Very Rarely
10/02/2026 15:31:17	Palak	20	2511570	BA	3	2023 - 2026	Instagram, Youtube	2	Youtube	2	Educational, Entertainment	Educational	Occasionally
10/02/2026 15:49:07	Shweta	21	255221	BCA	3	2023 - 2026	Instagram	5	Instagram	3	Entertainment, Motivational ,	News and current affairs	Very Rarely
10/02/2026 16:30:34	Neha Madhar	20	255222	BCA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	Instagram	2	Educational, Entertainment, `	Technology and gadgets	Very Often
10/02/2026 16:41:19	Loveleen	20	255353	BSc IT	3	2023 - 2026	Youtube, WhatsApp	2	Youtube	5	Educational, Entertainment, `	Educational	Occasionally
10/02/2026 16:51:15	Komalpreet kaur	22	255365	BSc IT	3	2023 - 2026	WhatsApp	1	Whatsapp	1	Educational, Motivational / Ir	Educational	Very Rarely
10/02/2026 17:07:54	Trisha	20	255204	BCA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	Youtube	2	Educational, Motivational / Ir	Educational	Often
10/02/2026 17:58:33	Jasleen Kaur	20	255228	BCA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	Instagram	2	Educational, Entertainment, I	Entertainment	Very Often
10/02/2026 18:00:52	Priya	22	255210	BCA	3	2023 - 2026	Instagram, Youtube	2	YouTube	2	Educational, Entertainment, `	Technology and gadgets	Very Rarely
10/02/2026 18:27:57	Kashish Sharma	21	255234	BCA	3	2023 - 2026	Instagram, LinkedIn, Youtube, WhatsApp	4	WhatsApp	3	Educational, Entertainment, I	Entertainment	Often
10/02/2026 18:50:18	Harnoor kaur	19	2514057	B Tech	1	2025-2028	Instagram	3	Instagram	4	Educational, Entertainment, I	Educational	Very Rarely
10/02/2026 19:36:14	Prabhjot	19	255235	BCA	3	2023 - 2026	Youtube	1	YouTube	2	Educational	Educational	Very Rarely
10/02/2026 20:01:56	Diya	20	255207	BCA	3	2024 - 2027	Instagram, Youtube, WhatsApp	3	Instagram	2	Educational, Entertainment, I	Entertainment	Very Rarely
10/02/2026 20:30:33	Chitman	21	10	Nutrition ar	3	2023-3027	Instagram, WhatsApp	2	Instagram	4	Educational, Entertainment, I	Entertainment	Very Rarely

Table 1: Dataset consisting of responses of the survey

b) Kaggle Dataset: The second method involves the use of secondary data obtained from Kaggle, a well-known platform for publicly available datasets. These datasets provide additional information related to social media usage and student behaviour, helping to increase the volume and diversity of the data. Incorporating Kaggle data not only strengthens the analysis but also allows for comparison and validation of trends observed in the primary dataset. The following is the description of the dataset taken from Kaggle.

Student Performance Dataset

A comprehensive dataset containing demographic, academic, and behavioral information.

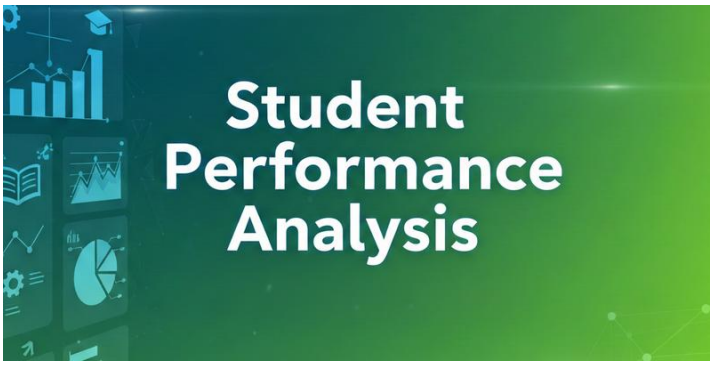


Figure 1: Student Performance Analysis

About Dataset

This dataset provides a comprehensive and realistic representation of various factors associated with student performance in an educational environment. It captures demographic details, family background, study habits, school-related characteristics, and academic scores across key subjects. The purpose of this dataset is to offer a structured and versatile resource for anyone interested in understanding patterns, behaviours, and influences that contribute to student outcomes.

The dataset includes rich information such as parental education, access to learning resources, attendance, study duration, extracurricular involvement, and preferred study methods. These aspects help outline a broader context of a student's academic journey, beyond mere scores. The subject-wise marks in Mathematics, Science, and English, along with a computed overall performance score and final grade, provide a complete view of student achievements.

This dataset can be used for:

- Educational research
- Institutional planning and analysis
- Understanding learning behaviour trends
- Examining the relationship between lifestyle and academic outcomes
- Exploratory studies on attendance, study habits
- Parental background
- General-purpose data exploration and visualization

About this file

- Filename: Student_Performance.csv
  Rows: 15,000
  Columns: 16

- This file contains individually structured student records, where each row represents a single student along with their demographic profile, educational background, learning habits, and

academic performance. The dataset combines behavioural, environmental, and academic factors, making it suitable for a wide range of educational and analytical applications.

- The file includes information on:

  Demographics: age, gender, school type
  Family background: parent education level
  Study-related habits: daily study hours, study method, internet access
  School engagement: attendance percentage, travel time, participation in extra activities
  Academic records: marks in Math, Science, and English
  Final outcomes: overall performance score and assigned grade
- All values follow consistent formatting, column naming conventions, and realistic ranges to ensure ease of use. The dataset is clean, balanced, and ready for immediate download and analysis.

- Depiction of Data taken from Kaggle using its first 10 rows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Student_ID	Age	Gender	Academic_	Country	Avg_Daily_	Most_Use	Affects_Ac	Sleep_Hou	Mental_He	Relationsh	Conflicts_(	Addicted_Score	
2	1	19	Female	Undergrad	Banglades	5.2	Instagram	Yes	6.5	6	In Relatior	3	8	
3	2	22	Male	Graduate	India	2.1	Twitter	No	7.5	8	Single	0	3	
4	3	20	Female	Undergrad	USA	6	TikTok	Yes	5	5	Complicate	4	9	
5	4	18	Male	High Schoc	UK	3	YouTube	No	7	7	Single	1	4	
6	5	21	Male	Graduate	Canada	4.5	Facebook	Yes	6	6	In Relatior	2	7	
7	6	19	Female	Undergrad	Australia	7.2	Instagram	Yes	4.5	4	Complicate	5	9	
8	7	23	Male	Graduate	Germany	1.5	LinkedIn	No	8	9	Single	0	2	
9	8	20	Female	Undergrad	Brazil	5.8	Snapchat	Yes	6	6	In Relatior	2	8	
10	9	18	Male	High Schoc	Japan	4	TikTok	No	6.5	7	Single	1	5	
11	10	21	Female	Graduate	South Kore	3.3	Instagram	No	7	7	In Relatior	1	4	

Table 2: Format of dataset taken from Kaggle

By combining manually collected data with externally sourced datasets, the project ensures a robust and reliable foundation for analysis. This dual approach enhances the overall quality of insights generated and supports a more comprehensive understanding of the impact of social media on students.

II. Cleaning of data:

After the successful collection of data from both primary and secondary sources, it was essential to perform data cleaning to ensure consistency, accuracy, and reliability for further analysis. Since the dataset was obtained from two different sources—manual data collection using Google Forms and external data from Kaggle—there were variations in structure, format, and representation. These differences made it necessary to standardize the data before proceeding with analysis. The manually collected data was carefully reviewed and modified to align with the structure and format of the Kaggle dataset. This involved organizing the data into consistent rows and columns, ensuring that similar attributes from both datasets were represented uniformly. Column names, data types, and value formats were adjusted so that both datasets could be merged seamlessly. For example, categorical values such as social media platforms or usage levels were standardized, and numerical fields such as usage hours, sleep duration, and scores were formatted consistently. In addition to structural alignment, the data cleaning process also included handling missing values, removing duplicate entries, and correcting inconsistencies or errors in the dataset. Any incomplete or irrelevant records were either corrected or excluded to maintain the quality of the dataset. This step ensured that the final dataset was accurate, complete, and suitable for meaningful analysis. By transforming and aligning the manually collected data according to the structure of the Kaggle dataset, a unified and well-organized dataset was created. This cleaned dataset serves as a strong foundation for applying analytical techniques, enabling accurate identification of patterns, trends, and correlations in the subsequent stages of the project. The following are all the changes made in the collected data according to the dataset taken from Kaggle:

- First 10 responses collected without changes:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Name	Age	RollNo	Course	Cur_year_of_gra	Session_of_grad	platforms_used_multiple	platforms_used_co	platform_used_one	hours_spent	content_preference_multiple	content_preference_one	engagement_frequency
2	Satwinder Kaur	21	255351	BSc IT	3	2023 - 2026	Instagram, LinkedIn, Youtube, Whats	4	WhatsApp	3	Educational, Entertainment, News and current	Entertainment	Very Rarely
3	Arpita	20	255220	BCA	3	2023 - 2026	Instagram	1	Instagram	1	Entertainment	Entertainment	Very Rarely
4	Kirti	21	255361	Bsc IT	3	2023 - 2026	Youtube, WhatsApp	2	You tube	3	Educational, Entertainment, Motivational / Insp	Entertainment	Very Rarely
5	Sharanjeet	18	255205	BCA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	instagram	1	Entertainment	Entertainment	Very Rarely
6	Vriddhi	20	255232	BCA	3	2023 - 2026	Instagram, WhatsApp	2	Instagram	2	Educational, Entertainment, Technology and g	Lifestyle and fashion	Occasionally
7	Jasmeet Kaur	20	..	BCA	3	2023 - 2026	Instagram, LinkedIn, Youtube, Whats	4	Instagram	6	Educational, Entertainment, Motivational / Insp	Entertainment	Often
8	Prabhjot kaur	20	255202	BCA	3	2023 - 2026	Youtube, WhatsApp	2	Instagram	2	Educational, Entertainment	Educational	Very Rarely
9	Niharika	20	255206	BCA	3	2023 - 2026	Instagram, Youtube, WhatsApp	3	Instagram	3	Educational, Entertainment, Motivational / Insp	Entertainment	Often
10	Amritjot kaur	21	2511563	BA	3	2023 - 2026	Youtube	4	YouTube	2	Educational, Entertainment, Motivational / Insp	Motivational / Inspirational	Occasionally
11	Paramveer	26	255362	BSc IT	3	2023 - 2026	Instagram, Youtube	5	Instagram	5	Educational, Entertainment	Entertainment	Often

Table 3: Format of manual responses as a dataset without any changes.

- Depiction of Data taken from Kaggle and its first 10 rows:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Student_ID	Age	Gender	Academic_	Country	Avg_Daily_	Most_Use	Affects_Ac	Sleep_Hou	Mental_He	Relationsh	Conflicts_(	Addicted_Score	
2	1	19	Female	Undergrad	Banglades	5.2	Instagram	Yes	6.5	6	In Relation	3	8	
3	2	22	Male	Graduate	India	2.1	Twitter	No	7.5	8	Single	0	3	
4	3	20	Female	Undergrad	USA	6	TikTok	Yes	5	5	Complicate	4	9	
5	4	18	Male	High Schoo	UK	3	YouTube	No	7	7	Single	1	4	
6	5	21	Male	Graduate	Canada	4.5	Facebook	Yes	6	6	In Relation	2	7	
7	6	19	Female	Undergrad	Australia	7.2	Instagram	Yes	4.5	4	Complicate	5	9	
8	7	23	Male	Graduate	Germany	1.5	LinkedIn	No	8	9	Single	0	2	
9	8	20	Female	Undergrad	Brazil	5.8	Snapchat	Yes	6	6	In Relation	2	8	
10	9	18	Male	High Schoo	Japan	4	TikTok	No	6.5	7	Single	1	5	
11	10	21	Female	Graduate	South Kore	3.3	Instagram	No	7	7	In Relation	1	4	

Table 4: Format of dataset taken from Kaggle.

- Changes were made in our own collected data according to the data format of that of data taken from Kaggle, to make sure both had the same format.
- Collected data after data cleaning.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_U	Most_Used_Platfor	Affects_Academic	Sleep_Hours_Per_Nig	Mental_Health_Score	Relationship_Status	Conflicts_Over_Social_	Addicted_Score
2	706	21	Female	Undergraduate	India	3	WhatsApp	Yes	8	6	Single	1	8
3	707	20	Female	Undergraduate	India	1	Instagram	No	7.5	5	Single	2	5
4	708	21	Female	Undergraduate	India	3	You tube	Yes	6	4	Single	0	2
5	709	18	Female	Undergraduate	India	1	instagram	Yes	7	3	Single	3	4
6	710	20	Female	Undergraduate	India	2	Instagram	Yes	7	5	Single	1	3
7	711	20	Female	Undergraduate	India	6	Instagram	Yes	8	6	Single	2	1
8	712	20	Female	Undergraduate	India	2	Instagram	Yes	7	4	Single	3	5
9	713	20	Female	Undergraduate	India	3	Instagram	Yes	7.5	5	Single	2	4
10	714	21	Female	Undergraduate	India	2	YouTube	Yes	6.5	6	Single	1	3
11	715	26	Female	Undergraduate	India	5	Instagram	No	6	5	Single	2	2

Table 5: Changes made to the manual responses of the survey.

- Above data has been referenced using the data taken from Kaggle. The following are its rows: 'Student_ID', 'Age', 'Gender', 'Academic_Level', 'Country', 'Avg_Daily_Usage_Hours', 'Most_Used_Platform', 'Affects_Academic_Performance', 'Sleep_Hours_Per_Night', 'Mental_Health_Score', 'Relationship_Status', 'Conflicts_Over_Social_Media', 'Addicted_Score'
- After cleaning the data and getting them in the same format, both the datasets were concatenated for obtaining the final dataset which would be further used for visualizing the data.

III. EDA (Exploratory Data Analysis) Coding:

Exploratory Data Analysis (EDA) is a crucial step in the project development process, as it helps in understanding the structure, patterns, and key characteristics of the dataset before applying advanced analytical techniques. In this project, EDA was performed using the Python programming language, which provides powerful libraries and tools for data analysis and visualization. All coding and implementation were carried out in Visual Studio Code, which served as the primary development environment for executing and managing the analysis. The main objective of performing EDA was to explore the cleaned dataset and gain meaningful insights into the relationships between different variables such as social media usage, mental health scores, sleep duration, addiction levels, and academic performance. Through this process, the data was examined in detail to identify trends, patterns, distributions, and any hidden relationships that could contribute to a better understanding of student behavior. Various Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to perform data manipulation and visualization. These tools enabled the generation of summary statistics, graphical representations, and correlation analysis, which are essential components of EDA. Techniques such as data visualization, distribution analysis, and comparison of variables were applied to interpret the dataset effectively. This section provides a detailed explanation of all the codes and methods used during the EDA process. It includes step-by-step descriptions of how the dataset was explored, analyzed, and visualized to extract meaningful insights. The outcomes of this analysis form the foundation for further stages of the project, including dashboard creation and interpretation of results. EDA - Exploratory Data Analysis of data was done using python programming language and all of this coding was done in Visual Studio code. The following are all the code snippets executed for performing analysis on the above taken dataset.

- Before starting with any of the preprocessing, all the required python libraries are imported into our jupyter notebook so that we could use them as per requirement.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

Python

Figure 2: Importing python libraries.

- After importing the required libraries, we start with reading the required datasets:

```python
# following steps comes under data profiling which we will use
# also the final data uses a combination of self collected data as well as online data
df1=pd.read_csv("Students Social Media Addiction.csv")
df2=pd.read_excel("SocialMedia_Project (Responses).xlsx")
```

Python

Figure 3: Reading datasets using pandas.read_csv().

- From now, data profiling starts, where the profile of data, its structure is analyzed and understood:

```python
df1.columns
```

Python

```
Index(['Student_ID', 'Age', 'Gender', 'Academic_Level', 'Country',
       'Avg_Daily_Usage_Hours', 'Most_Used_Platform',
       'Affects_Academic_Performance', 'Sleep_Hours_Per_Night',
       'Mental_Health_Score', 'Relationship_Status',
       'Conflicts_Over_Social_Media', 'Addicted_Score'],
      dtype='object')
```

```python
df2.columns
```

Python

```
Index(['Student_ID', 'Age', 'Gender', 'Academic_Level', 'Country',
       'Avg_Daily_Usage_Hours', 'Most_Used_Platform',
       'Affects_Academic_Performance', 'Sleep_Hours_Per_Night',
       'Mental_Health_Score', 'Relationship_Status',
       'Conflicts_Over_Social_Media', 'Addicted_Score'],
      dtype='object')
```

Figure 4: Checking columns of both datasets.

- Shape of both the datasets has been checked in the following cells, where 705 rows are of Kaggle dataset and 25 rows are that of responses of students which has been altered as per the format of Kaggle dataset.

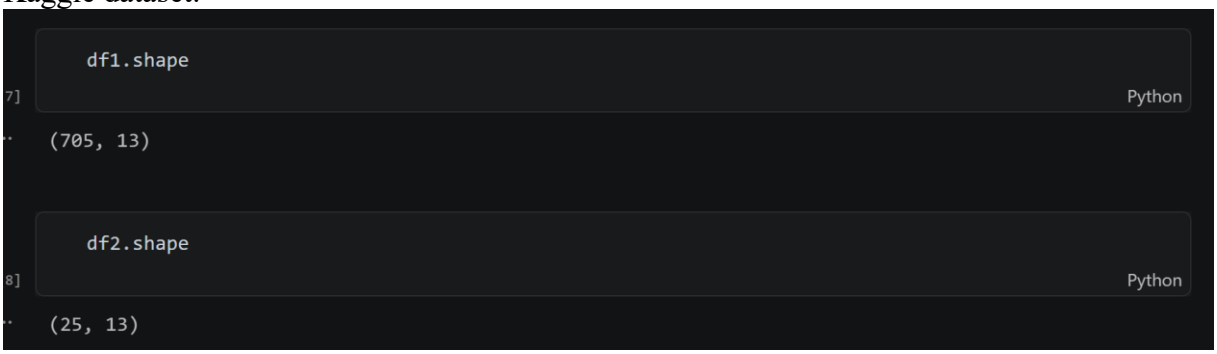
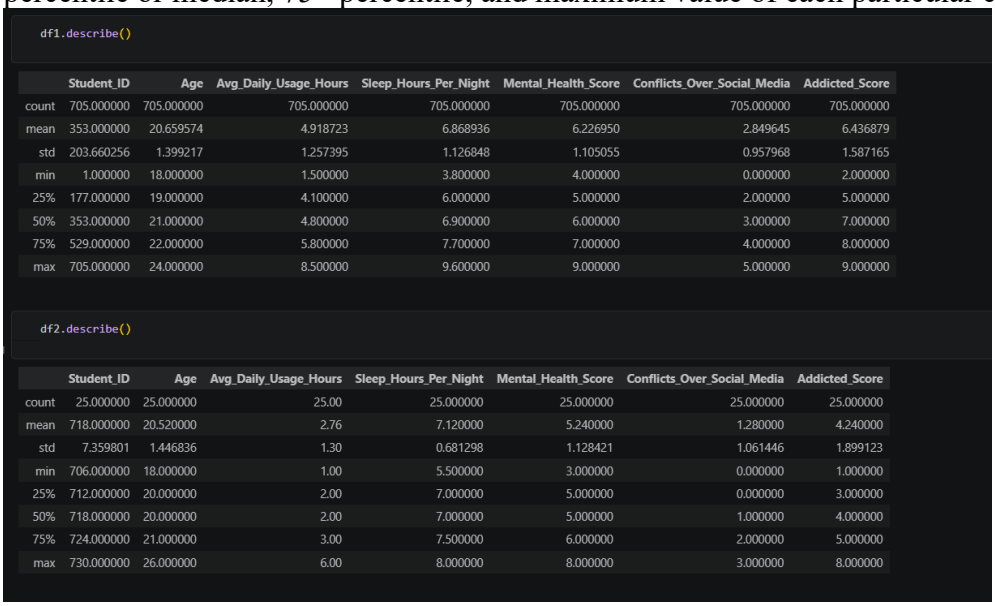


*Figure 5: Checking shapes of both the datasets(rows*columns).*

- These two cells describe the numerical columns of both the datasets, using the describe() function of python. It includes the statistical description of only numerical data as that of categorical data is not possible. It includes: count, mean, standard deviation, minimum value, 25th percentile, 50th percentile or median, 75th percentile, and maximum value of each particular column.



df1.describe()

	Student_ID	Age	Avg_Daily_Usage_Hours	Sleep_Hours_Per_Night	Mental_Health_Score	Conflicts_Over_Social_Media	Addicted_Score
count	705.000000	705.000000	705.000000	705.000000	705.000000	705.000000	705.000000
mean	353.000000	20.659574	4.918723	6.868936	6.226950	2.849645	6.436879
std	203.660256	1.399217	1.257395	1.126848	1.105055	0.957968	1.587165
min	1.000000	18.000000	1.500000	3.800000	4.000000	0.000000	2.000000
25%	177.000000	19.000000	4.100000	6.000000	5.000000	2.000000	5.000000
50%	353.000000	21.000000	4.800000	6.900000	6.000000	3.000000	7.000000
75%	529.000000	22.000000	5.800000	7.700000	7.000000	4.000000	8.000000
max	705.000000	24.000000	8.500000	9.600000	9.000000	5.000000	9.000000

df2.describe()

	Student_ID	Age	Avg_Daily_Usage_Hours	Sleep_Hours_Per_Night	Mental_Health_Score	Conflicts_Over_Social_Media	Addicted_Score
count	25.000000	25.000000	25.00	25.000000	25.000000	25.000000	25.000000
mean	718.000000	20.520000	2.76	7.120000	5.240000	1.280000	4.240000
std	7.359801	1.446836	1.30	0.681298	1.128421	1.061446	1.899123
min	706.000000	18.000000	1.00	5.500000	3.000000	0.000000	1.000000
25%	712.000000	20.000000	2.00	7.000000	5.000000	0.000000	3.000000
50%	718.000000	20.000000	2.00	7.000000	5.000000	1.000000	4.000000
75%	724.000000	21.000000	3.00	7.500000	6.000000	2.000000	5.000000
max	730.000000	26.000000	6.00	8.000000	8.000000	3.000000	8.000000

Figure 6: Describes the statistics of all the numerical columns of both the datasets using describe() of python.

- These two cells describe that no column of any dataset has any null value. This is seen using .isnull() function which tells if a dataset has any null values or not, and value_counts() function counts the values of any null values present in the dataset. This returns false as there is no any null values in the datasets.

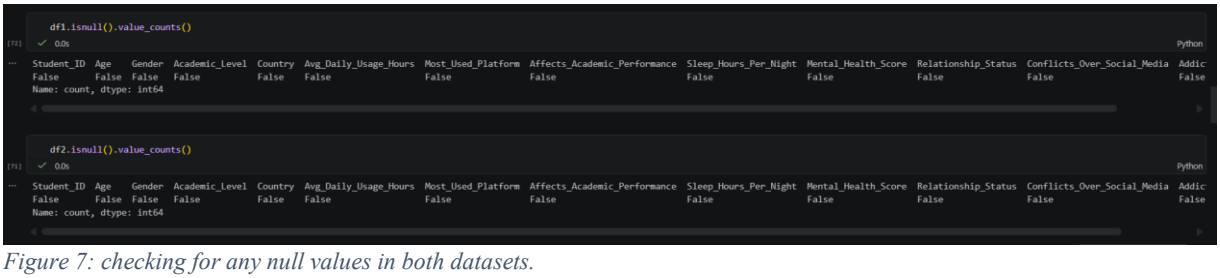


Figure 7: checking for any null values in both datasets.

- The following two cells tells about the data types of each of the columns of both datasets. These data types are accurate as the data is cleaned previously, if it wouldn't be clean, we would require to clean it in python as well. It uses the function dtypes() of python for checking the data types of the columns.

```
df1.dtypes
# all the dtypes of df1 is accurate
✓ 0.0s

Student_ID                      int64
Age                             int64
Gender                         object
Academic_Level                 object
Country                        object
Avg_Daily_Usage_Hours         float64
Most_Used_Platform             object
Affects_Academic_Performance   object
Sleep_Hours_Per_Night         float64
Mental_Health_Score             int64
Relationship_Status            object
Conflicts_Over_Social_Media     int64
Addicted_Score                  int64
dtype: object
```

```
df2.dtypes
✓ 0.0s

Student_ID                      int64
Age                             int64
Gender                         object
Academic_Level                 object
Country                        object
Avg_Daily_Usage_Hours           int64
Most_Used_Platform             object
Affects_Academic_Performance   object
Sleep_Hours_Per_Night         float64
Mental_Health_Score             int64
Relationship_Status            object
Conflicts_Over_Social_Media     int64
Addicted_Score                  int64
dtype: object
```

Figure 8: checking datatypes of both data frames.

- Here, in the following cells the final dataset has been created using concat() of pandas library which concatenates two of the datasets. Further, final datasets profile has been checked, which checks the shape of the dataset, column names, data types of the columns, null values in the column which are null.

```
final_df=pd.concat([df1,df2],ignore_index=True)
✓ 0.0s
```

```
final_df.shape
# our final df is ready to get be modelled.
✓ 0.0s

(730, 13)
```

```
final_df.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 730 entries, 0 to 729
Data columns (total 13 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Student_ID                    730 non-null    int64
 1   Age                           730 non-null    int64
 2   Gender                        730 non-null    object
 3   Academic_Level                730 non-null    object
 4   Country                       730 non-null    object
 5   Avg_Daily_Usage_Hours         730 non-null    float64
 6   Most_Used_Platform            730 non-null    object
 7   Affects_Academic_Performance  730 non-null    object
 8   Sleep_Hours_Per_Night         730 non-null    float64
 9   Mental_Health_Score           730 non-null    int64
 10  Relationship_Status           730 non-null    object
 11  Conflicts_Over_Social_Media   730 non-null    int64
 12  Addicted_Score                730 non-null    int64
dtypes: float64(2), int64(5), object(6)
memory usage: 74.3+ KB
```

Figure 9: Concatenation of both data frames and checking the final data frame's profile.

- This cell checks if final dataset has any duplicated values, whose sum is returned as 0, which means it has no duplicated values.

```
final_df.duplicated().sum()
# no duplicated values are present in this data
✓ 0.0s

np.int64(0)
```

Figure 10: Checking for any duplicated values in the final_df.

- As the data is clean and its data types are also accurately classified, so we now move forward to Exploratory data analysis. For this we classify the columns into categorical and numerical columns using list comprehension of python.

```
# as the data is clean and its dtypes are also accurately classified, so we now move forward to Exploratory data anylysis
columns=final_df.columns
categorical_features=[feature for feature in columns if final_df[feature].dtype=="O"]
numerical_features=[feature for feature in columns if final_df[feature].dtype!="O"]
```
✓ 0.0s

```
categorical_features
```
✓ 0.0s

```
'Gender',
'Academic_Level',
'Country',
'Most_Used_Platform',
'Affects_Academic_Performance',
'Relationship_Status']
```

```
numerical_features
```
✓ 0.0s

```
'Student_ID',
'Age',
'Avg_Daily_Usage_Hours',
'Sleep_Hours_Per_Night',
'Mental_Health_Score',
'Conflicts_Over_Social_Media',
'Addicted_Score']
```

Figure 11: dividing the columns into list of categorical and numerical columns using list comprehension.

- For analyzing categorical columns, we use the following code which prints the column name followed by its values it has and percentage of value counts.

```
# categorical feature analysis
for col in categorical_features:
    print(f"{col} : {final_df[col].value_counts(normalize=True)*100}\n ")
```
✓ 0.0s

Figure 12: Performing categorical feature analysis by checking each one's values and value counts.

This cell prints each column name followed by all of its unique categories and their value counts of each category. This is used to analyze values of each of the columns. The result can be seen as follows:

```
Gender : Gender
Female    51.780822
Male      48.219178
Name: proportion, dtype: float64

Academic_Level : Academic_Level
Undergraduate    51.780822
Graduate         44.520548
High School       3.698630
Name: proportion, dtype: float64

Country : Country
India          10.684932
USA             5.479452
Canada          4.657534
France          3.698630
Mexico          3.698630
                  ...
Oman            0.136986
Afghanistan     0.136986
Syria           0.136986
Yemen           0.136986
```

Figure 13: This figure shows the value counts of "Gender", "Academic_Level" and "Country".

```
Bhutan          0.136986
Name: proportion, Length: 110, dtype: float64

Most_Used_Platform : Most_Used_Platform
Instagram     34.520548
TikTok        21.095890
Facebook      16.849315
WhatsApp       7.397260
Twitter        4.109589
LinkedIn       2.876712
WeChat         2.054795
Snapchat       1.780822
KakaoTalk      1.643836
VKontakte      1.643836
LINE           1.643836
YouTube        1.369863
Instagram      1.369863
YouTube        0.547945
Youtube        0.410959
WhatsApp       0.273973
You tube       0.136986
instagram      0.136986
Whatsapp       0.136986
Name: proportion, dtype: float64

Affects_Academic_Performance : Affects_Academic_Performance
Yes    64.109589
No     35.890411
Name: proportion, dtype: float64

Relationship_Status : Relationship_Status
Single             56.027397
In Relationship    39.589041
Complicated         4.383562
Name: proportion, dtype: float64
```

Figure 14: This figure shows the value counts of "Most_used_platform", "Affects_academic_Performance", and "Relationship_status"

- From the following cells, Univariate analysis can be seen. In these cells, we have taken over each column of the dataset and represented them graphically.

```
sns.countplot(x=final_df['Gender'])
✓ 0.1s

<Axes: xlabel='Gender', ylabel='count'>
```

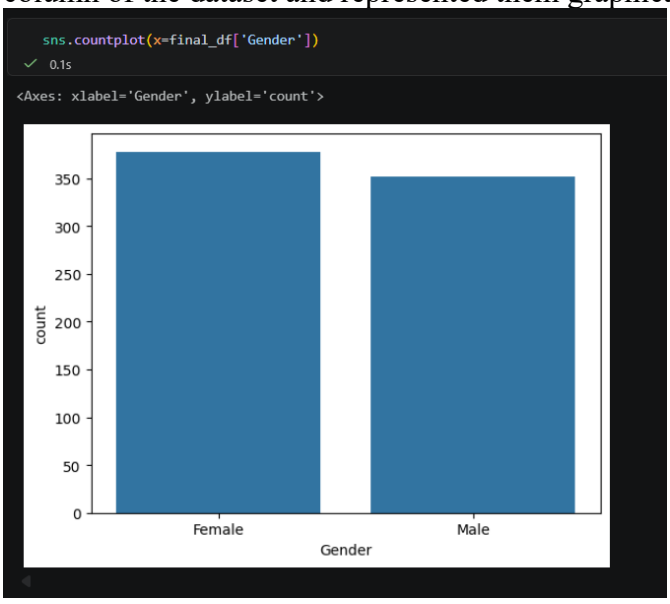


Figure 15: Shows the number of females and males individually.

In this we can see the "Gender" column and its count plot, this shows that there are over 350 students who are female and approximately 350 students are male.

- This cell checks the column – “Academic_Level”, in which more than 350 students are Undergraduate, more than 300 students are graduate and 20-30 students are in high school.

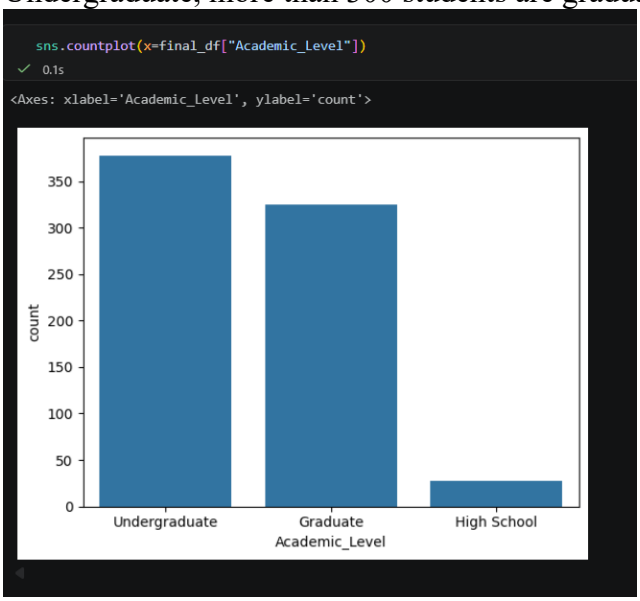


Figure 16: Counts the number of students at each level of Academics.

- Country column is checked using pie chart, which depicts that highest number of students are from India, then USA, Canada and as it can be seen in the following picture.

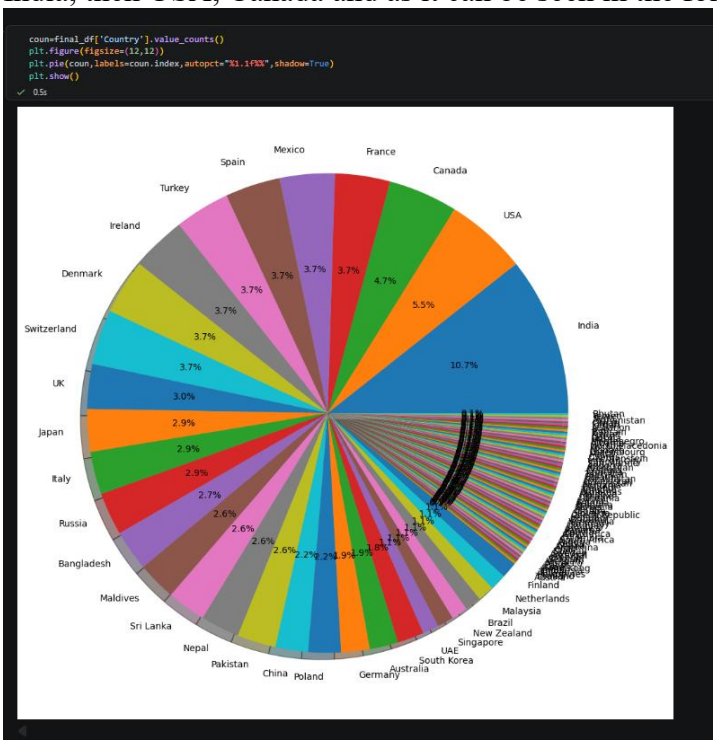


Figure 17: Shows the distribution of countries by using a pie chart.

- Further, the column platforms used is graphically represented, which shows that Instagram is used the most, followed by Tiktok, facebook, WhatsApp and as follows.

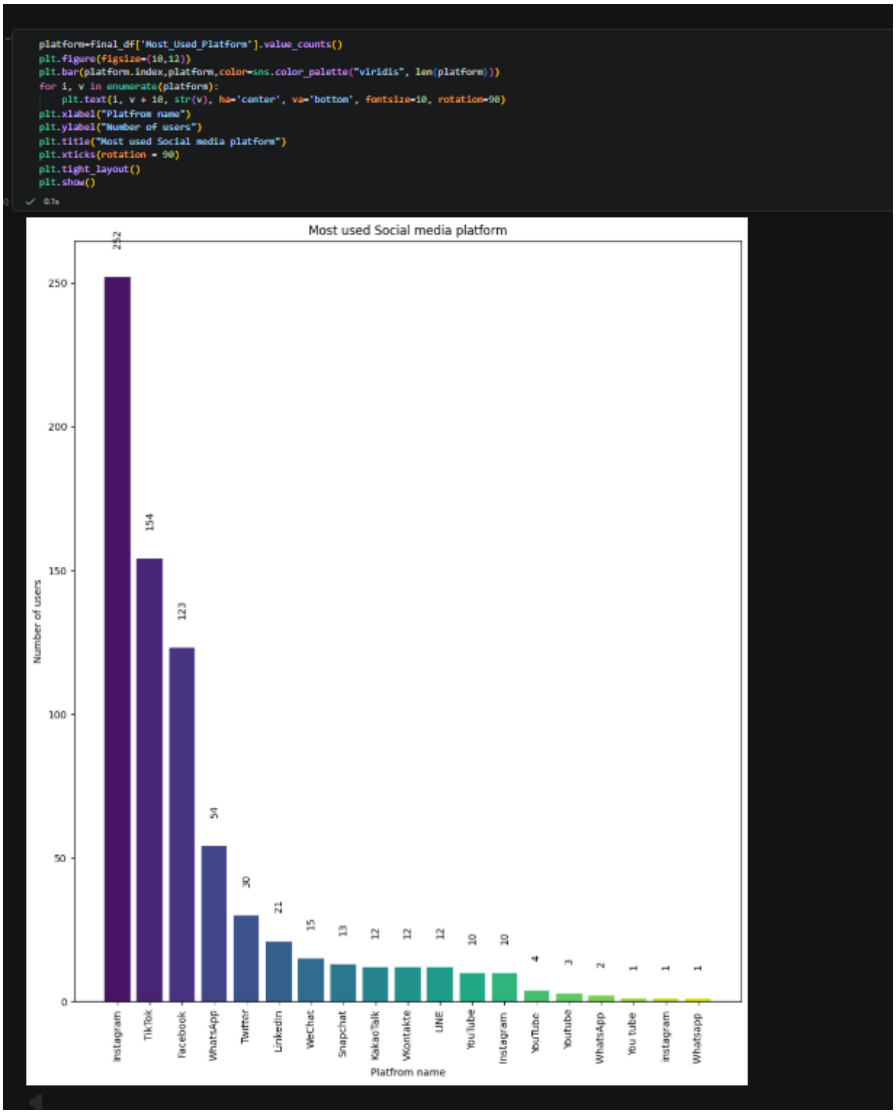


Figure 18: Depicts most used platforms using a bar plot.

- The analysis reveals that 64.1% of students believe social media impacts their academic performance, indicating a significant perceived influence on learning outcomes, while only 35.9% report no noticeable effect.

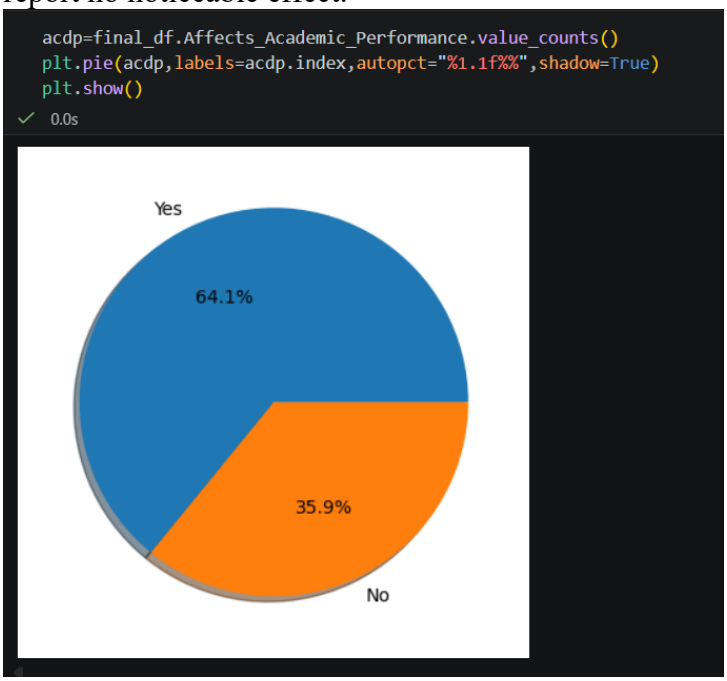


Figure 19: Depicts the values counts of effects on academic performance according on students.

- The insights reveals that 400 students out of 730 students are single, almost 300 of them are in relationship and 30 of the students say their relationship is complicated.

```
sns.countplot(final_df['Relationship_Status'])
✓ 0.0s
<Axes: xlabel='count', ylabel='Relationship_Status'>
```

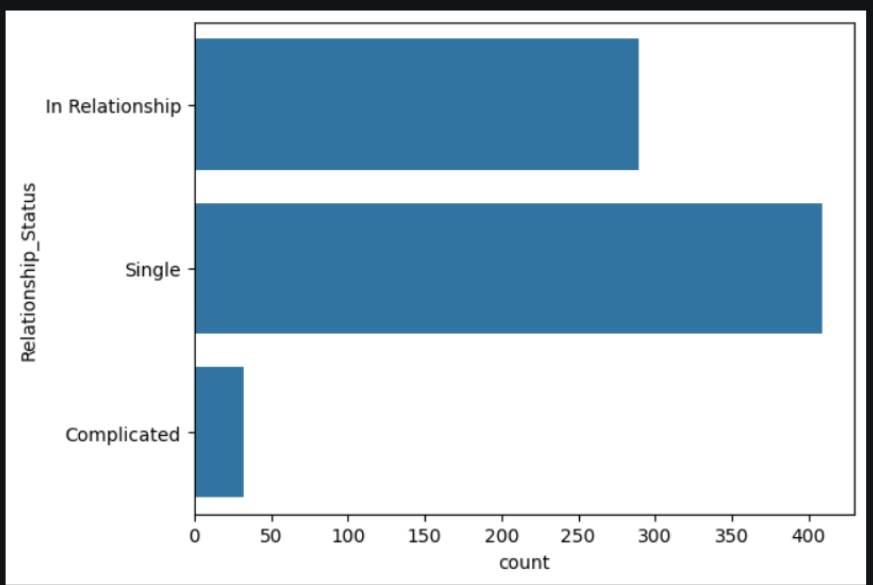


Figure 20: Shows the count plot of relationship status.

- Till here, all the categorical columns are covered. After this the numerical columns are to be covered and represented graphically.
- The analysis reveal, that the age of students range in between 18 – 26. More closed range is in between 19 – 22 years of age.

```
sns.histplot(final_df['Age'],bins=10,kde=True)
✓ 0.1s
<Axes: xlabel='Age', ylabel='Count'>
```

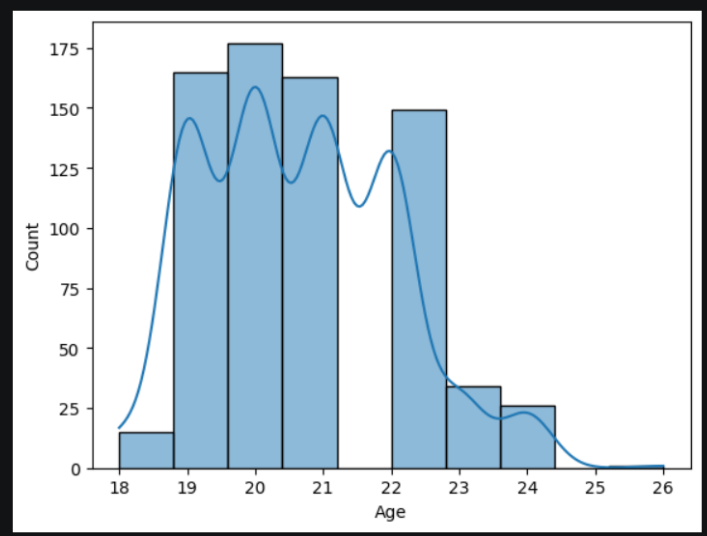


Figure 21: Shows the histogram of "Age" feature, which ranges from 18 to 26.

- The Average daily usage hours of the students range in 1-8 hours. Average hours spent og Highest number of students is between 4 – 5 hours per day. This cell uses a distplot for representing this graph.

```
sns.displot(final_df['Avg_Daily_Usage_Hours'],kde=True)
✓ 0.1s
<seaborn.axisgrid.FacetGrid at 0x1f574fb1550>
```

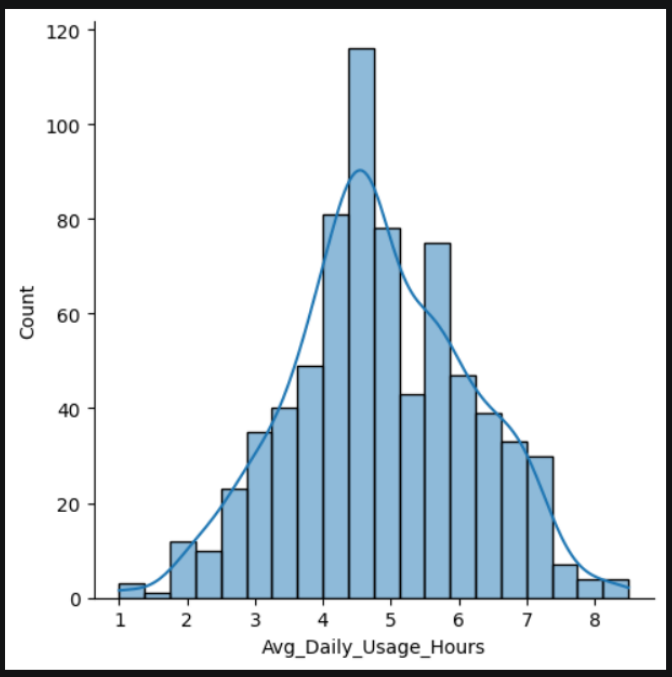


Figure 22: Distribution plot of Average Daily Usage hours of students.

- Sleep hours per night is also a numerical column graphically represented using distplot. The range of sleep hours is from 4 hours to 9 hours and more. Highest number of students have sleep hours between 7-8 hours per night.

```
sns.displot(final_df['Sleep_Hours_Per_Night'],kde=True)
✓ 0.1s
<seaborn.axisgrid.FacetGrid at 0x1f5750b5a90>
```

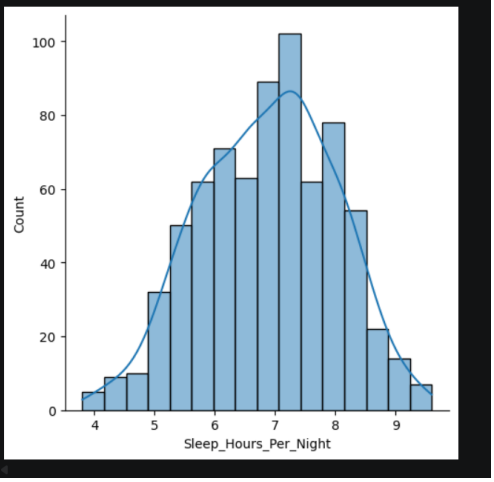


Figure 23: Distribution plot of sleep hours per night of students.

- Mental health Score can be visualized using boxplot whose range is between 3-9 and median value is 6. The box plot indicates that most students fall within a moderate to high mental health score range, with a median around 7. This suggests that the majority experience noticeable levels of the measured factor, with some reaching very high levels.

```
sns.boxplot(final_df["Mental_Health_Score"])
✓ 0.0s
<Axes: ylabel='Mental_Health_Score'>
```

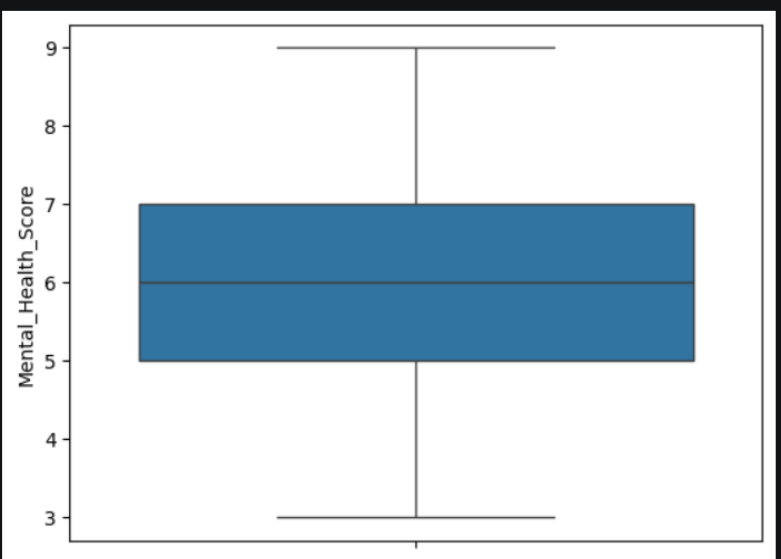


Figure 24: Box plot depicting mental health score with median value of 6.

- The chart below showcases conflict over social media using a histogram, which reveals that conflicts of students ranges from 0 to 5 and highest number of students face 3 conflicts over social media.

```
sns.histplot(final_df["Conflicts_Over_Social_Media"],bins=10,kde=True)
✓ 0.0s
<Axes: xlabel='Conflicts_Over_Social_Media', ylabel='Count'>
```

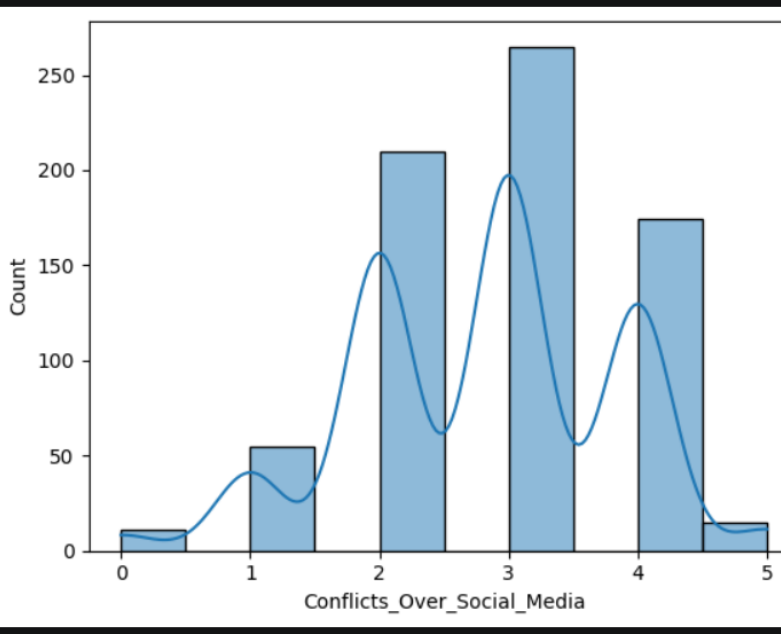


Figure 25: Histogram depicting Conflicts over social media.

- Addiction of social media is rated from 0 to 9 score. This score is graphically presented using the following distribution plot which shows that maximum students have an average addiction score of 7, few have higher scores, and rest has lesser score.

```
sns.displot(final_df["Addicted_Score"],kde=True)
✓ 0.1s
<seaborn.axisgrid.FacetGrid at 0x1f5783b9950>
```

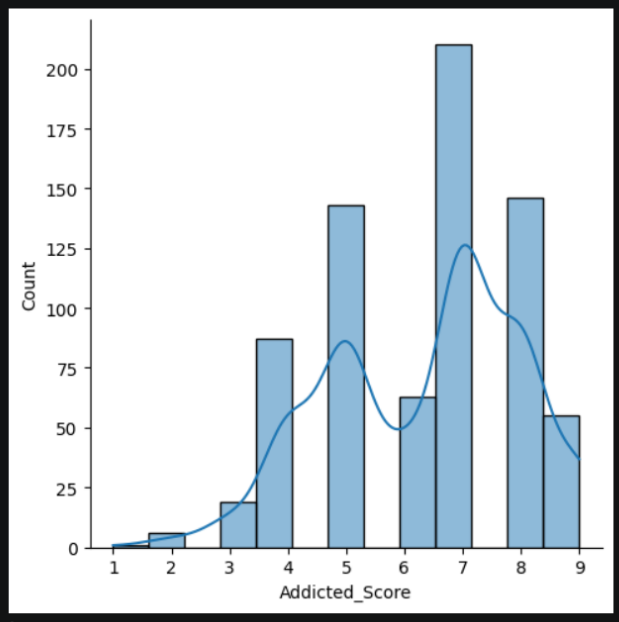


Figure 26: Distribution plot of Addiction score of students.

- Till here was a graphical depiction of univariate analysis of the student performance dataset. Now we will take a look on the graphical representation of bivariate analysis and understand them more.
- This chart is a scatter plot showing the relationship between Age and Sleep Hours Per Night. X-axis shows age of students (around 18–26). Y-axis shows Number of hours students sleep. Each dot represents one student's data. Most students sleep between 5 to 8 hours per night. Ages 19–22 have the highest concentration of data points more students in this range. There is no strong upward or downward trend → sleep hours don't significantly increase or decrease with age. A few students sleep less than 5 hours, which could indicate poor sleep habits. There is no strong correlation between age and sleep duration.
  Sleep patterns appear fairly consistent across different age groups. The scatter plot analysis reveals that sleep duration remains relatively consistent across age groups, with most students getting 5–8 hours of sleep. No strong relationship between age and sleep patterns was observed, suggesting that factors other than age influence students' sleep habits.

```
# age vs sleep hours per night
sns.scatterplot(x='Age',y='Sleep_Hours_Per_Night',data=final_df)
✓ 0.0s
<Axes: xlabel='Age', ylabel='Sleep_Hours_Per_Night'>
```

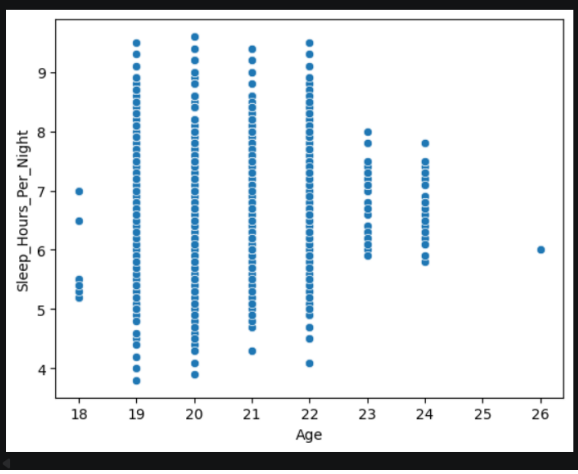


Figure 27: Scatter plot depicting relationship between Age on the x axis and sleep hours per nights on the y axis.

- This is a Seaborn pair plot, which shows relationships between multiple variables in your dataset, while also separating them by whether social media affects academic performance (Yes/No). This pair plot includes:
  - Variables used:
    - Age
    - Avg_Daily_Usage_Hours
    - Addicted_Score
    - Coloured by: Affects_Academic_Performance (Yes/No)
  - How to read it:
    - Diagonal graphs → Distribution (spread) of each variable
    - Off-diagonal graphs → Relationship between two variables
    - Colours:
      Pink - Yes (affects academics)
      Teal - No (does not affect academics)
  - Key Insights:

A. Usage Hours vs Academic Impact

Students who said "Yes" mostly have higher daily usage (5–8 hours)
Students who said "No" are concentrated around lower usage (2–5 hours)
Higher screen time is strongly linked to negative academic impact

B. Addiction Score vs Academic Impact

"Yes" group shows higher addiction scores (6–10)
"No" group has lower scores (2–6)
Higher addiction = more impact on academics

C. Usage Hours vs Addiction Score

Clear positive relationship, As usage increases, addiction score also increases

D. Age vs Other Factors

Age is mostly between 18–24
No strong pattern between age and usage/addiction
Age is not a major influencing factor

- Final Conclusion:
  The pair plot clearly shows that:
  Higher social media usage → higher addiction → greater academic impact
  Age does not significantly influence these patterns.
  "The pair plot analysis reveals a strong relationship between daily social media usage and addiction levels, both of which are significantly higher among students who report negative academic impacts. However, age does not appear to play a major role in influencing these factors."



Figure 28: Pairplot depiction relationship between multiple pair of features namely: "Age","Addiction score" and Avg_daily_usage_hours".

- **The following graph depicts correlation between various important attributes of the dataset; the following are the various notable important correlations:**
  - Mental health and daily usage hours are negatively correlated
  - Sleep hours per night and daily usage hours are negatively correlated
  - Conflicts over social media and daily usage hours are negatively correlated
  - Addiction score and daily usage hours are negatively correlated
  - Mental health score and sleep hours per night are positively correlated
  - Conflicts over social media and sleep hours per night are negatively correlated
  - Addicted score and sleep hours per night are negatively correlated
  - Conflicts over social media and mental health score are negatively correlated
  - Addiction score and mental health score are negatively correlated
  - Conflicts over social media and addicted score are highly positively correlated

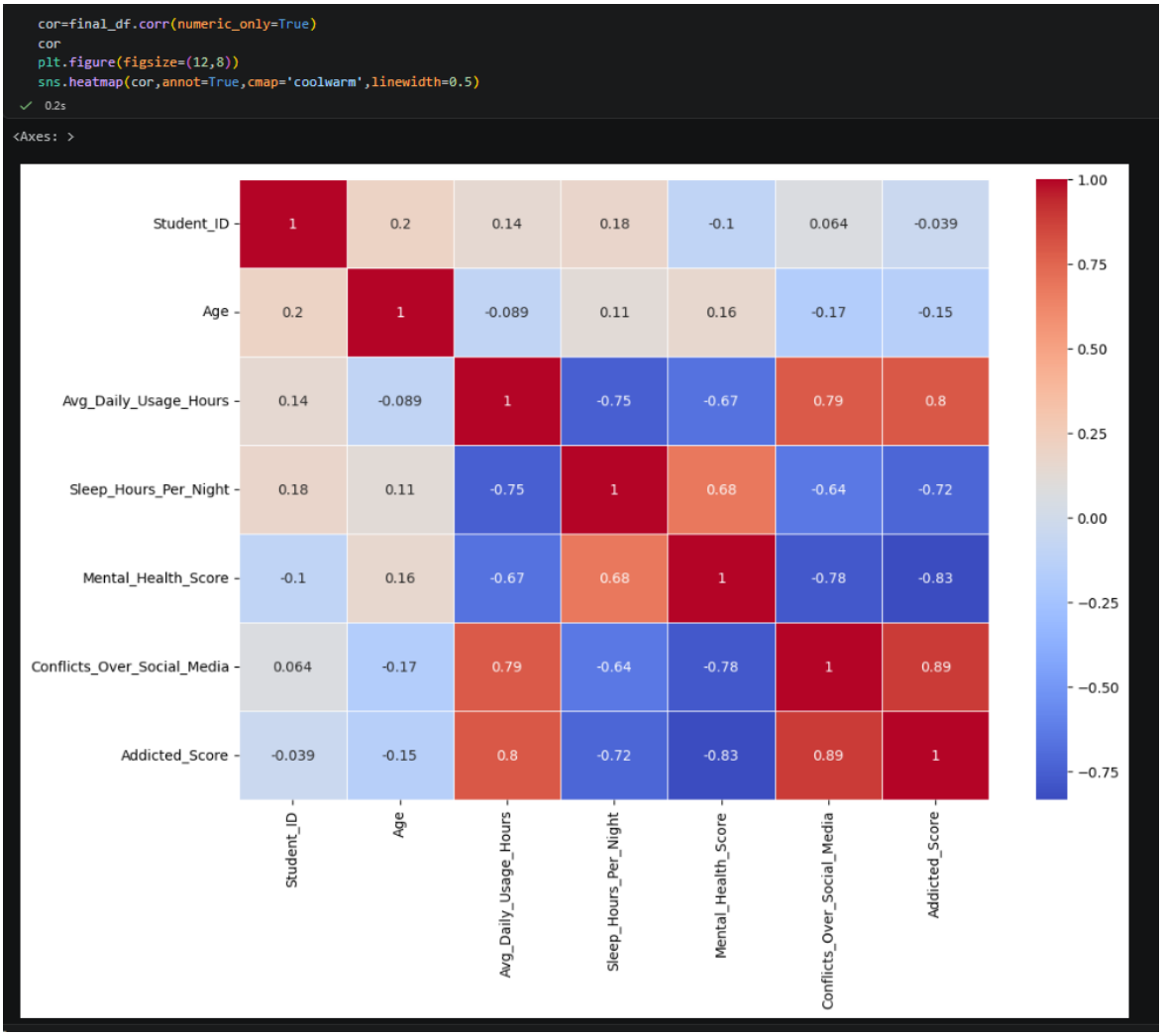


Figure 29: Correlation between multiple features

- Now that we know which columns are highly correlated, we will see each correlated columns as follows via various charts.
- The following graph shows mental health score on the x axis and daily usage hours on the y axis. These two columns has a negative correlation between each other that is, if mental health score increases (is better), then the daily usage hours should be less and vice versa.

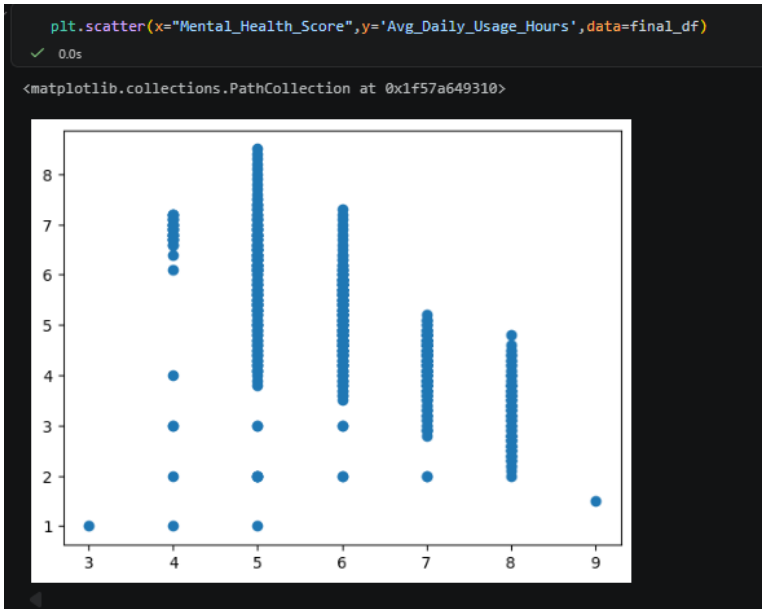


Figure 30: Scatter plot between mental health score and Avg daily usage hours. This graph shows a negative relation between the two features.

- The following graph shows sleep hours per night on the x axis and daily usage on the y axis. These two columns have a negative correlation between each other that is, if sleep hours per night increases (is better), then the daily usage hours should be less and vice versa.

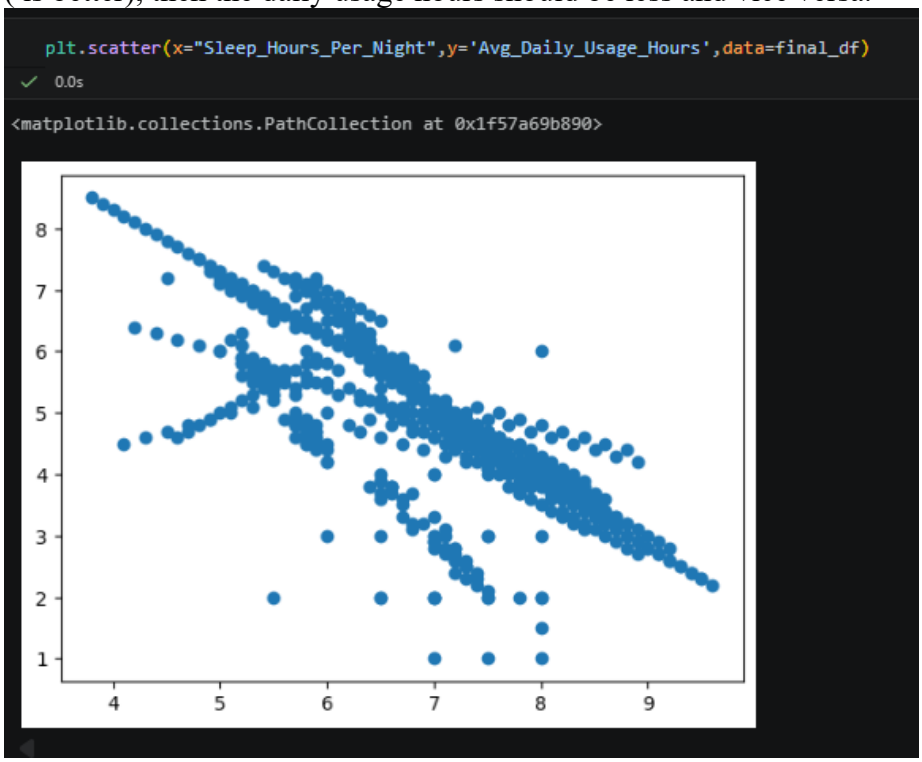


Figure 31: Scatter plot between sleep hours per night and Avg daily usage hours. This graph shows a negative relation between the two features.

- The following graph shows sleep hours per night on the x axis and Addiction score on the y axis. These two columns have a negative correlation between each other that is, if sleep hours per night increases (is better), then the addiction score becomes less and vice versa.

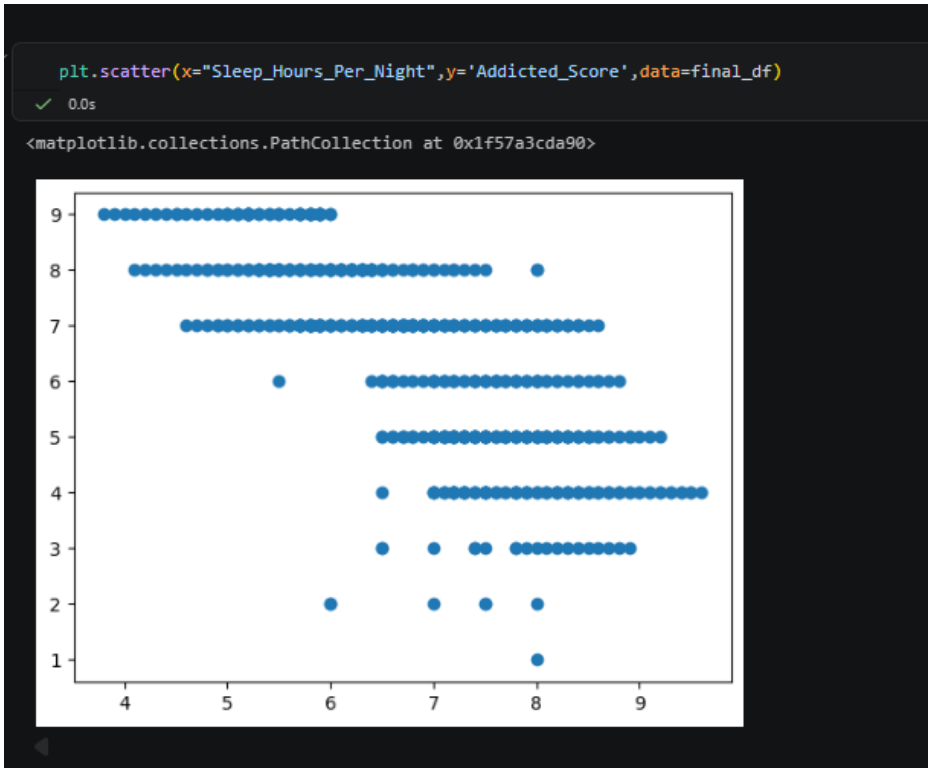


Figure 32: Scatter plot between sleep hours per night and addiction score. This graph shows a negative relation between the two features.

- The following graph shows conflicts over social media on the x axis and daily usage on the y axis. These two columns have a positive correlation between each other that is, if conflicts over social media increases, then the daily usage hours increases and vice versa.

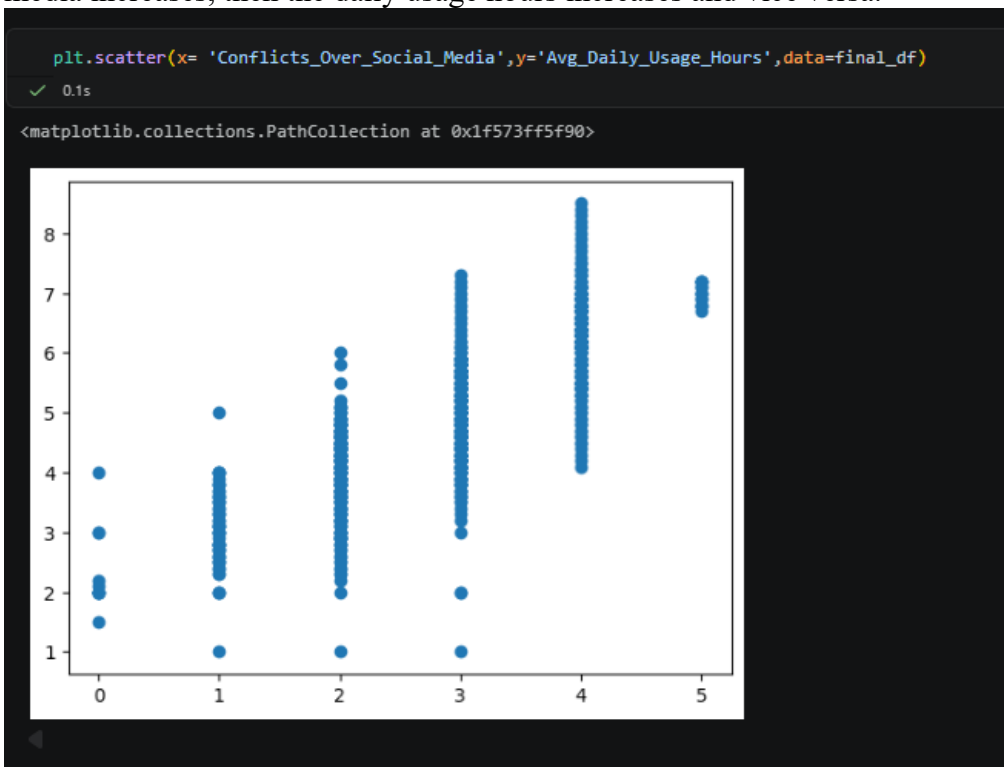


Figure 33: Scatter plot between conflicts over social media and Avg daily usage hours. This graph shows a positive relation between the two features.

- The following graph shows addiction score on the x axis and daily usage on the y axis. These two columns have a positive correlation between each other that is, if addiction score increases, then the daily usage hours increases and vice versa.

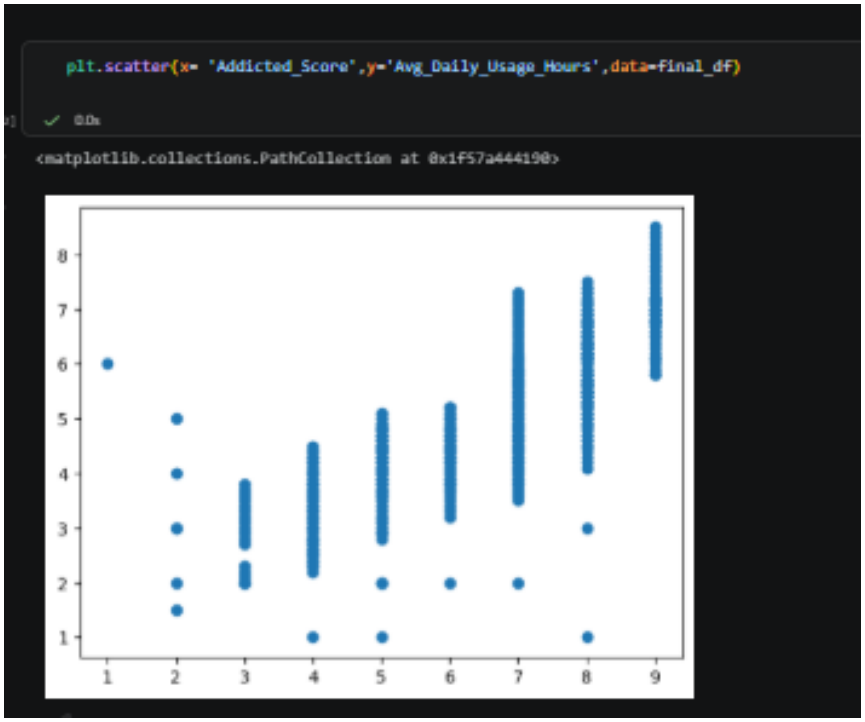


Figure 34: Scatter plot between addiction score and Avg daily usage hours. This graph shows a positive relation between the two features.

- The following graph shows mental health score on the x axis and sleep hours per night on the y axis. These two columns have a positive correlation between each other that is, if mental health score increases, then the sleep hours per night increases and vice versa.

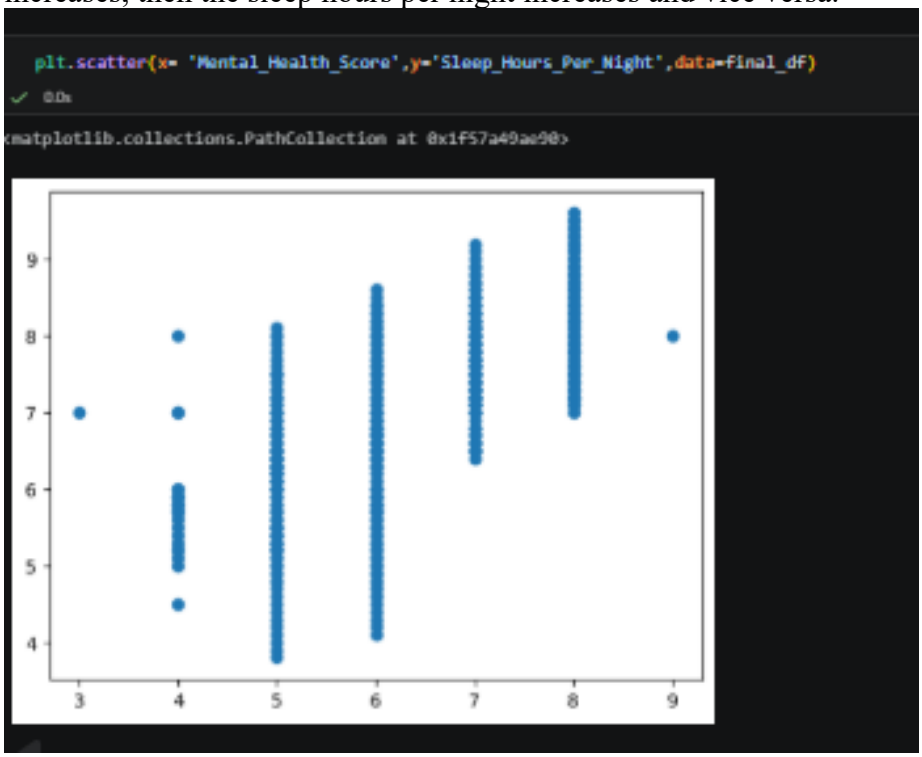


Figure 35: Scatter plot between mental health score and sleep hours per nights. This graph shows a positive relation between the two features.

- The following graph shows conflicts over social media on the x axis and sleep hours per night on the y axis. These two columns have a negative correlation between each other that is, if conflicts over social media increases, then the sleep hours per night could decrease and vice versa.

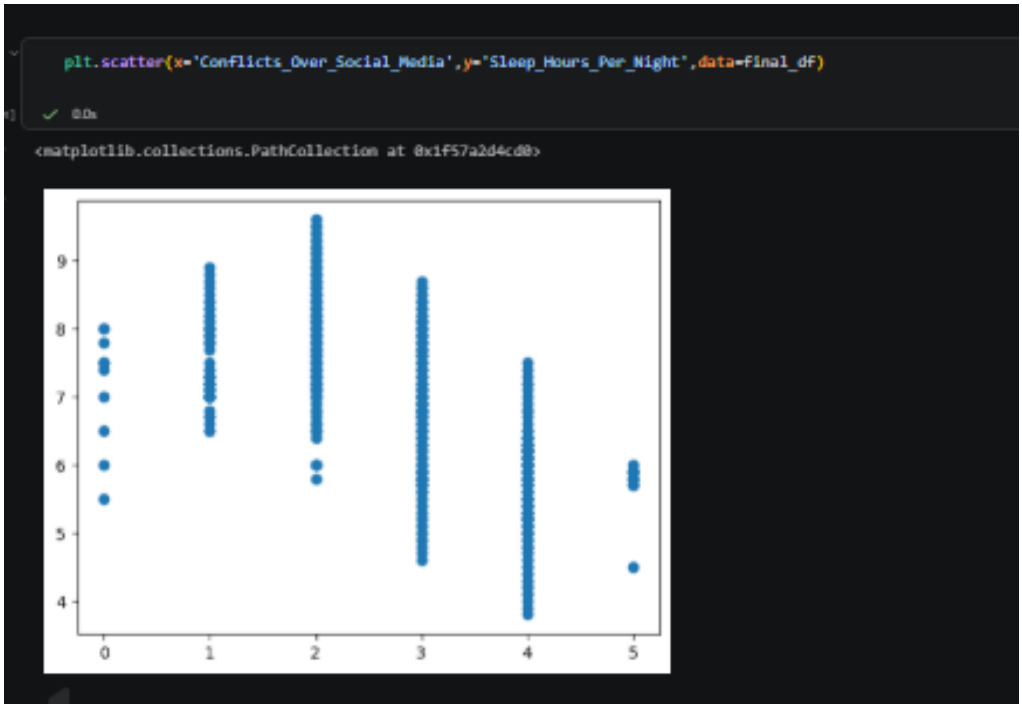


Figure 36: Scatter plot between conflicts over social media and sleep hours per night. This graph shows a positive relation between the two features.

- The following chart shows a bar plot with mental health score on the x axis and conflicts over social media on the y axis. They show a negative slope or negative correlation that means if conflicts over social media increases, then the mental health score decreases and vice-versa.

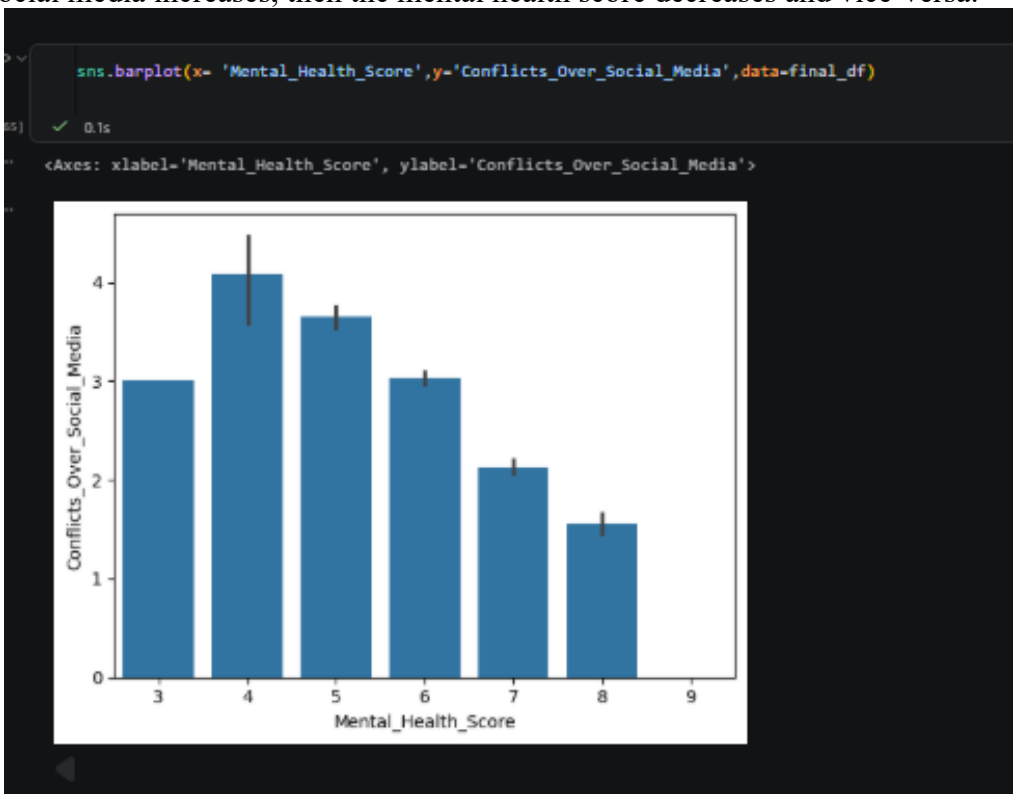


Figure 37: Bar plot depicting relationship between conflicts over social media and mental health score.

- The following chart shows a bar plot with mental health score on the x axis and conflicts over social media on the y axis. They show a negative slope or negative correlation that means if mental health score increases, then the conflicts over social media decreases and vice-versa.

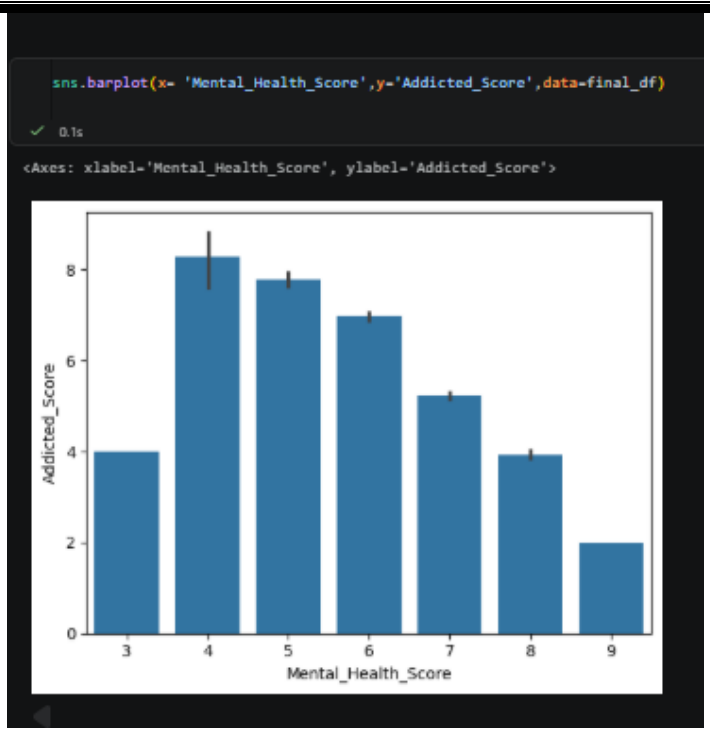


Figure 38: Bar plot depicting relationship between mental health score and addiction score.

- The following chart shows a bar plot with conflicts over social media on the x axis and addiction score on the y axis. They show a positive slope or positive correlation that means if conflicts over social media increases, then the addiction score increases and vice-versa.

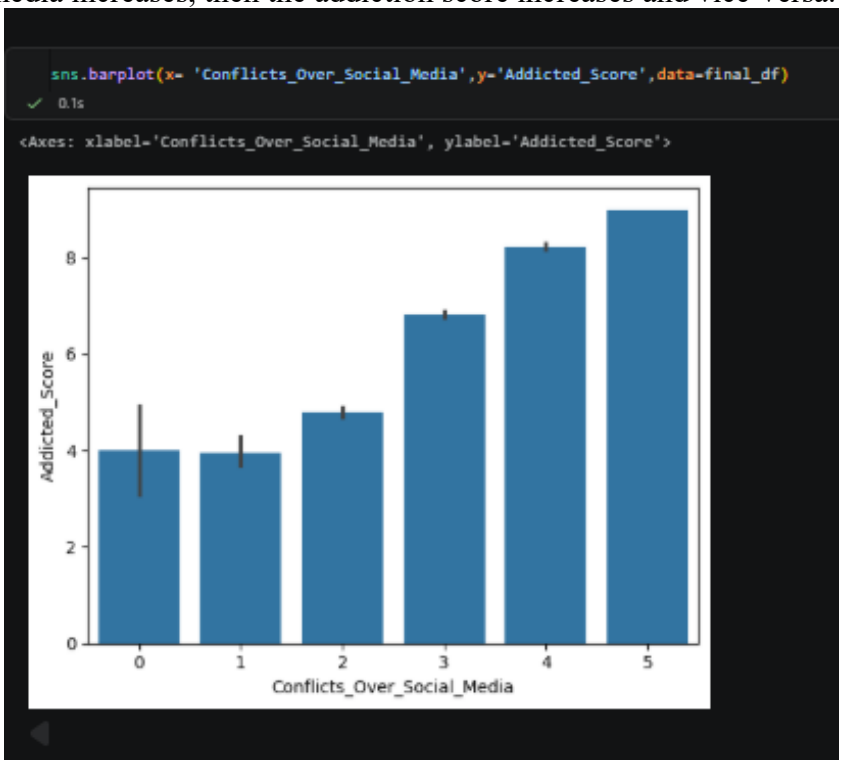


Figure 39: Barplot depicting relationship between conflicts over social media and addiction score.

- This was all about data profiling, its exploratory data analysis including its statistical analysis as well as graphical analysis. It also included univariate as well as bivariate analysis. Now lets performs machine learning models on this final dataset and check its accuracy as per the models.

IV. Machine Learning Models (Coding):

This section focuses on the development and application of machine learning models to analyze the impact of social media usage on students' academic performance and well-being. After completing data cleaning, exploratory data analysis, and feature preparation, various predictive models are implemented to identify patterns and relationships within the dataset.The primary objective of this stage is to build models that can accurately predict outcomes such as academic performance impact based on factors like daily usage hours, addiction levels, sleep patterns, and mental health scores. Different machine learning algorithms are explored and compared to determine the most effective approach for this problem.

Additionally, this section evaluates model performance using appropriate metrics, ensuring that the selected model is both reliable and generalizable. Through this process, meaningful insights are derived, helping to understand how different factors contribute to students' academic outcomes.

Overall, this section bridges data analysis and predictive intelligence, transforming insights into actionable, data-driven conclusions.

- The basic machine learning pipeline used for model building and evaluation is used as follows:
  - **Data Ingestion:** This is the first step where data is collected and loaded into the system. It can come from sources like CSV files, databases, or APIs.
    In our project, this would be loading the dataset into Python (e.g., using Pandas).
  - **Exploratory Data Analysis (EDA):** EDA is used to **understand the data** by analysing patterns, trends, and relationships.
    Checking distributions
    Finding correlations
    Visualizing data using graphs: Helps in identifying issues and insights before modelling.
  - **Data Preparation:**
    In this step, the data is **cleaned and transformed** to make it suitable for modeling.
    Handling missing values
    Encoding categorical variables
    Scaling/normalizing data: Ensures better model performance.
  - **Feature & Target Split (X, Y)**
    The dataset is divided into:
    X (Features): Input variables (e.g., age, usage hours)
    Y (Target): Output variable (e.g., academic performance impact): This defines what the model will learn and predict.
  - **Train-Test Split**
    The data is split into:
    Training set: Used to train the model
    Testing set: Used to evaluate performance: Prevents overfitting and ensures the model works on unseen data.
  - **Model Building:**
    A machine learning algorithm is selected and trained on the training data.
    Examples:
    Logistic Regression
    Decision Tree
    Random Forest: The model learns patterns from the data.
  - **Model Evaluation**
    The model's performance is measured using evaluation metrics such as:
    Accuracy
    Precision
    Recall
    F1-score: Helps determine how well the model is making predictions.

- Possible target variables for model building we have are:

  A. Addicted_Score(1-10) - Regression model,
  B. Affects_Academic_Performance(Yes, No) - classification model,
  C. Mental_Health_Score(1-10) - Regression model

- **The following is a depiction of code snippets used to create models using the above pipeline:**

1. Model – 1: Simple Linear Regression Model:

This model has one independent variable: Addiction score and one dependent variable: Average Daily Usage Hours, using these variables a simple linear regression model is made shown as follows:

- The following image depicts the code for first model, where the features x – independent variable is taken as Avg_Daily_Usage_Hours and y – dependent variable is taken as Addicted_Score, are selected and further are split using "train_test_split" under sklearn.model_selection. This splitting result in x_test, x_train , y_train and y_ test. Test size is taken as 0.3. Further, LinearRegression is imported from sklearn.linear_model and is used to fit the model using x_train and y_train, i.e., using the training data from the splitted data.



Figure 40: Simple Linear Model is trained and fitted.

- Once the model is fitted, its coefficient and intercept are checked which turns out to be 1.82 and 1.39 respectively. This gives the equation to predict values for the data as:
- Y = 1.39 + 1.82*X
- This means that with 1 unit change in number of hours spent on social media on dialy basis, there is 1.8 units change in addiction score. Also, this means that average addiction score/ starting point od addiction score is 1.39 units.
- Further, y_pred is calculated using model.predict, and this prediction is being made on the x_test data.
- After the model has been built and used for prediction, it needs to be evaluated. But before evaluating the model's performance we visualise the data as a linear model as follows:
- The linear relationship between the two independent and dependent variables can be seen in the following chart.

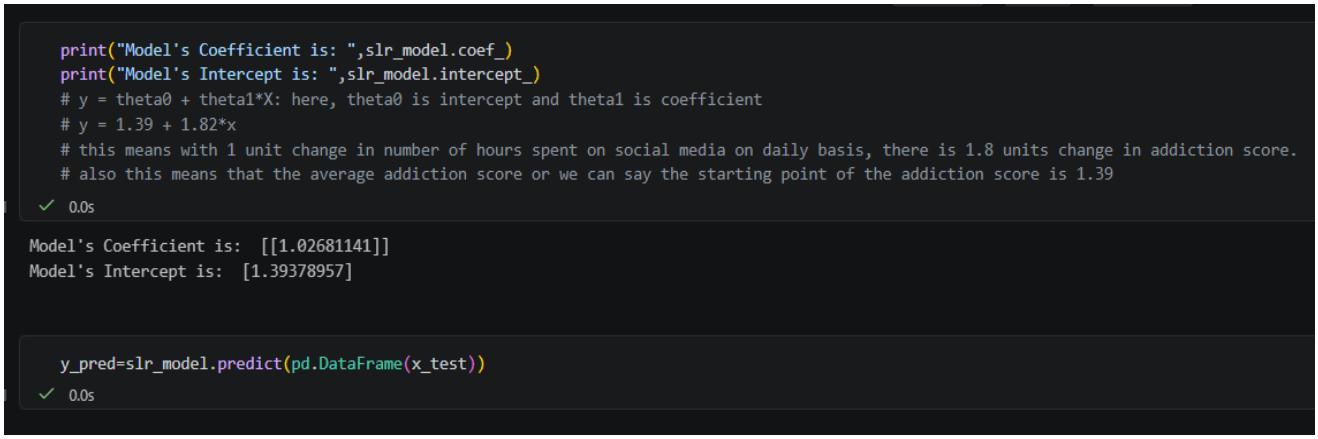


Figure 41: Model's Coefficient and intercept is calculated.

```
# visualise the result
plt.scatter(x_test,y_test,color='black',label='Actual data')
plt.plot(x_test,y_pred,color='blue',linewidth=3,label='Predicted Linear Regression Line')
plt.xlabel("Average hours spent on daily basis on social media")
plt.ylabel("Addiction score")
plt.title("Linear regression on Addiction score of students on social media")
plt.legend()
plt.show()
```

0.1s

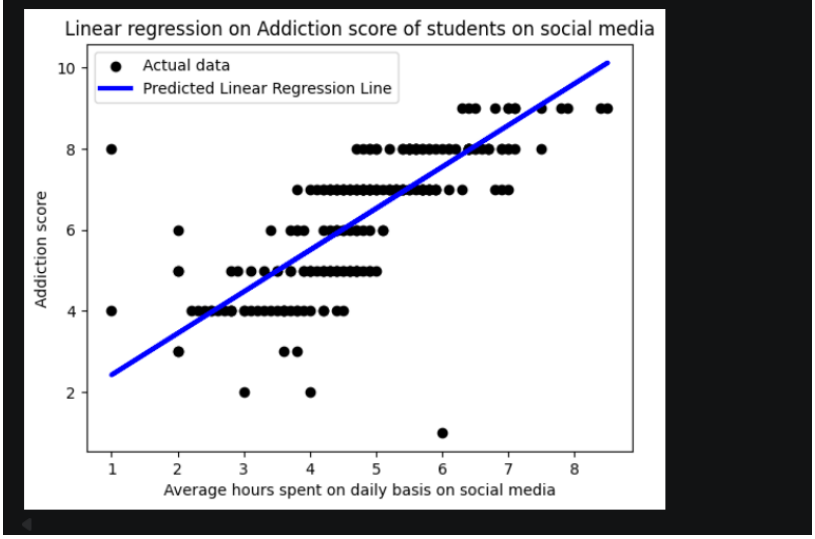


Figure 42: Linear regression plot on addiction score.

- Now that we have seen the chart, we can do the last step of an Ml pipeline, i.e., model evaluation, this can be seen as follows:

```
# evaluation
from sklearn.metrics import r2_score
r2_score(y_test,y_pred)
```

0.0s

```
0.5615984041890609
```

Figure 43: R2 Score of the SLR model is checked.

As the above model is just 56% accurate, we will try fitting Multiple linear regression, as in simple linear regression, it is not predicting/ performing that well. So, we need to take more independent variables to predict the dependent variable, hence using multiple linear regression

2. Model – 2: Multiple Linear Regression Model - I:

- For making the dataset ready for multiple linear regression, we have dropped all the unnecessary variables and fitted the model on the important variables only. For doing so, a copy of original dataset has been made. And all the non-requiring attributes has been dropped.

```
# we will build 2 MLR models : 1. with target "Addiction score", 2. "Mental health score"
# Multiple linear regression: 1. with target "Addiction score"
X=mlr_df.drop(["Addicted_Score"],axis=1)
Y=mlr_df["Addicted_Score"]
```

0.0s

```
x_train,x_test,y_train,y_test=train_test_split(X,Y,test_size=0.3,random_state=1)
mlr_model=LinearRegression()
mlr_model.fit(x_train,y_train)
```

0.0s

LinearRegression

▸ Parameters

Figure 44: Multiple linear regression model – 1 on addiction score.

- This is a Multiple linear regression model and has six independent variables: 'Academic_Level', 'Avg_Daily_Usage_Hours','Affects_Academic_Performance','Sleep_Hours_Per_Night',

  'Mental_Health_Score', 'Conflicts_Over_Social_Media' and one dependent variable: Addiction score.
- The following image depicts the code for this model, where the features x – six independent variables and y – dependent variable is taken as Addicted_Score, are selected and further are split using "train_test_split" under sklearn.model_selection. This splitting result in x_test, x_train , y_train and y_ test. Test size is taken as 0.3. Further, LinearRegression is imported from sklearn.linear_model and is used to fit the model using x_train and y_train, i.e., using the training data from the splitted data.

```
print("Model's Coefficient is: ",mlr_model.coef_)
print("Model's Intercept is: ",mlr_model.intercept_)
# y = theta0 + theta1*X: here, theta0 is intercept and theta1 is coefficient
# y = 5.15 + (-0.05*x1) + (0.23*x2) + (0.64*x3) + (-0.06*x4) + (-0.32*x5) + (0.66*x6)
```

✓ 0.0s

```
Model's Coefficient is:  [-0.05514538  0.2377898   0.64592512 -0.06735377 -0.32919975  0.66094955]
Model's Intercept is:  5.515524520867293
```

Figure 45: MLR model's coefficient and intercept are calculated.

- After the model has been fitted, its coefficient and intercept are calculated and equation to predict y is shaped. The intercept value is 5.15 and six of the coefficient's values are: -0.05, 0.23, 0.64, -0.06, -0.32, 0.66. These six coefficients represent each of the independent attributes respectively.
- The intercept is the baseline value of the dependent variable (Addiction Score) when all independent variables are zero. This acts as a starting point of the model. Even without any influence from other factors, the addiction score begins at 5.15.
- The Coefficients:
  - Academic_Level (-0.05): A slight increase in academic level leads to a small decrease in addiction score.
  - Avg_Daily_Usage_Hours (+0.23): Higher daily usage results in an increase in addiction score.
  - Affects_Academic_Performance (+0.64): If social media impacts academics, the addiction score increases significantly.
  - Sleep_Hours_Per_Night (-0.06): More sleep is associated with a slight reduction in addiction score.
  - Mental_Health_Score (-0.32): Better mental health leads to a noticeable decrease in addiction score.
  - Conflicts_Over_Social_Media (+0.66): Increased conflicts due to social media strongly raise the addiction score.
- Regression model can be written as:

- $Y = 5.15 – 0.05*X1 + 0.23*X2 + 0.64X3 – 0.06*X4 – 0.32*X5 + 0.66X6$

- Now that the model is built, it is evaluated using r2 score:
- This models' performance is very good with a percentage of 80.9. this means that predicting addiction score by using only one independent variable vas not affective, whereas predicting the same using multiple variables is much more affective and this multiple linear regression model performs much more well than the previous simple linear regression model.

```
y_pred=mlr_model.predict(x_test)
from sklearn.metrics import r2_score
r2_score(y_test,y_pred)
# the model has imporved very much
```

✓ 0.0s

```
0.8097001
```

Figure 46: Model's R2 score is calculated.

- The following scatter plot compares actual values (x-axis) with predicted values (y-axis) from the Multiple Linear Regression model.
- Each point represents how close the model's prediction is to the true value.
- Points closer to an imaginary diagonal line (perfect prediction line) indicate better accuracy.
- The plot shows a positive trend, meaning predictions generally increase with actual values.

- However, some scattered points indicate prediction errors, showing the model is not perfectly accurate but reasonably aligned.
- The model performs fairly well, with predictions roughly matching actual values, though some deviations exist.

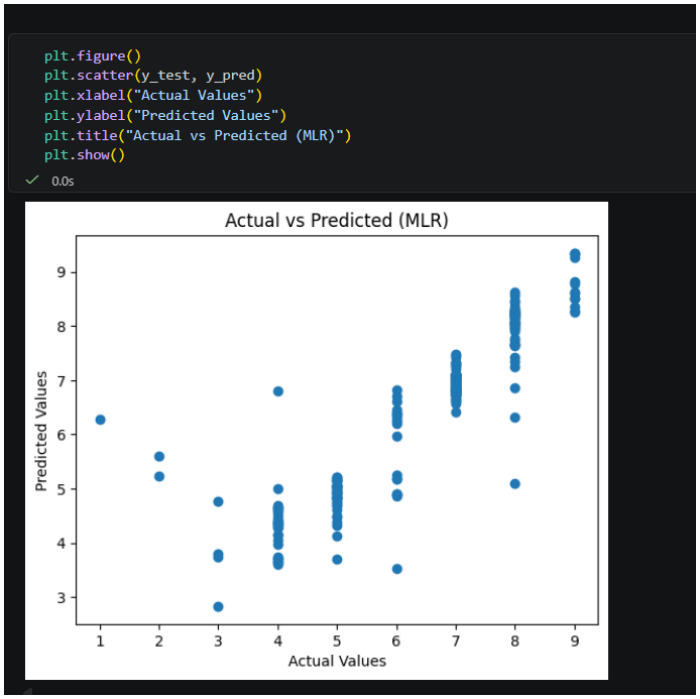


Figure 47: MLR model's Actual values distinguished with the predicted values.

- This scatter plot compares actual addiction scores (x-axis) with predicted scores (y-axis) from the model.
- Points close to a diagonal line indicate accurate predictions.
- The plot shows a positive trend, meaning predicted values generally follow actual values.
- Some scattered points indicate prediction errors, especially at certain values.
- Summary: The model shows good accuracy, with predictions reasonably aligned.

3. Model – 3: Multiple Linear Regression Model – II

- The previous model was also a MLR model with target variable as addiction score , this is also a MLR model with target variable as mental health score.
- This is a Multiple linear regression model and has six independent variables: 'Academic_Level', 'Avg_Daily_Usage_Hours','Affects_Academic_Performance','Sleep_Hours_Per_Night', 'Addiction score', 'Conflicts_Over_Social_Media' and one dependent variable: 'Mental_Health_Score'.
- The following image depicts the code for this model, where the features x – six independent variables and y – dependent variable is taken as Addicted_Score, are selected and further are split using "train_test_split" under sklearn.model_selection. This splitting result in x_test, x_train , y_train and y_ test. Test size is taken as 0.3. Further, LinearRegression is imported from sklearn.linear_model and is used to fit the model using x_train and y_train, i.e., using the training data from the splitted data.
- After the model has been fitted, its coefficient and intercept are calculated and equation to predict y is shaped. The intercept value is 7.5 and six of the coefficient's values are-0.068, 0.086, -0.328, 0.19, -0.228, -0.34.
- These six coefficients represent each of the independent attributes respectively.
- The intercept represents the **baseline Mental Health Score** when all independent variables are zero. This means the predicted mental health score starts at **around 7.5** in the absence of other influencing factors.

```
X=mlr_df.drop(["Mental_Health_Score"],axis=1)
Y=mlr_df["Mental_Health_Score"]
x_train,x_test,y_train,y_test=train_test_split(X,Y,test_size=0.3,random_state=1)
mlr_model=LinearRegression()
mlr_model.fit(x_train,y_train)
print("Model's Coefficient is: ",mlr_model.coef_)
print("Model's Intercept is: ",mlr_model.intercept_)
# y = theta0 + theta1*X: here, theta0 is intercept and theta1 is coefficient
y_pred=mlr_model.predict(x_test)
from sklearn.metrics import r2_score
print("R2 Score: ",r2_score(y_test,y_pred))
# the model has imporved very much
# y = theta0 + theta1*X: here, theta0 is intercept and theta1 is coefficient
# y = 7.152 + (-0.06*x1) + (0.86*x2) + (-0.32*x3) + (0.19*x4) + (-0.22*x5) + (-0.34*x6)
```

✓ 0.0s

```
Model's Coefficient is:  [-0.06832127  0.08654808 -0.32893315  0.19580107 -0.22879215 -0.34016597]
Model's Intercept is:  7.521746317036159
R2 Score:  0.6330658017158894
```

Figure 48: MLR model on mental health score is trained and fitted. Its coefficients and intercept values are calculated. Further its R2 scoe is being calculated.

- Each coefficient shows how the Mental Health Score changes with a 1-unit increase in that variable (keeping others constant):
  - Academic_Level (-0.068): Higher academic level slightly reduces mental health score.
  - Avg_Daily_Usage_Hours (+0.086): More usage slightly improves mental health score (very weak effect).
  - Affects_Academic_Performance (-0.328): If academics are affected, mental health decreases significantly.
  - Sleep_Hours_Per_Night (+0.19): More sleep improves mental health.
  - Addiction_Score (-0.228): Higher addiction leads to lower mental health.
  - Conflicts_Over_Social_Media (-0.34): More conflicts strongly reduce mental health.
  - Summary: Positive coefficients improve mental health, while negative coefficients worsen it.
- The R^2 score indicates how well the model explains the data. This means the model explains about 63% of the variation in Mental Health Score. The model has moderate accuracy. It captures a goodportion of the relationship, but some variation is still unexplained

```
plt.figure()
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted (MLR)")
plt.show()
```

✓ 0.1s

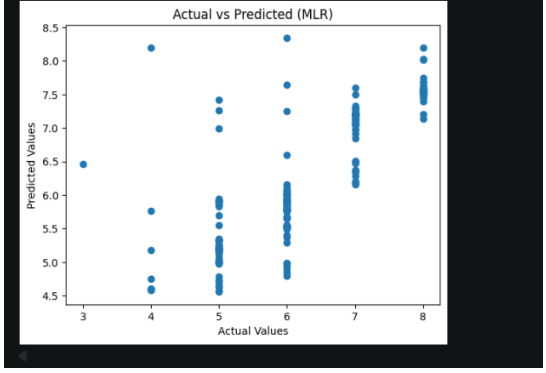


Figure 49: Actual values of this MLR model is being distinguished to its predicted values.

- This scatter plot compares actual addiction scores (x-axis) with predicted scores (y-axis) from the model.
- Points close to a diagonal line indicate accurate predictions.
- The plot shows a positive trend, meaning predicted values generally follow actual values.
- Some scattered points indicate prediction errors, especially at certain values.
- Summary: The model shows moderate accuracy, with predictions reasonably aligned but not perfectly precise.

4. Model - 4: Classification Model:

The last model is a classification model taking multiple variables as independent variables and dependent variable as "Affects_Academic_performace" whose values are in the form of yes/no. The following image depicts the feature selection of this model:

```
4th model: classification model. Dependent/target variable: 'Affects_Academic_Performance' independent variables:

    mlr_df.columns

Index(['Academic_Level', 'Avg_Daily_Usage_Hours',
       'Affects_Academic_Performance', 'Sleep_Hours_Per_Night',
       'Mental_Health_Score', 'Conflicts_Over_Social_Media', 'Addicted_Score'],
      dtype='object')

    X=mlr_df.drop(['Affects_Academic_Performance'],axis=1)
    Y=mlr_df['Affects_Academic_Performance']
```

Figure 50: independent and dependent variables are divided for developing a classification model.

- The features taken are further split into training and testing data using train_test_split under sklearn.model_selection. Further this model uses LogisticRegression under sklearn.Linear_model for performing classification model. This classifier uses train data to fit the model and uses test data to predict the model.
- Furthermore, the performance of this model is evaluated using confusion matrix which resukts in minimal false values. Accuracy score is checked which turns out to be 98%, which is highly accurate. Classification report is checked, which includes values of 0's, 1's, accuracy, precision, recall, f1-score, support.

```
    from sklearn.model_selection import train_test_split
    x_train,x_test,y_train,y_test=train_test_split(X,Y,test_size=0.20,random_state=1)
    from sklearn.linear_model import LogisticRegression
    classifier=LogisticRegression()
    classifier.fit(x_train,y_train)
    y_pred=classifier.predict(x_test)
    y_pred_proba=classifier.predict_proba(x_test)[:,1]
    from sklearn.metrics import confusion_matrix,accuracy_score,classification_report
    print("Confusion matrix: \n",confusion_matrix(y_test,y_pred))
    print("Accuracy score: ",accuracy_score(y_test,y_pred))
    print("Classification report",classification_report(y_test,y_pred))
✓ 0.0s

Confusion matrix:
 [[55  0]
 [ 3 88]]
Accuracy score:  0.9794520547945206
Classification report              precision    recall  f1-score   support

           0       0.95      1.00      0.97        55
           1       1.00      0.97      0.98        91

    accuracy                           0.98       146
   macro avg       0.97      0.98      0.98       146
weighted avg       0.98      0.98      0.98       146
```

Figure 51: a classification model is being trained, fitted and predicted values on the basis of trained model. Further, its classification metrics has been calculated which generates 97% of accuracy.

- After the model has been evaluated, lets visualize its ROC-AUC curve, which is used to showcase the

accuracy of a classification model. The dark orange line on the true positive rate represents the roc curve, that means it is really close to '1', so it is highly accurate.

```
from sklearn.metrics import roc_curve,auc
fpr,tpr, thresholds=roc_curve(y_test,y_pred_proba)
roc_auc=auc(fpr,tpr)

# plot of roc_auc

plt.figure(figsize=(8, 6))  # Create a new figure with a specified size
plt.plot(fpr, tpr, color='darkorange', linewidth=2, label='ROC curve (area = %0.2f)' % roc_auc)  # Plot ROC curve,
plt.plot([0, 1], [0, 1], color='navy', linewidth=2, linestyle='--')  # Plot the diagonal line representing random guessing
plt.xlim([0.0, 1.0])  # Set x-axis limits
plt.ylim([0.0, 1.05])  # Set y-axis limits
plt.xlabel('False Positive Rate')  # Set x-axis label
plt.ylabel('True Positive Rate')  # Set y-axis label
plt.title('Receiver Operating Characteristic (ROC) Curve')  # Set plot title
plt.legend(loc="lower right")  # Add legend to the plot
plt.show()  # Show the plot
```

0.3s

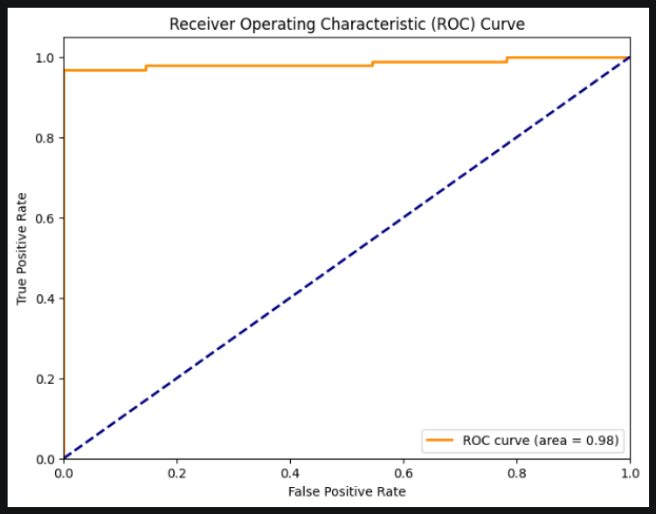


Figure 52: ROC Curve for the classification model.

Key Insights from the above Models:

- Social media usage alone does influence addiction, but it is not sufficient to explain addiction behaviour completely.
- Addiction is a multi-factor problem, not dependent on a single variable.
- Addiction is influenced by a combination of behavioural, academic, and psychological factors.
- Conflicts and academic impact are the strongest contributors to addiction.
- Including multiple variables makes predictions much more accurate and realistic.
- Mental health is strongly affected by addiction and social conflicts.
- Sleep plays a crucial positive role in improving mental health.
- Social media usage alone has minimal direct impact, but indirect effects (via addiction/conflicts) are significant.
- Social media behaviour patterns are strong indicators of academic performance issues.

V. Visualization

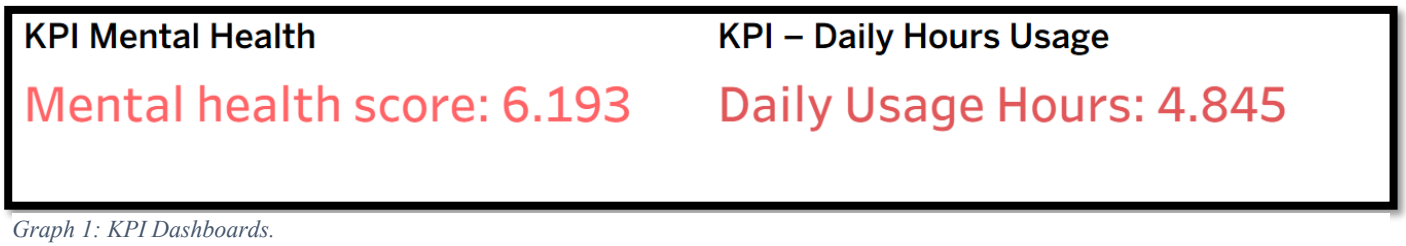
Data visualization is a vital component of the Student Social Media Analytics project, as it transforms complex analytical results into clear, interactive, and easily interpretable insights. After performing data cleaning and exploratory data analysis (EDA), the processed data was visualized using Tableau to present meaningful patterns, trends, and relationships in a user-friendly manner. Visualization not only simplifies the understanding of large datasets but also helps in effective communication of findings to both technical and non-technical users.

In this project, dashboards were created to provide a structured and layered view of the analysis. These dashboards allow users to interact with the data, apply filters, and explore insights dynamically. The visualizations include charts, graphs, and key performance indicators (KPIs) that collectively highlight the impact of social media usage on students' academic performance, mental health, sleep patterns, and addiction levels.

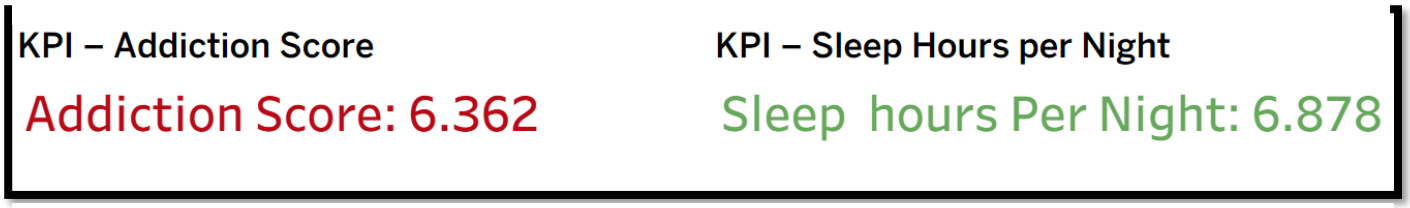
To ensure clarity and depth of analysis, the visualizations were divided into three main dashboards: KPI Dashboard, Basic Insights Dashboard, and Advanced Insights Dashboard. Each dashboard serves a specific purpose and provides different levels of analytical detail.

1) KPI Dashboard:

The KPI (Key Performance Indicators) Dashboard provides a high-level summary of the most important metrics derived from the dataset. It is designed to give users a quick overview of the overall trends and key statistics related to student behaviour.



Graph 1: KPI Dashboards.



Graph 1: KPI Dashboards.

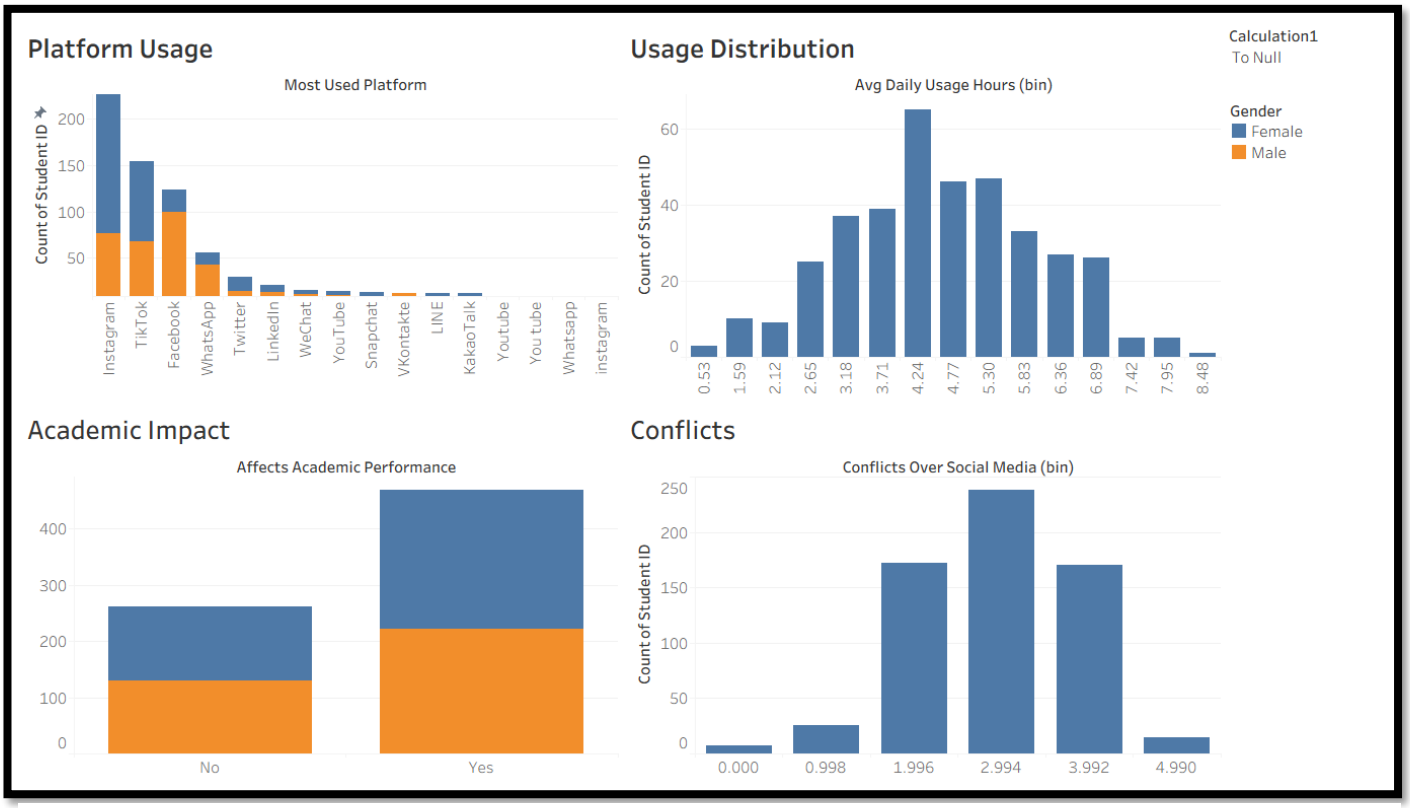
- Displays aggregated metrics such as:
  - Average daily social media usage
  - Average sleep duration
  - Average mental health score
  - Average addiction level
- Helps in quickly understanding the general condition of students without going into detailed analysis
- Uses simple visual elements such as number cards and summary indicators for easy interpretation
- Serves as the starting point for users to get an overall snapshot of the dataset

This dashboard is particularly useful for administrators and decision-makers who require quick insights without exploring detailed visualizations.

2) Basic Insights Dashboard:

The Basic Insights Dashboard focuses on fundamental relationships and patterns within the dataset. It provides a deeper understanding of how different variables interact with each other.

- Includes visualizations such as:
  - Bar charts showing distribution of social media usage
  - Pie charts representing platform preferences
  - Comparisons between usage levels and academic performance

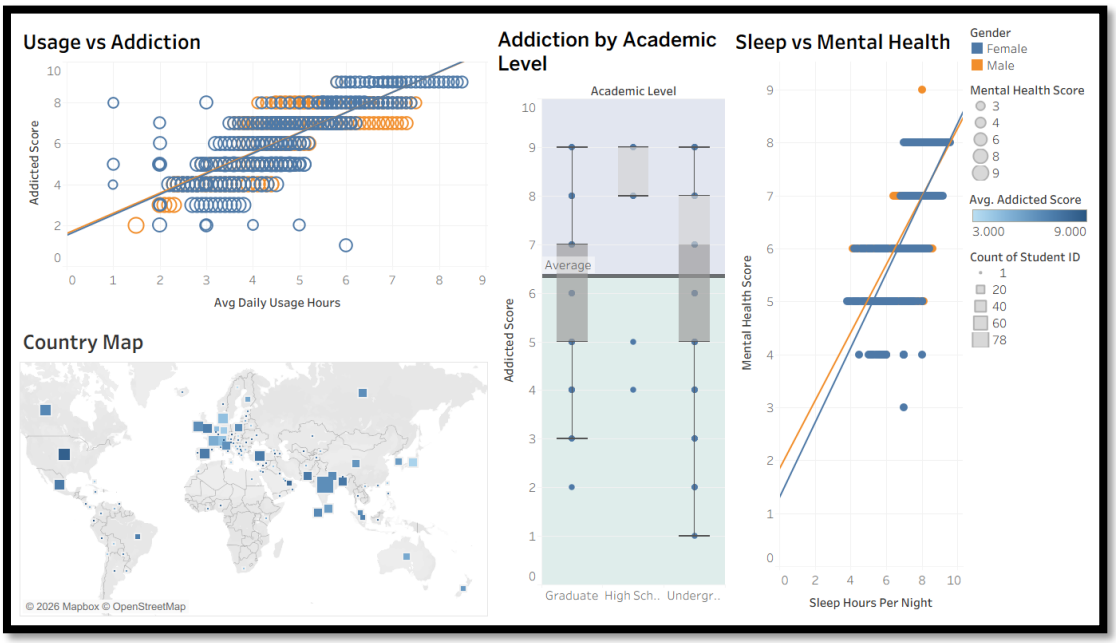


Graph -2: Usage Analysis Dashboard.

- Highlights general trends such as:
  - How many students fall into different usage categories
  - Which platforms are most popular among students
- Provides a clear and simple interpretation of data for users with minimal technical knowledge

This dashboard helps users move beyond summary statistics and begin exploring relationships between key variables. The dashboard is as follows:

3) Advanced Insights Dashboard:



Graph -3 : Advanced insights Dashboard.

The Advanced Insights Dashboard provides a detailed and in-depth analysis of the dataset by focusing on complex relationships and correlations.

- Includes advanced visualizations such as:
    - Correlation heatmaps
    - Multi-variable comparisons
    - Trend analysis graphs
- Analyses relationships such as:
    - Impact of social media usage on mental health and sleep
    - Connection between addiction levels and academic performance
    - Combined effect of multiple factors on student well-being
- Allows users to apply filters and drill down into specific data segments for deeper insights

This dashboard is designed for users who want a comprehensive understanding of the data and are interested in detailed analytical findings.

Conclusion

The visualization component of the project plays a crucial role in converting raw and analyzed data into actionable insights. By organizing the visualizations into KPI, Basic Insights, and Advanced Insights dashboards, the project ensures a structured and progressive exploration of data. These dashboards not only enhance the clarity of the findings but also make the system interactive, user-friendly, and effective for decision-making. Ultimately, data visualization bridges the gap between analysis and understanding, making the results of the project more impactful and accessible.

VI. Implementation of website:

- The implementation of the website is a crucial phase in the development of the **Student Social Media Analytics** project, as it serves as the final platform for presenting the analysis, insights, and visualizations to users. After completing data collection, cleaning, analysis, and dashboard creation, the website was developed to integrate all components into a single, accessible, and user-friendly interface.
- The primary objective of the website is to provide an interactive medium through which users—such as students, faculty, and administrators—can explore the results of the project in an organized and visually appealing manner. It acts as a bridge between complex data analysis and end-users by displaying insights through structured sections, dashboards, and key findings.
- The website was developed using modern web technologies such as HTML, CSS, and JavaScript, ensuring a responsive and visually engaging design. It includes multiple sections such as the homepage, project overview, dashboards, key insights, and other relevant pages, allowing users to navigate easily through the content. Additionally, Tableau dashboards were embedded into the website to enable real-time interaction with the visualizations.
- This section of the project focuses on explaining how the website was designed, structured, and implemented to effectively showcase the outcomes of the analysis. It highlights the tools, technologies, and design approach used to create a seamless and informative user experience.
- The following is the code for the website:
- **Index.html (Html code):**

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Analytics Portfolio</title>
<link rel="stylesheet" href="style.css">
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css">
</head>
<script type="module">
import { initializeApp } from "https://www.gstatic.com/firebasejs/10.7.1/firebase-app.js";
import { getAuth, onAuthStateChanged, signOut }
from "https://www.gstatic.com/firebasejs/10.7.1/firebase-auth.js";
const firebaseConfig = {
  apiKey: "AIzaSyDVaLxUbdPTFR1BE7w_js72Z_M8IpIpoX8",
  authDomain: "student-social-media-analysis.firebaseapp.com",
  projectId: "student-social-media-analysis",
  storageBucket: "student-social-media-analysis.firebasestorage.app",
  messagingSenderId: "939621840778",
  appId: "1:939621840778:web:6b205f509ff16e0b03fa73",
  measurementId: "G-T9QHTV60SH"
};
const app = initializeApp(firebaseConfig);
const auth = getAuth(app);
window.addEventListener("DOMContentLoaded", () => {
    onAuthStateChanged(auth, (user) => {
        if (user) {
            const emailSpan = document.getElementById("userEmail");
            if (emailSpan) {
                emailSpan.innerText = "👤 " + user.email;
            }
        } else {
            window.location.href = "login.html";
        }
    });
});
```

Figure 53: Index.html code

```
window.logout = function () {
    signOut(auth).then(() => {
        window.location.href = "login.html?logout=success";
    });
};
import { setPersistence, browserSessionPersistence } from "https://www.gstatic.com/firebasejs/10.7.1/firebase-auth.js";
setPersistence(auth, browserSessionPersistence);
</script>
<body>
<section class="hero">
    <div class="hero-content">
        <img src="images/logo.png" alt="Logo" class="logo">
        <div>
            <h1>Analysis of Social Platforms in Students</h1>
            <p>Analyzing the impact of Social Media Usage on Mental Health, Addiction, and Academic Performance</p>
        </div>
    </div>
</section>
<nav>
    <button onclick="showSection('aboutProject')">About</button>
    <button onclick="showSection('kpi')">KPI's</button>
    <button onclick="showSection('usage')">Basic Insights</button>
    <button onclick="showSection('insights')">Advanced Insights</button>
    <button onclick="showSection('keyInsights')">Findings</button>
    <button onclick="showSection('aboutMe')">About Me</button>
    <button onclick="window.location.href='contact.html'">Contact Us</button>
    <span id="userEmail" style="margin-left:auto; padding:10px;"></span>
    <button onclick="logout()" title="Logout">Logout <i class="fas fa-sign-out-alt"></i></button>
</nav>
<div id="aboutProject" class="section active">
    <section class="tools">
        <h2>Tools & Technologies Used</h2>
        <div class="tools-container">
            <img src="images/tableau_img.jpg" alt="Tableau">
            <img src="images/python_img.jpg" alt="Python">
            <img src="images/kaggle_img.svg" alt="Kaggle">
            <img src="images/google_forms_img.webp" alt="Google Forms">
            <img src="images/web.jpg" alt="Web Development">
        </div>
    </section>
```

Figure 53: Index.html code

```
    <div class="card">
        <div class="insight-box">
            <h2>About the Project</h2>
            <p>
                Student Social Media Analytics Project analyzes the impact of Social Media usage on Students' Mental Health, Addiction levels, Sleep patterns, and Academic Performance.
                This project focuses on analyzing the growing influence of social media usage on students' overall well-being and academic performance. With the rapid rise of platforms
                like Instagram, Facebook, and others, students are spending increasing amounts of time online, which may significantly affect their mental health, sleep quality,
                productivity, and daily routines.
                <br>The study leverages data analytics and visualization techniques to uncover patterns and correlations between social media usage and key lifestyle factors such as
                mental health scores, sleep duration, addiction levels, and academic outcomes. By transforming raw data into meaningful insights through dashboards and visualizations,
                the project aims to provide a clear understanding of how digital habits shape student life.
                <br> Ultimately, this project serves as a data-driven approach to highlight potential risks, encourage responsible usage, and support informed decision-making for
                students, educators, and researchers.
            </p>
        </div>
        <div class="insight-box">
            <h3>Objectives</h3>
            <ul>
                <li>Understand usage patterns across platforms:
                    <p>Examine how frequently students use social media, which platforms are most popular, and how usage varies across different demographics such as age, academic
                    level, and region.</p>
                </li>
                <li>Analyze relationship between usage and addiction:
                    <p>Examine how frequently students use social media, which platforms are most popular, and how usage varies across different demographics such as age, academic
                    level, and region.</p>
                </li>
                <li>Evaluate impact on mental health:
                    <p>Study how social media usage correlates with mental health indicators, identifying whether higher usage leads to increased stress, anxiety, or lower well-being
                    scores.</p>
                </li>
                <li>Identify Addiction trends:
                    <p>Investigate the relationship between time spent on social media and addiction scores to understand how excessive usage contributes to dependency behaviors.</p>
                </li>
                <li>Assess Effects on Sleep Patterns:
                    <p>Analyze how screen time, especially late-night usage, affects sleep duration and quality among students.</p>
                </li>
                <li>Measure Academic Performance Impact:
                    <p>Explore whether increased social media engagement negatively or positively influences students' academic performance and productivity.</p>
                </li>
```

Figure 53: index.html code

```
                <li>Generate Data-Driven Insights:
                    <p>Use interactive dashboards (KPI, Usage Analysis, and Advanced Insights) to present clear, actionable insights that help users easily interpret trends and
                    patterns.</p>
                </li>
                <li>Support Awareness and Better Decision-Making:
                    <p>Provide meaningful conclusions that can help students adopt healthier digital habits and assist educators in understanding behavioral trends.</p>
                </li>
            </ul>
        </div>
        <div class="insight-box">
            <h3>Tools Used</h3>
            <ul>
            <li>Data Collection:
                <ol>
                    <li>Google Forms:
                        <p>Google Forms was used to design and distribute surveys for collecting primary data directly from students. It enabled structured data gathering on factor
                        such as daily social media usage, sleep patterns, mental health scores, and academic performance. The tool ensured ease of access, real-time responses, and
                        efficient data export for further analysis.</p>
                    </li>
                    <li>Kaggle:
                        <p>Kaggle was utilized to obtain additional datasets and explore publicly available data related to social media usage and student behavior. It also served
                        platform for understanding data structures, preprocessing techniques, and benchmarking insights with similar projects.</p>
                    </li>
                </ol>
                    </li>
            <li>Data Profiling and Analysis:
                <p>Visual Studio Code was used as the development environment, while Python was employed for data cleaning, preprocessing, and exploratory data analysis (EDA).
                Libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to handle missing values, analyze patterns, and generate preliminary visualizations.</p>
            </li>
            <li>Data Visualization:
                <p>Tableau was used to create interactive dashboards, including KPI dashboards, usage analysis, and advanced insights. It helped transform complex datasets into
                visually appealing charts and graphs, making it easier to identify trends, correlations, and actionable insights.</p>
            </li>
            <li>Web Development:
                <p>These technologies were used to design and develop the project website. HTML structured the content, CSS enhanced the visual styling and layout, and JavaScript a
                interactivity, allowing users to navigate dashboards and insights seamlessly.</p>
            </li>
        </ul>
        </div>
```

Figure 53: index.html code

```
    </div>
</div>
<div id="kpi" class="section">
    <div class="card">
        <h2>KPI Dashboard</h2>
        <img src="images/dash1A.png" class="dashboard-img" onclick="openModal(this)">
        <img src="images/dash1B.png" class="dashboard-img" onclick="openModal(this)">
        <div class="insight-box">
            <h3>Key Insights</h3>
            <ul>
                <li>Moderate mental health score (~6.19):
                    <p>The average mental health score indicates a moderate level of well-being among students. While not critically low, it suggests that a significant portion of
                    students may be experiencing mild stress, anxiety, or emotional strain—potentially influenced by lifestyle factors such as screen time and online engagement.</p>
                </li>
                <li>Noticeable addiction level (~6.36):
                    <p>The addiction score reflects a relatively high dependency on social media platforms. This suggests that many students exhibit habitual or compulsive usage
                    patterns, which could interfere with daily activities, focus, and overall productivity.</p>
                </li>
                <li>High usage (~4.8 hrs/day):
                    <p>Students are spending a considerable amount of time on social media each day. This level of usage goes beyond casual interaction and indicates prolonged screen
                    exposure, which may contribute to reduced physical activity, lower attention spans, and increased digital fatigue.</p>
                </li>
                <li>Reduced sleep (~6.8 hrs):
                    <p>The average sleep duration is slightly below the recommended 7-9 hours for healthy individuals. This suggests that excessive screen time—especially late-night
                    usage—may be negatively affecting sleep quality and recovery.</p>
                </li>
                <li>Interconnected Impact of Usage Patterns:
                    <p>A clear relationship emerges between high social media usage, increased addiction levels, and declining well-being indicators. Students who spend more time
                    online are more likely to experience reduced sleep and moderate mental health challenges.</p>
                </li>
                <li>Conclusion: Behavioral and Health Implications
                    <p> The combined insights suggest that increased social media usage is closely linked with higher addiction levels, reduced sleep duration, and moderate mental
                    health concerns. This highlights the importance of promoting balanced digital habits and raising awareness about the long-term impact of excessive screen time.
                    Increased usage impacts mental health and sleep quality.</p>
                </li>
            </ul>
        </div>
    </div>
</div>
```

Figure 53: index.html code

```
<div id="usage" class="section">
    <div class="card">
        <h2>Usage Analysis</h2>
        <div class="dashboard-layout">
            <div class="graph-container">
                <img src="images/dash2A.png" class="graph active-graph">
                <img src="images/dash2B.png" class="graph">
                <img src="images/dash2C.png" class="graph">
                <img src="images/dash2D.png" class="graph">
            </div>
            <div class="side-panel">
                <div class="filters-panel" id="filtersPanel">
                    <img src="images/dash2_filters.png" class="filters-img">
                </div>
                <div class="analysis-text" id="analysisText">
                    <p><b>Takeaways:</b><br> This graph depicts the most used platforms by students. The bars in orange colour for males and in blue colour for females. Here, Instagram is the platform used by most students, followed by Tiktok, Facebook and so on.</p>
                </div>
            </div>
        </div>
        <div class="pagination">
            <span onclick="showGraph(0)" class="active-page">1</span>
            <span onclick="showGraph(1)">2</span>
            <span onclick="showGraph(2)">3</span>
            <span onclick="showGraph(3)">4</span>
        </div>
        <div class="insight-box" id="usageInsights">
            <h3>Key Insights</h3>
            <ul>
                <li>Instagram dominates usage:
                    <p>Among all social media platforms, Instagram emerges as the most widely used by students. Its visually engaging content, reels, and constant updates make it highly addictive and time-consuming. This dominance suggests that students are more inclined toward entertainment-driven platforms rather than informational or professional ones.</p>
                </li>
                <li>Most users spend 3-6 hours daily:
                    <p>A significant proportion of students fall within the 3-6 hours daily usage range, indicating moderate to high engagement levels. This amount of screen time goes beyond casual browsing and reflects habitual usage, which may reduce time available for academic work, physical activity, and real-world social interactions.</p>
                </li>
                <li>Academic performance is affected:
                    <p>The data indicates a noticeable impact of social media usage on students academic performance. Excessive time spent online can lead to reduced concentration, procrastination, and lower productivity, ultimately affecting grades and learning outcomes.</p>
```

Figure 53: index.html code

```
                </li>
                <li>Moderate conflicts observed:
                    <p>Students report experiencing moderate levels of conflict due to social media usage, including disagreements with family or peers and distractions during important activities. While not extremely severe, these conflicts highlight the social and behavioral implications of excessive usage.</p>
                </li>
                <li>Conclusion:
                    <p>Heavy usage impacts academics and social life. These insights suggest that while social media is deeply integrated into students daily lives, excessive usage—especially on platforms like Instagram—can lead to reduced academic focus, increased dependency, and noticeable social conflicts. Maintaining a balanced approach to usage is essential for overall well-being and productivity.</p>
                </li>
            </ul>
        </div>
    </div>
</div>
<div id="insights" class="section">
    <div class="card">
        <h2>Advanced Insights</h2>
        <div class="dashboard-layout">
            <div class="graph-container">
                <img src="images/dash3A_male.png" class="insights-graph active-insight">
                <img src="images/dash3A_female.png" class="insights-graph">
                <img src="images/dash3B.png" class="insights-graph">
                <img src="images/dash3C.png" class="insights-graph">
                <img src="images/dash3D_male.png" class="insights-graph">
                <img src="images/dash3D_female.png" class="insights-graph">
            </div>
            <div class="side-panel">
                <div class="filters-panel">
                    <img id="insightFilter" src="images/dash3_filters1.png" class="filters-img">
                </div>
                <div class="analysis-text" id="insightAnalysis">
                    <p><b>Takeaways:</b><br> This graph impact of usage hours on addiction levels on MALE students. A strong positive correlation is visible. This explains that with increase in usage hours there is increase in Addiction score as well. Usage hours extends from 1.5 to 7.5 hours of daily usage. Alternatively, addiction score extends from 3.0 to 9.0, 3 being the lowest and 9 being the highest.</p>
                </div>
            </div>
        </div>
    </div>
```

Figure 53: index.html code

```
    <div class="pagination">
        <span onclick="showInsightGraph(0)" class="active-page">1</span>
        <span onclick="showInsightGraph(1)">2</span>
        <span onclick="showInsightGraph(2)">3</span>
        <span onclick="showInsightGraph(3)">4</span>
        <span onclick="showInsightGraph(4)">5</span>
        <span onclick="showInsightGraph(5)">6</span>
        </div>
        <div class="insight-box">
            <h3>Key Insights</h3>
            <ul>
                <li>Usage strongly correlates with addiction:
                    <p>The analysis reveals a clear positive relationship between the number of hours spent on social media and the level of addiction among users. As daily usage increases, dependency behaviors such as compulsive checking, difficulty staying offline, and reduced self-control become more prominent. This indicates that prolonged exposure significantly contributes to addictive patterns.</p>
                </li>
                <li>Sleep improves mental health:
                    <p>A strong connection is observed between adequate sleep and better mental health scores. Users who maintain consistent and sufficient sleep schedules tend to report lower stress levels, improved mood, and better emotional stability. This suggests that limiting late-night social media usage can positively impact overall well-being.</p>
                </li>
                <li>Undergraduates show more variation:
                    <p>Undergraduate students display a wider range of behaviors in terms of social media usage, addiction levels, and mental health outcomes. This variability may be due to differences in lifestyle, academic pressure, social environment, and personal habits, making this group more diverse and less predictable compared to others.
                    </p>
                </li>
                <li> Global usage trends observed:
                    <p>The data highlights consistent patterns of social media usage across different countries and regions. Despite cultural and geographical differences, users exhibit similar behaviors such as high engagement, preference for visual platforms, and increasing screen time, indicating that social media influence is globally widespread.</p>
                </li>
                <li>Conclusion:
                    <p>The findings demonstrate that behavioral patterns such as excessive usage and poor sleep habits have a direct and measurable impact on mental health outcomes. Promoting balanced usage and healthier routines is essential to reduce addiction risks and improve overall well-being.</p>
                </li>
            </ul>
        </div>
    </div>
</div>
```

Figure 53: index.html code

```
<div id="keyInsights" class="section">
    <div class="card insight-box">
        <h2>Key Findings</h2>
        <ul>
            <li>Instagram Dominates Usage:
                <p>Among all social media platforms, Instagram emerges as the most widely used by students. Its visually engaging content, reels, and constant updates make it highly addictive and time-consuming. This dominance suggests that students are more inclined toward entertainment-driven platforms rather than informational or professional ones.</p>
            </li>
            <li>Higher social media usage leads to increased addiction:
                <p>The data shows a strong positive relationship between the amount of time spent on social media and the level of user addiction. Individuals who spend more hours online are more likely to develop compulsive usage habits, including frequent checking, dependency, and difficulty disengaging from platforms.</p>
            </li>
            <li>Reduced sleep negatively impacts mental health:
                <p>Users who experience lower sleep durations tend to report poorer mental health outcomes. Insufficient sleep contributes to increased stress, anxiety, and reduced emotional stability, highlighting the importance of maintaining healthy sleep patterns alongside controlled social media usage.</p>
            </li>
            <li>Students spending 3-6 hours are most affected:
                <p>A significant portion of students fall within the 3-6 hours daily usage range, where the impact of social media becomes most noticeable. This group shows higher levels of distraction, reduced productivity, and a greater tendency toward addiction compared to lower usage groups.</p>
            </li>
            <li>Academic performance is significantly influenced:
                <p>Excessive engagement with social media platforms can negatively affect academic outcomes. Students often face issues such as reduced concentration, procrastination, and lower study efficiency, which ultimately impact their grades and overall academic performance.</p>
            </li>
            <li>Social media impact is globally consistent:
                <p>The patterns observed in the dataset are consistent across different countries and regions. Regardless of geographic location, users demonstrate similar trends in usage behavior, addiction levels, and mental health effects, indicating the widespread and universal influence of social media.</p>
            </li>
            <li>Moderate Conflicts Observed:
                <p>Students report experiencing moderate levels of conflict due to social media usage, including disagreements with family or peers and distractions during important activities. While not extremely severe, these conflicts highlight the social and behavioral implications of excessive usage</p>
            </li>
            <li>Conclusion:
                <p>Overall, the insights suggest that while social media is deeply integrated into students daily lives, excessive usage—especially on platforms like Instagram—can lead to reduced academic focus, increased dependency, and noticeable social conflicts. Maintaining a balanced approach to usage is essential for overall well-being and productivity.</p>
            </li>
        </ul>
    </div>
</div>
```

Show hidden icons

Figure 53: index.html code

```
<div id="aboutMe" class="section">
    <div class="card">
        <h2>About Me</h2>
        <p><b>Name:</b> Kulpreet Kaur Sembhi</p>
        <p><b>Skills:</b> Data Analysis, Tableau, Python, Power BI</p>
        <p><b>Project:</b> Student Social Media Analytics</p>
        <p><b>Email:</b> kulpreetsembhi@email.com</p>
        <p><b>LinkedIn:</b>Kulpreet Sembhi</p>
        <p><b>Contact:</b>977957XXXX</p>
    </div>
</div>
<div id="modal" class="modal">
    <span class="close" onclick="closeModal()">&times;</span>
    <img id="modal-img">
</div>
<footer>
© 2026 Student Analytics Portfolio
</footer>
<script>
function showSection(id) {
    let sections = document.querySelectorAll('.section');
    sections.forEach(sec => sec.classList.remove('active'));
    document.getElementById(id).classList.add('active');
}
function openModal(img) {
    document.getElementById("modal").style.display = "block";
    document.getElementById("modal-img").src = img.src;
}
function closeModal() {
    document.getElementById("modal").style.display = "none";
}
function showGraph(index) {
    let graphs = document.querySelectorAll('.graph');
    let pages = document.querySelectorAll('#usage .pagination span');
    let filtersPanel = document.getElementById('filtersPanel');
    let analysisText = document.getElementById('analysisText');
    graphs.forEach(g => g.classList.remove('active-graph'));
    pages.forEach(p => p.classList.remove('active-page'));
    graphs[index].classList.add('active-graph');
    pages[index].classList.add('active-page');
    if (index === 0 || index === 2) {
```

Figure 53: index.html code

```
        filtersPanel.style.display = "flex";
    } else {
        filtersPanel.style.display = "none";
    }
    if (index === 0) {
        analysisText.innerHTML = `
        <p><b>Takeaways:</b><br> This graph depicts the most used platforms used by students. Here, Instagram is the platform used by most students, followed by Tiktok, Facebook and so on.</p>`;
    }
    else if (index === 1) {
        analysisText.innerHTML = `
        <p><b>Takeaways:</b><br> This Graph shows average daily hours of social media usage by stundents. This average ranges from 0.5 hours to 8.5 hours of daily usage. The highest number of students use social media for about 4.5 hours on daily basis. 40% of students usage hours are below 4.5 hours of usage and rest 60% are more than 4.5 hours.</p>`;
    }
    else if (index === 2) {
        analysisText.innerHTML = `
        <p><b>Takeaways:</b><br> This graph shows student perception of social media's affect on academics. in this, majority of students say that social media affects academics and minority of students say that it doesnot. The blue bars represents males and orange bars represnt females. We can conclude that academics are affected by social media as surveyed by students.</p>`;
    }
    else if (index === 3) {
        analysisText.innerHTML = `
        <p><b>Takeaways:</b><br> This graph depicts number of conflicts faced by students on social media. the range of conflicts are from 0 to 5. Highest number of students face 3 conflicts. </p>`;
    }
}
function showInsightGraph(index) {
    let graphs = document.querySelectorAll('.insights-graph');
    let pages = document.querySelectorAll('#insights .pagination span');
    let analysisText = document.getElementById('insightAnalysis');
    let filterImg = document.getElementById('insightFilter');
    graphs.forEach(g => g.classList.remove('active-insight'));
    pages.forEach(p => p.classList.remove('active-page'));
    graphs[index].classList.add('active-insight');
    pages[index].classList.add('active-page');
    const texts = [
        "This graph impact of usage hours on addiction levels on MALE students. A strong positive correlation is visible. This explains that with increase in usage hours there is increase in Addiction score as well. Usage hours extends from 1.5 to 7.5 hours of daily usage. Alternatively, addiction score extends from 3.0 to 9.0, 3 being the lowest and 9 being the highest. ",
```

Figure 53: index.html code

```
        "This graph shows usage vs addiction for FEMALE students. The trend is similar but slightly less steep. This also explains that with increase in usage hours there is increase in Addiction score as well. Usage hours extends from 1 to 9 hours of daily usage. Alternatively, addiction score extends from 3.0 to 9.0, 3 being the lowest and 9 being the highest. Here, Females has a higher rate of daily hours of usage than Males, leading to higher addiction scores.",
        "This graph highlights Global Analysis of Student Addiction Scores by Country. The more the addiction score of students is, darker is the country highlighted with. Higher the number of students of a country, bigger is its square with which the country is showcased. ",
        "This graph shows variation of addiction scores across academic levels, with undergraduates showing wider spread, Graduates showing moderate spread and students of high school showing least spread. Average addiction score is depicted as 6.5. ",
        "This graph shows Impact of Sleep Hours on Mental Health on Male students Sleep hours ranges from 4 to 9.5. Mental healt score extends from 4 to 8. This graph also shows a positive correlation between Sleep hours and mental health score, that means, more sleep is linked to better mental health outcomes. It strongly supports the idea that sleep quality and duration are important factors in mental well-being.",
        "This graph shows Impact of Sleep Hours on Mental Health on Female students Sleep hours ranges from 4 to 9.5. Mental healt score extends from 4 to 8. This graph also shows a positive correlation between Sleep hours and mental health score, that means, more sleep is linked to better mental health outcomes. It strongly supports the idea that sleep quality and duration are important factors in mental well-being."
    ];
    const filters = [
        "images/dash3_filters1.png",
        "images/dash3_filters2.png",
        "images/dash3_filters3.png",
        "images/dash3_filters4.png",
        "images/dash3_filters5.png",
        "images/dash3_filters6.png"
    ];
    analysisText.innerHTML = `<p><b>Takeaways:</b><br> ${texts[index]}</p>`;
    filterImg.src = filters[index];
}
</script>
</body>
</html>
```

Figure 53: index.html code

- # login.html (html code)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Login | Student Analytics</title>
<style>
body {
    margin: 0;
    font-family: 'Segoe UI', sans-serif;
    background: linear-gradient(135deg, #0f2027, #203a43, #2c5364);
    color: white;
}
.login-container {
    width: 350px;
    margin: 120px auto;
    padding: 25px;
    background: rgba(255,255,255,0.08);
    border-radius: 15px;
    box-shadow: 0px 10px 30px rgba(0,0,0,0.5);
    backdrop-filter: blur(12px);
    text-align: center;
}
.login-container h2 {
    margin-bottom: 20px;
    color: #00ffff;
}
.login-container input {
    width: 90%;
    padding: 10px;
    margin-top: 10px;
    border: none;
    border-radius: 8px;
    outline: none;
}
.login-container button {
    width: 50%;
    padding: 10px;
    margin-top: 10px;
    border: none;
    border-radius: 8px;
    background: #00ffff;
    color: black;
    font-weight: bold;
    cursor: pointer;
    transition: 0.3s;
```

Figure 54: login.html code

```
}
.login-container button:hover {
    background: #00e6e6;
}
.message {
    margin-top: 10px;
    font-size: 14px;
}
</style>
</head>
<body>
<div class="login-container">
    <h2>Admin Login</h2>

```
<input type="email" id="email" placeholder="Enter Email">
<input type="password" id="password" placeholder="Enter Password">
<button onclick="login()">Login</button>

```
</div>
<script type="module">
import { initializeApp } from "https://www.gstatic.com/firebasejs/10.7.1/firebase-app.js";
import { getAuth, createUserWithEmailAndPassword, signInWithEmailAndPassword }
from "https://www.gstatic.com/firebasejs/10.7.1/firebase-auth.js";
const firebaseConfig = {
  apiKey: "AIzaSyDVaLxUbdPTFR1BE7w_js72Z_M8IplpoX8",
  authDomain: "student-social-media-analysis.firebaseapp.com",
  projectId: "student-social-media-analysis",
  storageBucket: "student-social-media-analysis.firebasestorage.app",
  messagingSenderId: "939621840778",
  appId: "1:939621840778:web:6b205f509ff16e0b03fa73",
  measurementId: "G-T9QHTV60SH"
};
const app = initializeApp(firebaseConfig);
const auth = getAuth(app);
window.signup = function () {
    const email = document.getElementById("email").value;
    const password = document.getElementById("password").value;
    createUserWithEmailAndPassword(auth, email, password)
    .then(() => {
```

Figure 54: login.html code

```
        document.getElementById("msg").innerText = "Signup Successful ✅";
    })
    .catch(error => {
        document.getElementById("msg").innerText = error.message;
    });
};
window.login = function () {
    const email = document.getElementById("email").value;
    const password = document.getElementById("password").value;
    signInWithEmailAndPassword(auth, email, password)
    .then(() => {
        document.getElementById("msg").innerText = "Login Successful ✅";
        setTimeout(() => {
            window.location.href = "index.html";
        }, 1000);
    })
    .catch(error => {
        document.getElementById("msg").innerText = error.message;
    });
};
const params = new URLSearchParams(window.location.search);
if (params.get("logout") === "success") {
    document.getElementById("msg").innerText = "Logged out successfully ✅";
}
</script>
</body>
</html>
```

Figure 54: login.html code

- # contact.html (html code)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Contact Us | Student Analytics</title>
<style>
body {
    font-family: 'Segoe UI', sans-serif;
    background: linear-gradient(135deg, #0f2027, #203a43, #2c5364);
    color: white;
}
.container {
    width: 400px;
    margin: 80px auto;
    padding: 25px;
    background: rgba(255,255,255,0.08);
    border-radius: 15px;
    text-align: center;
}
input, textarea {
    width: 90%;
    padding: 10px;
    margin: 10px 0;
    border-radius: 8px;
    border: none;
}
button {
    width: 50%;
    padding: 10px;
    background: #00ffff;
    border: none;
    border-radius: 8px;
    font-weight: bold;
    cursor: pointer;
}
#msg {
    margin-top: 10px;
}
</style>
</head>
<body>
<div class="container">
    <h2>Contact Us</h2>
```<br>
```
<form onsubmit="sendMessage(event)">
```

Figure 55: contact.html code

```
    <input id="name" type="text" placeholder="Your Name" required>
    <input id="phone" type="tel" placeholder="Phone Number"
        pattern="[0-9]{10}" required>
    <input id="email" type="email" placeholder="Email Address" required>
    <textarea id="message" placeholder="Your Message" required></textarea>
    <button type="submit">Send Message</button>
</form>
<p id="msg"></p>
```
```
</div>
<script type="module">
import { initializeApp } from "https://www.gstatic.com/firebasejs/10.7.1/firebase-app.js";
import { getFirestore, collection, addDoc }
from "https://www.gstatic.com/firebasejs/10.7.1/firebase-firestore.js";
const firebaseConfig = {
  apiKey: "AIzaSyDVaLxUbdPTFRlBE7w_js72Z_M8IplpoX8",
  authDomain: "student-social-media-analysis.firebaseapp.com",
  projectId: "student-social-media-analysis"
};
const app = initializeApp(firebaseConfig);
const db = getFirestore(app);
window.sendMessage = async function (event) {
    event.preventDefault();
    const name = document.getElementById("name").value;
    const phone = document.getElementById("phone").value;
    const email = document.getElementById("email").value;
    const message = document.getElementById("message").value;
    if (!name || !email || !message) {
        document.getElementById("msg").innerText = "Please fill required fields ❌";
        return;
    }
    try {
        await addDoc(collection(db, "messages"), {
            name: name,
            phone: phone,
            email: email,
            message: message,
            time: new Date()
        });
        document.getElementById("msg").innerHTML = "✅ Message sent successfully!";
```

Battery status: 43% avai

Figure 55: contact.html code

```
        document.getElementById("msg").innerHTML = "✅ Message sent successfully!";
        document.getElementById("name").value = "";
        document.getElementById("phone").value = "";
        document.getElementById("email").value = "";
        document.getElementById("message").value = "";
    } catch (error) {
        document.getElementById("msg").innerText = error.message;
    }
};
</script>
</body>
</html>
```

Figure 55: contact.html code

- **style.css (CSS code)**

```
body {
  display: flex;
  flex-direction: column;
  min-height: 100vh;
  margin: 0;
  font-family: "Segoe UI", sans-serif;
  background: linear-gradient(135deg, #0f2027, #203a43, #2c5364);
  color: white;
  scroll-behavior: smooth;
}
.hero {
  padding: 40px;
  text-align: center;
}
.hero-content {
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 20px;
}
.logo {
  width: 150px;
  height: 150px;
  object-fit: contain;
  filter: drop-shadow(0 0 6px #00e6e6);
}
@media (max-width: 768px) {
  .hero-content {
    flex-direction: column;
  }
  .logo {
    width: 70px;
  }
}
.hero h1 {
  margin: 0;
  font-size: 28px;
}
.hero p {
  margin: 5px 0 0;
  font-size: 14px;
}
header {
  text-align: center;
  padding: 30px;
```

Figure 56: style.css code

```
    font-size: 28px;
    font-weight: bold;
    letter-spacing: 1px;
  }
  nav {
    display: flex;
    justify-content: center;
    background: rgba(255, 255, 255, 0.08);
    backdrop-filter: blur(12px);
    position: sticky;
    top: 0;
    z-index: 100;
    padding-left: 200px;
    padding-right: 200px;
  }
  nav button {
    background: none;
    border: none;
    color: white;
    margin: 15px;
    font-size: 16px;
    cursor: pointer;
    transition: all 0.3s ease;
    padding: 8px 15px;
    border-radius: 8px;
  }
  nav button:hover {
    background: rgba(0, 255, 255, 0.2);
    color: #00ffff;
  }
  .section {
    display: none !important;
    padding: 40px 20px;
    text-align: center;
    animation: fadeIn 0.5s ease;
    opacity: 0;
    transition: 0.3s;
  }
  .section.active {
    display: block !important;
    flex: 1;
  }
```

Figure 56: style.css code

```
  .active {
    display: block;
    opacity: 1;
  }
  .card {
    background: rgba(255, 255, 255, 0.07);
    padding: 25px;
    border-radius: 18px;
    width: 90%;
    min-height: unset;
    max-width: 1100px;
    margin: auto;
    margin-bottom: 20px;
    box-shadow: 0px 10px 30px rgba(0, 0, 0, 0.5);
    backdrop-filter: blur(15px);
  }
  .dashboard-layout {
    display: flex;
    gap: 20px;
    align-items: stretch;
  }
  .dashboard-img {
    width: 100%;
    max-width: 1000px;
    margin-top: 20px;
    border-radius: 12px;
    cursor: pointer;
    transition: transform 0.3s ease;
  }
  .dashboard-img:hover {
    transform: scale(1.03);
  }
```

Figure 56: style.css code

```css
.dashboard-img:hover {
  transform: scale(1.03);
}
.insight-box {
  margin-top: 30px;
  text-align: left;
  background: rgba(255, 255, 255, 0.08);
  padding: 20px;
  border-radius: 12px;
  border-left: 4px solid #00ffff;
  width: 100%;
  box-sizing: border-box;
}
.insight-box h3 {
  margin-top: 0;
  color: #00ffff;
}
.insight-box ul {
  padding-left: 20px;
  line-height: 1.8;
}
.insight-box li {
  margin-bottom: 8px;
}
.modal {
  display: none;
  position: fixed;
  z-index: 200;
  background: rgba(0, 0, 0, 0.95);
  top: 0;
  left: 0;
  width: 100%;
  height: 100%;
}
.modal img {
  display: block;
  margin: auto;
  margin-top: 60px;
  width: 80%;
  max-width: 1200px;
  border-radius: 10px;
}
```

Figure 56: style.css code

```css
.close {
  position: absolute;
  right: 40px;
  top: 20px;
  font-size: 40px;
  cursor: pointer;
}
footer {
  margin-top: auto;
  text-align: center;
  padding: 20px;
  opacity: 0.7;
  font-size: 14px;
}
@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(15px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}
```

Figure 56: style.css code

```css
@media (max-width: 768px) {
  header {
    font-size: 22px;
  }

  nav {
    flex-wrap: wrap;
  }

  nav button {
    margin: 8px;
  }

  .dashboard-img {
    width: 100%;
  }
}
.hero {
  text-align: center;
  padding: 60px 20px;
  background: linear-gradient(135deg, #7cdbf5, #66a8fa);
  color: white;
}
.hero h1 {
  font-size: 36px;
  margin-bottom: 10px;
}
.hero p {
  font-size: 18px;
  opacity: 0.9;
}
.graph-container {
  position: relative;
  text-align: center;
}
.pagination {
  text-align: center;
  margin-top: 15px;
}
```

Figure 56: style.css code

```css
.pagination span {
  display: inline-block;
  margin: 0 10px;
  padding: 8px 14px;
  cursor: pointer;
  background: #2f4f4f;
  color: white;
  border-radius: 6px;
  transition: 0.3s;
}
.pagination span:hover {
  background: #00e6e6;
  color: black;
}
.active-page {
  background: #00e6e6;
  color: black;
  font-weight: bold;
}
.usage-layout {
  display: flex;
  gap: 20px;
  align-items: flex-start;
  flex-wrap: nowrap;
}
.graph-container {
  flex: 70%;
}
.filters-image {
  flex: 30%;
}
.graph {
  width: 100%;
  border-radius: 10px;
  display: none;
}
.active-graph {
  display: block;
}
.filters-image img {
  width: 100%;
  max-width: 250px;
  border-radius: 10px;
}
```

Figure 56: style.css code

```css
.dashboard-layout {
  display: flex;
  gap: 20px;
  align-items: flex-start;
}
.graph-container {
  flex: 3;
}
.filters-image {
  flex: 1;
}
.filters-image img {
  width: 100%;
}
.pagination span {
  cursor: pointer;
  padding: 5px 10px;
  margin: 5px;
  border: 1px solid #ccc;
}
.active-page {
  background-color: #333;
  color: white;
}
.tools {
  text-align: center;
  padding: 40px 20px;
  background: #0f2027;
}
.tools h2 {
  color: #00e6e6;
  margin-bottom: 20px;
}
.tools-container {
  display: flex;
  justify-content: center;
  gap: 40px;
  flex-wrap: wrap;
}
.tools-container img {
  width: 70px;
  height: 70px;
  transition: transform 0.3s ease;
}
```

Figure 56: style.css code

```css
.tools-container img:hover {
  transform: scale(1.2);
}
.usage-layout {
  display: flex;
  gap: 20px;
  align-items: flex-start;
}
.graph-container {
  flex: 70%;
}
.filters-panel {
  flex: 30%;
  display: flex;
  flex-direction: column;
  gap: 15px;
}
.filters-img {
  width: 100%;
  max-width: 250px;
  border-radius: 10px;
}
```

Figure 56: style.css code

```
.filters-img {
  width: 100%;
  max-width: 250px;
  border-radius: 10px;
}
.analysis-text {
  background: #2c3e50;
  padding: 12px;
  border-radius: 8px;
  font-size: 14px;
  color: white;
  text-align: justify;
}
.side-panel {
  flex: 30%;
  display: flex;
  flex-direction: column;
  gap: 15px;
}
.filters-panel {
  display: block;
}
.insights-graph {
  display: none;
  width: 100%;
  border-radius: 10px;
}
.active-insight {
  display: block;
}
#aboutMe {
  text-align: center;
}
#aboutMe .card {
  max-width: 500px;
  padding: 20px;
}
#aboutMe p {
  margin: 8px 0;
  font-size: 15px;
}
#aboutMe .card {
  text-align: left;
  border-left: 4px solid #00ffff;
}
```

Figure 56: style.css code

```
nav {
  display: flex;
  justify-content: flex-start;
  align-items: center;
}
nav button:last-child {
  margin-left: auto;
}
#userEmail {
  font-size: 14px;
  color: #00ffff;
}
@media (max-width: 768px) {
  .hero {
    padding: 20px 10px !important;
  }
  .hero-content {
    flex-direction: column;
    gap: 10px;
  }
```

Figure 56: Style.css code

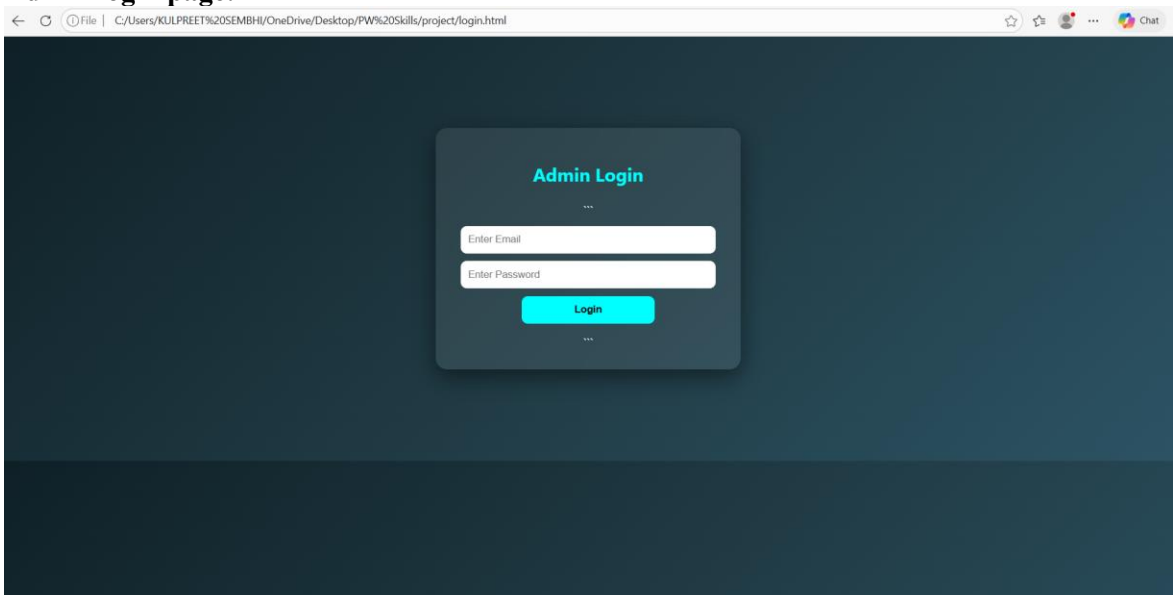
```
392     .logo {
393       width: 60px !important;
394       height: 60px !important;
395     }
396     .hero h1 {
397       font-size: 18px !important;
398       line-height: 1.3;
399     }
400     .hero p {
401       font-size: 13px !important;
402     }
403   }
404   @media (max-width: 768px) {
405     nav button {
406       font-size: 13px;
407       padding: 6px 10px;
408     }
409   }
410   @media (max-width: 768px) {
411     .dashboard-layout,
412     .usage-layout {
413       flex-direction: column !important;
414       align-items: stretch !important;
415     }
416     .graph-container,
417     .side-panel,
418     .filters-panel {
419       width: 100% !important;
420     }
421     .graph-container {
422       order: 1;
423     }
424     .filters-panel {
425       order: 2;
426     }
427     .analysis-text {
428       order: 3;
429     }
430     .side-panel {
431       gap: 10px;
432     }
433     .graph,
434     .insights-graph {
435       width: 100% !important;
```

Figure 56: style.css code

```
436       height: auto;
437     }
438     .filters-img {
439       max-width: 100% !important;
440     }
441   }
442   @media (max-width: 768px) {
443     .card {
444       padding: 15px !important;
445     }
446   }
```

Figure 56: style.css code

- **The above code resulted in the following website pages:**
- **Admin login page:**

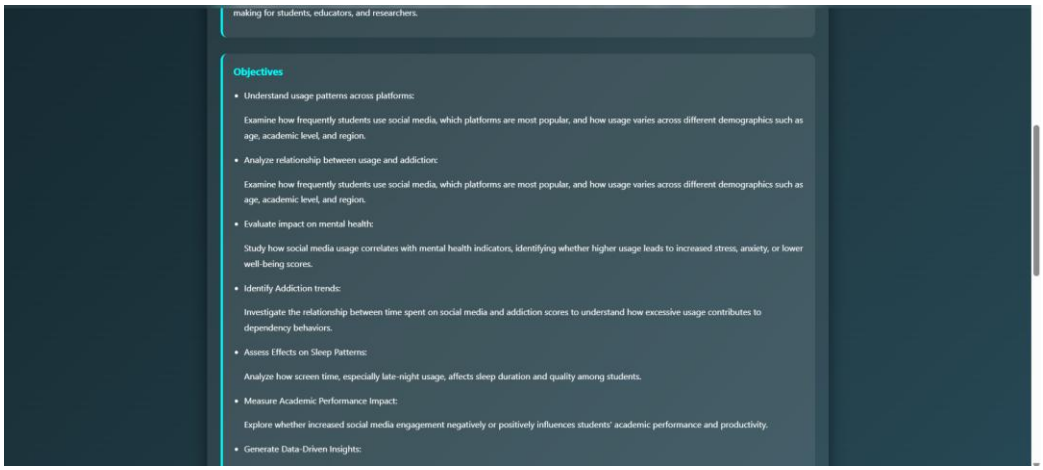


Webpage 1: Admin login page

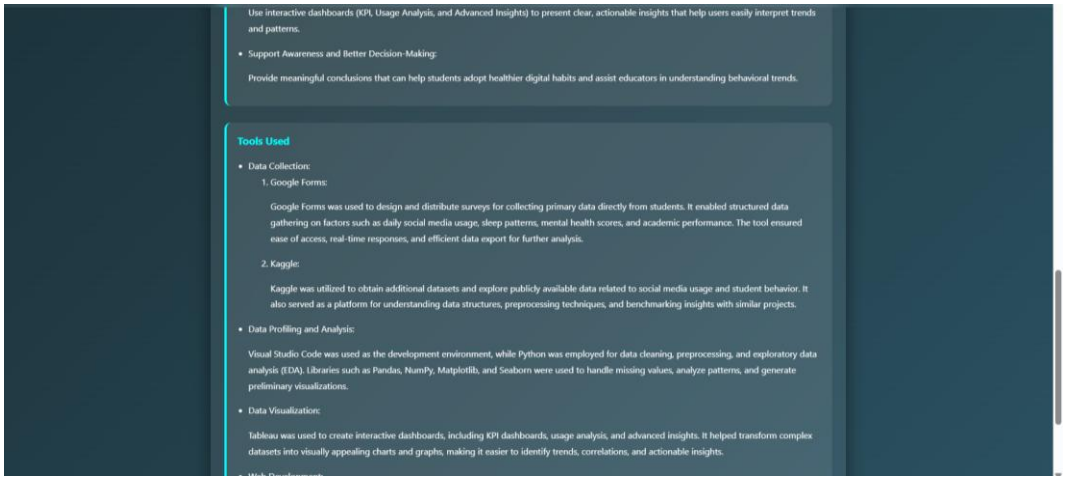
- **About Page:**



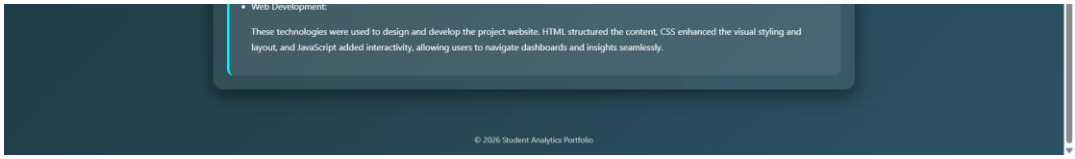
Webpage 2: About page



Webpage 2: About page

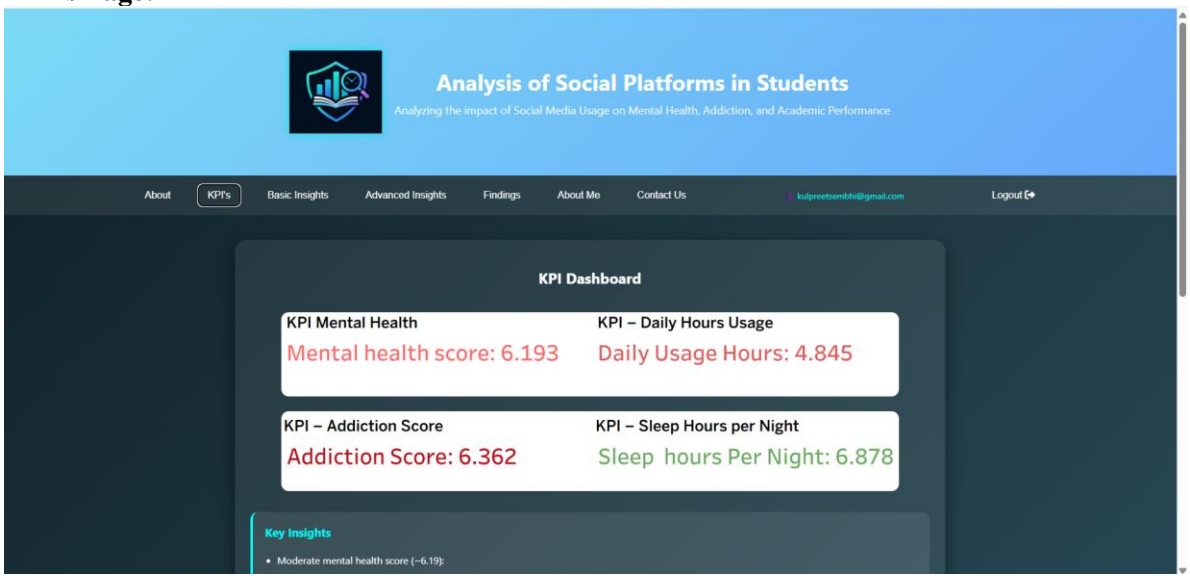


Webpage 2: About page

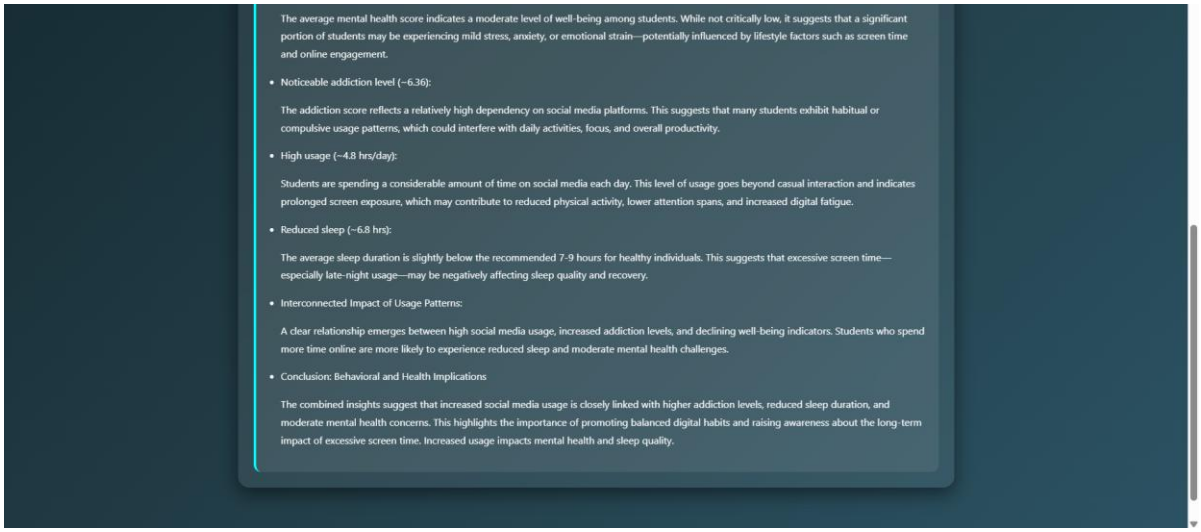


Webpage 2: About Page

- **KPI's Page:**



Webpage 3: KPI page



Webpage 3: KPI page

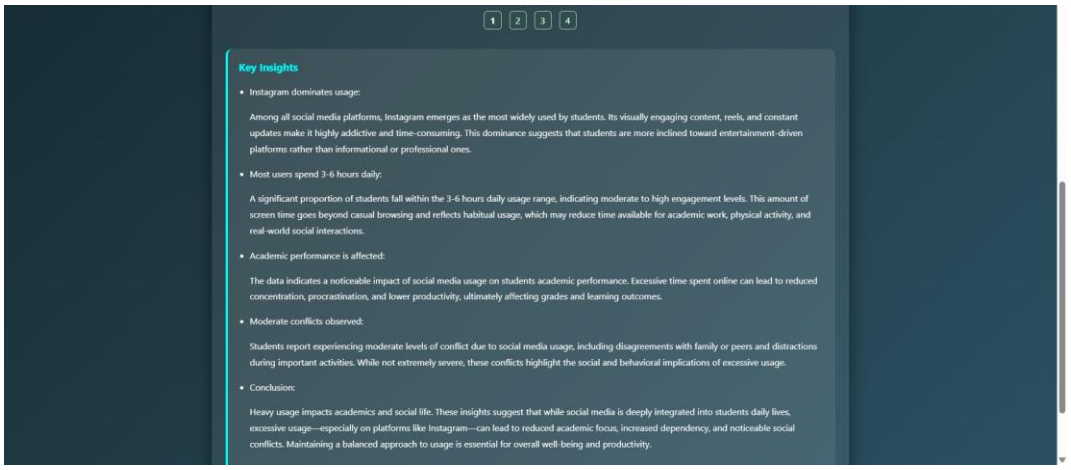


Webpage 3: KPI page

- **Basic Insights page (Pagination-1):**



Webpage 4: Basic insights (Pagination-1)

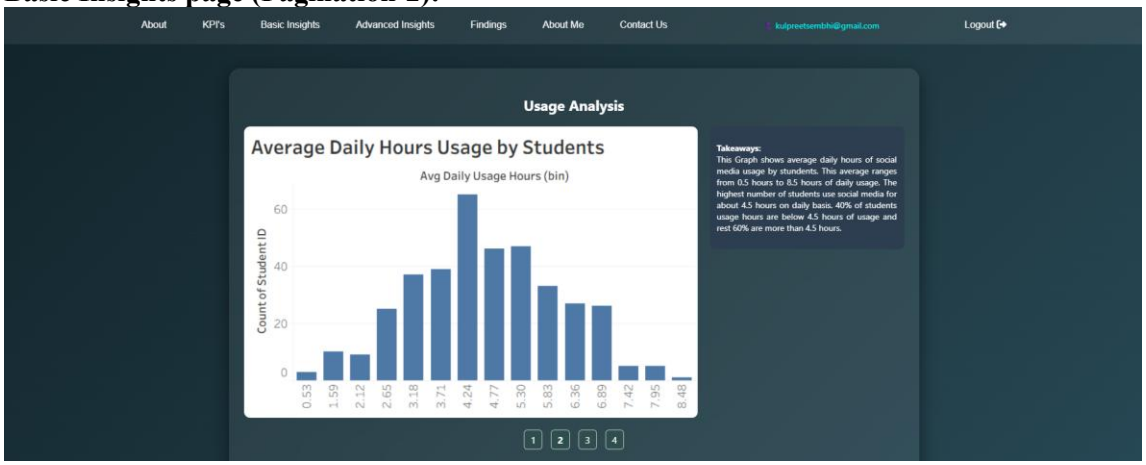


Webpage 4: Basic insights (Pagination-1)



Webpage 4: Basic insights (Pagination-1)

- **Basic Insights page (Pagination-2):**



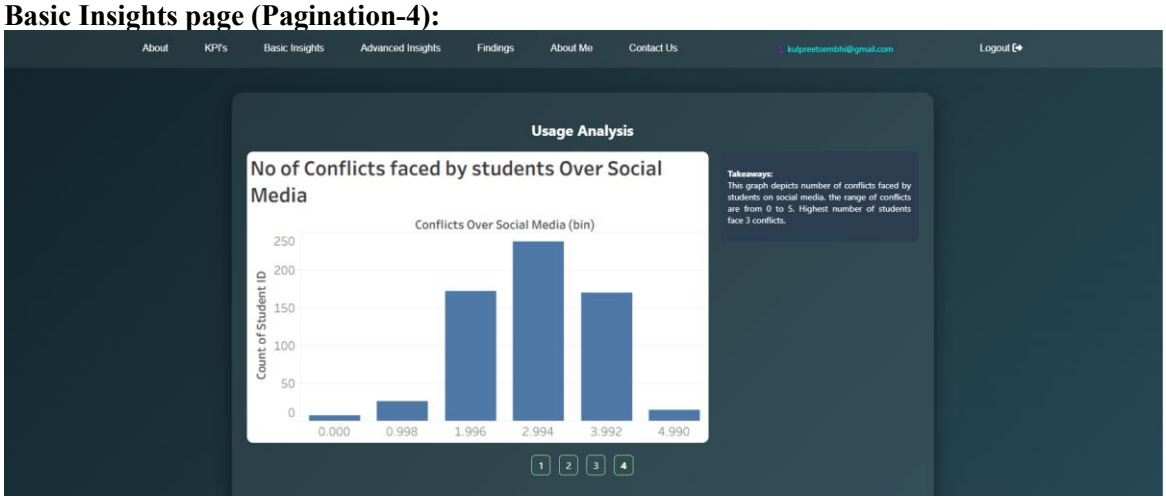
Webpage 4: Basic insights (Pagination-2)

- **Basic Insights page (Pagination-3):**



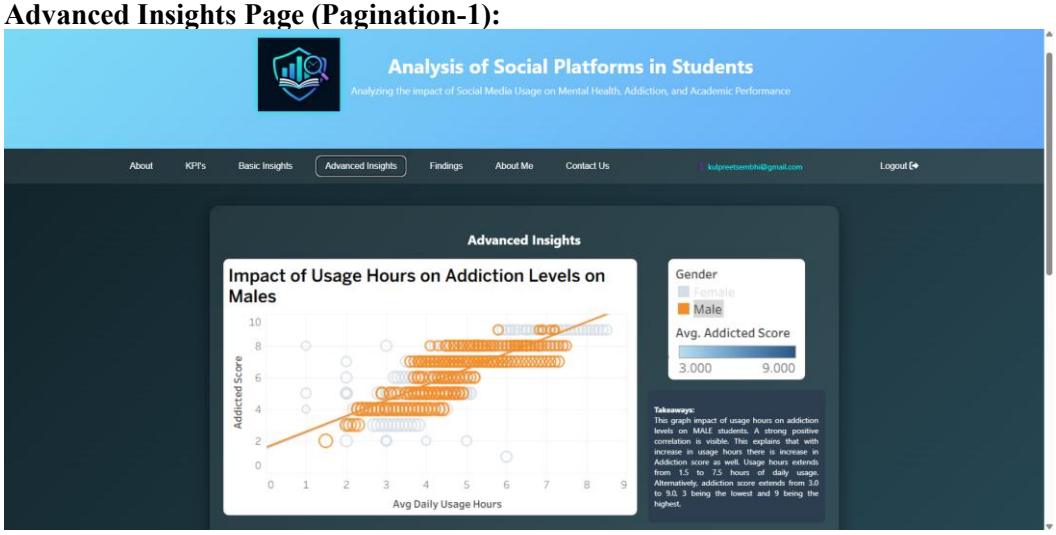
Webpage 4: Basic insights (Pagination-3)

- **Basic Insights page (Pagination-4):**

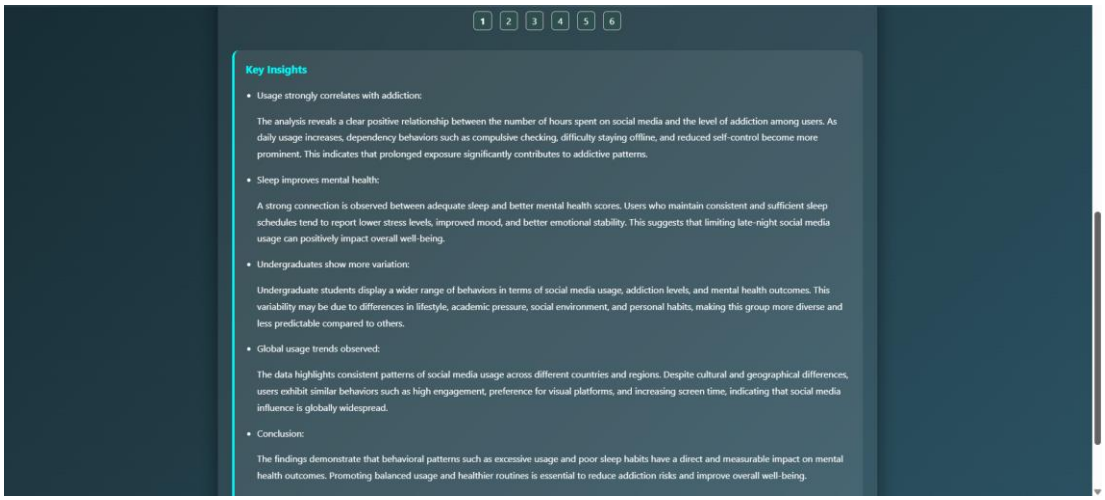


Webpage 4: Basic insights (Pagination-4)

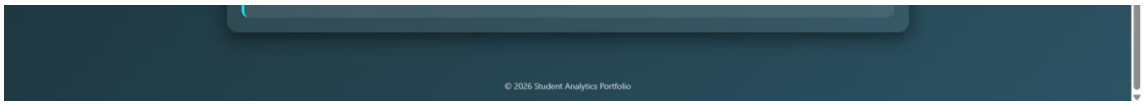
- **Advanced Insights Page (Pagination-1):**



Webpage 5: Advanced insights page

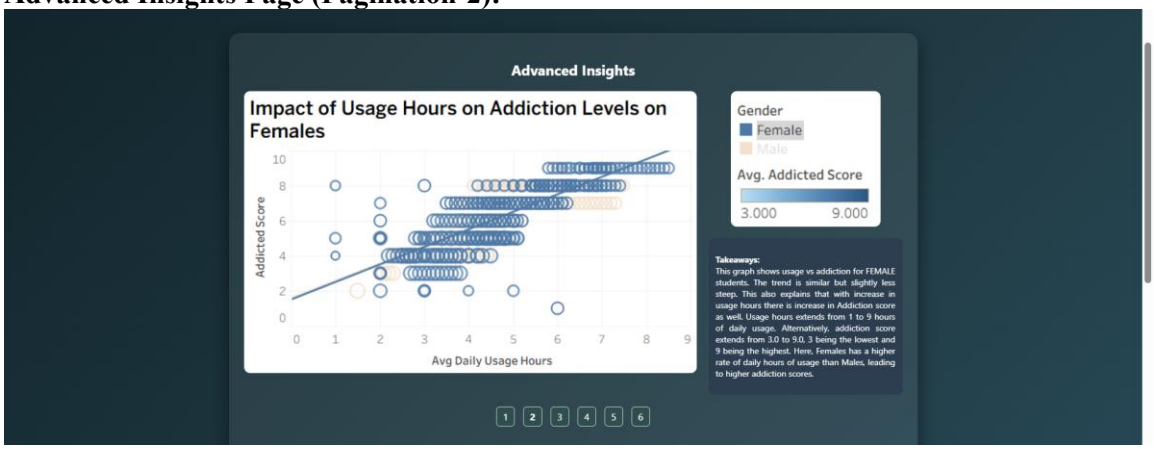


Webpage 5: Advanced insights page



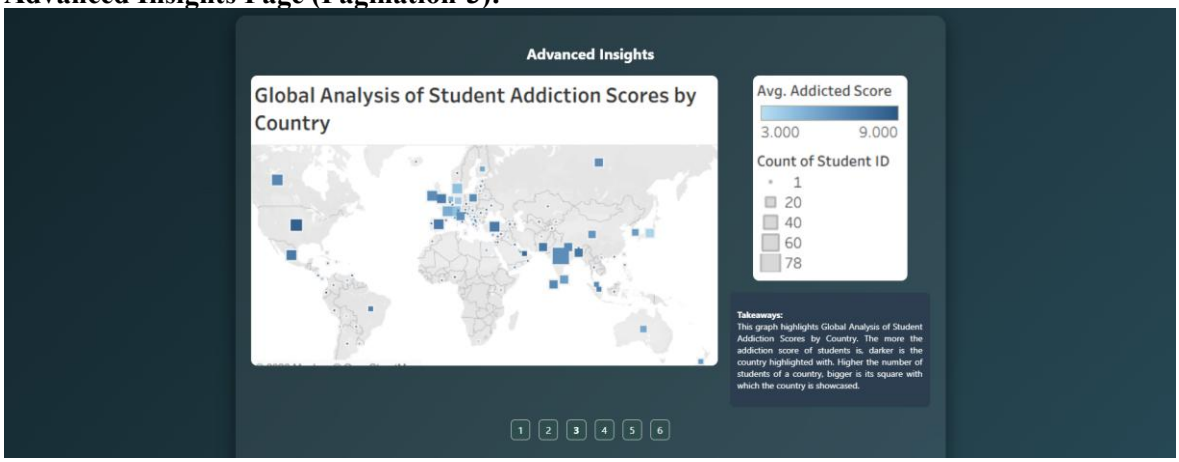
Webpage 5: Advanced insights page

- **Advanced Insights Page (Pagination-2):**



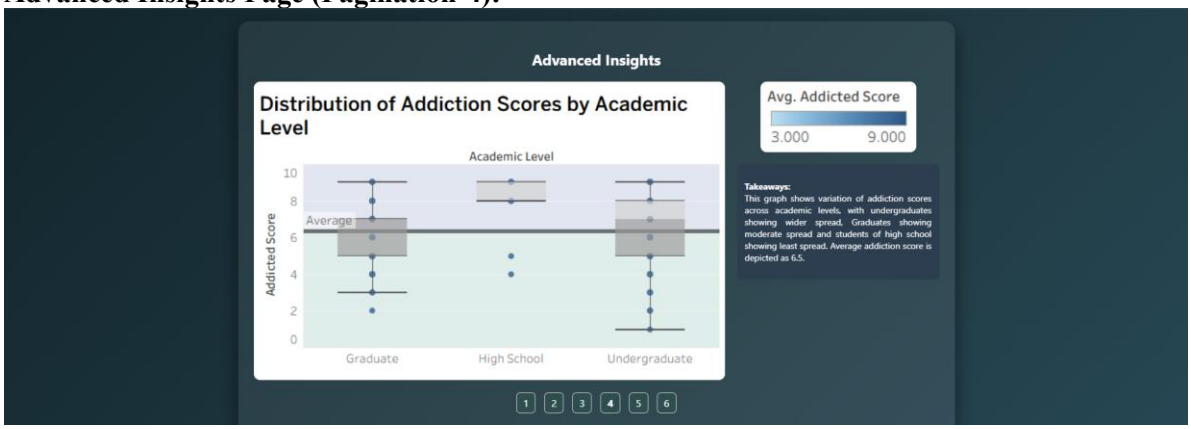
Webpage 5: Advanced Insights page (pagination-2)

- **Advanced Insights Page (Pagination-3):**



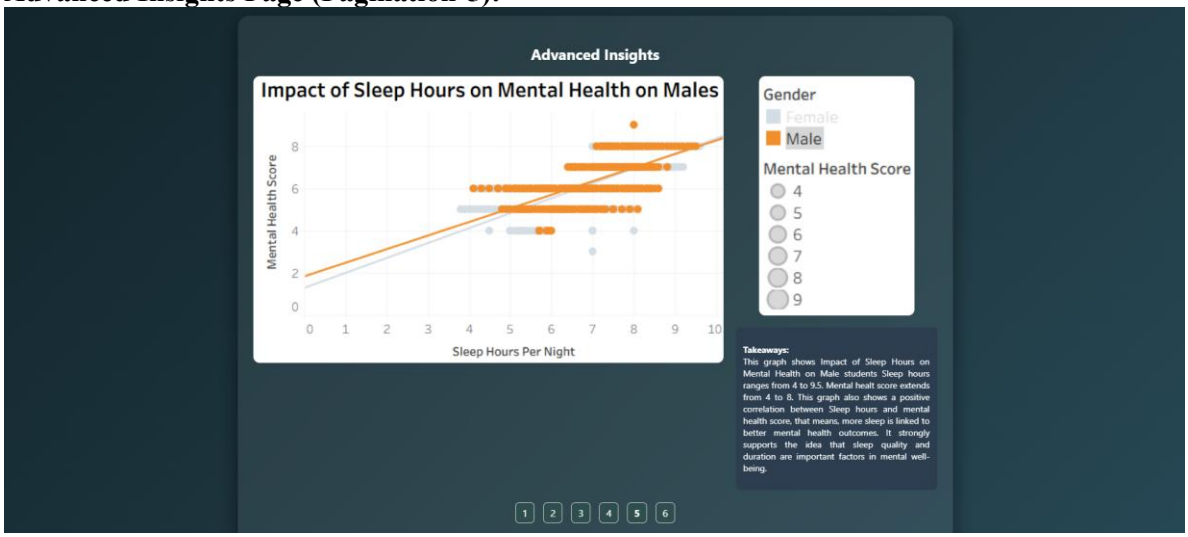
Webpage 5: Advanced Insights page (pagination-3)

- **Advanced Insights Page (Pagination-4):**



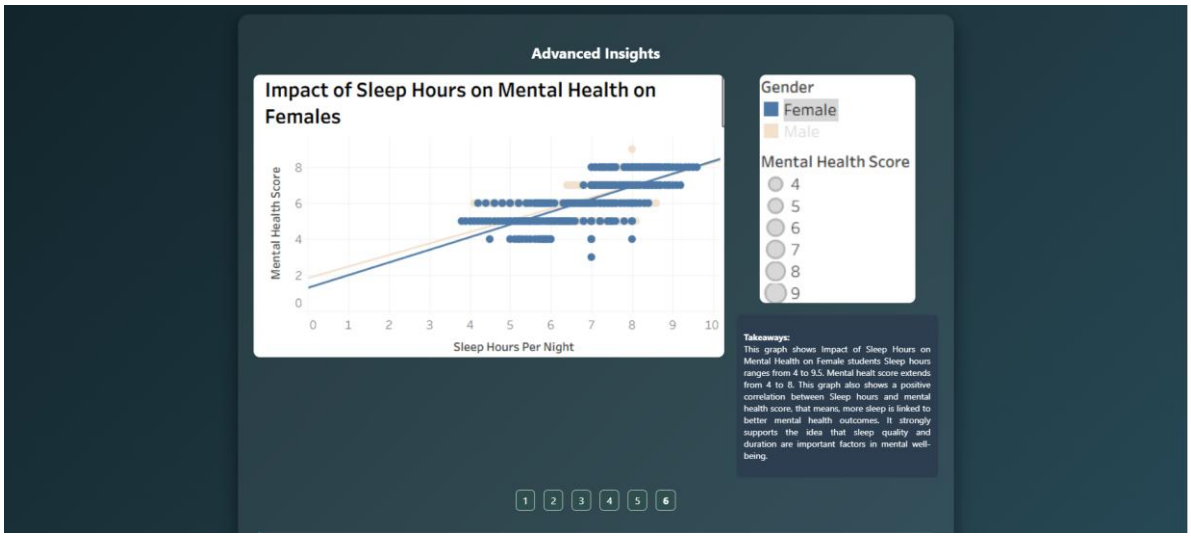
Webpage 5: Advanced Insights page (pagination-4)

- **Advanced Insights Page (Pagination-5):**



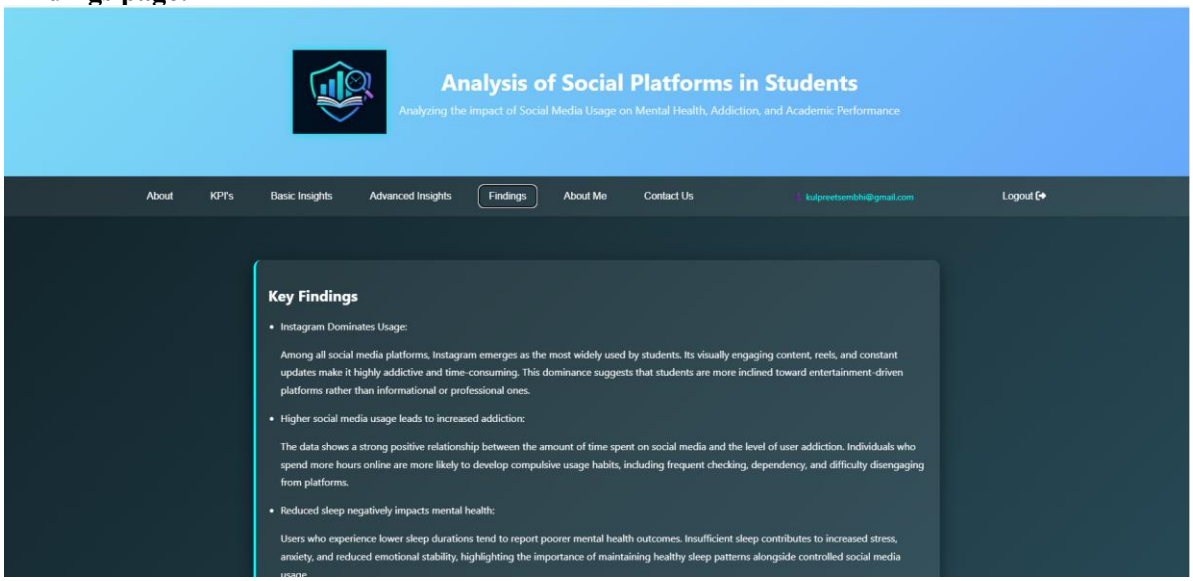
Webpage 5: Advanced Insights page (pagination-5)

- **Advanced Insights Page (Pagination-6:**

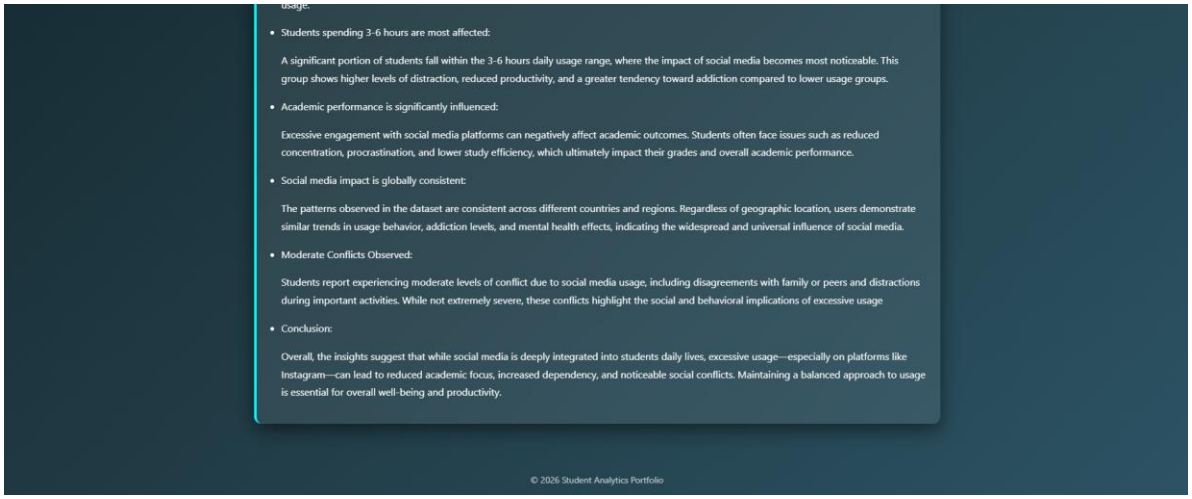


Webpage 5: Advanced Insights page (pagination-6)

- **Findings page:**

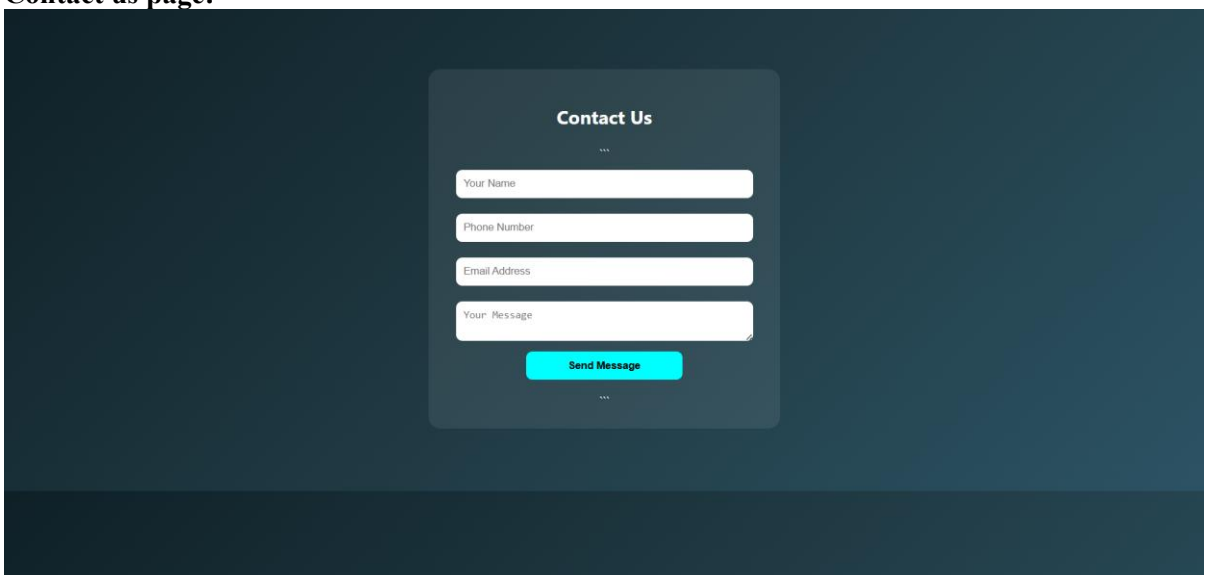


Webpage 6: Findings page



Webpage 6: Findings page

- **Contact us page:**



Webpage 7: Contact Us page

VII. Deployment of the website using Firebase Hosting:

- **Overview of Deployment:**
  Deployment refers to the process of making a developed web application available for public access over the internet. In this project, Analysis of Social Platforms in Students website was deployed using Firebase Hosting, a cloud-based hosting service provided by Google Firebase. It is specifically designed for hosting static and dynamic web applications with high speed, reliability, and security.
- **Introduction to Firebase Hosting:**
  Firebase Hosting is a production-grade web hosting service that allows developers to deploy web applications quickly with minimal configuration. It provides features such as global content delivery through CDN (Content Delivery Network), automatic SSL certification, and easy version control. Firebase supports static websites built using technologies such as HTML, CSS, and JavaScript, which makes it highly suitable for this project**.**
- **Objectives of Deployment:**
  The deployment phase aimed to achieve the following:
  - Make the website accessible globally via the internet
  - Ensure fast loading and responsiveness
  - Provide a secure browsing experience using HTTPS
  - Enable easy updates and maintenance
  - Present the project in a professional and shareable format
- **Tools and Technologies Used:**
  - Firebase CLI (Command Line Interface) – Used to configure and deploy the project
  - Node.js & npm – Required for installing Firebase tools
  - Web Browser – Used to access Firebase Console and deployed website
  - HTML, CSS, JavaScript – Core technologies used in the website
- **Step-by-Step Deployment Process:**
  - **Step 1: Installation of Firebase CLI**
    The Firebase CLI was installed globally using Node Package Manager (npm):
    npm install -g firebase-tools.

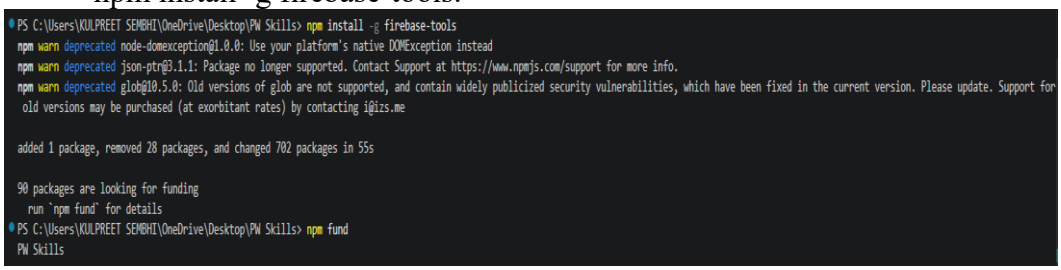


Figure 57: Showcasing VS-Code terminal for installing firebase tools for deploying the project.

This tool allows interaction with Firebase services directly from the command line.

  - **Step 2: Login to Firebase:**
    For this step, first we needed to create an account on the firebase website.
    Further, we logged into our Firebase account into our terminal using:
    firebase login

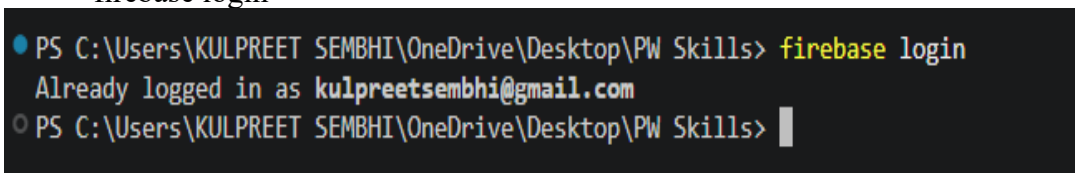


Figure 58: Login to firebase from your terminal for providing connectivity to your code.

This step connects the local system to the Firebase account.

- **Step 3: Project Creation:**
  A new project was created in the Firebase console:
  Project Name: *Student Social Media Analytics*
  Hosting service was enabled for the project

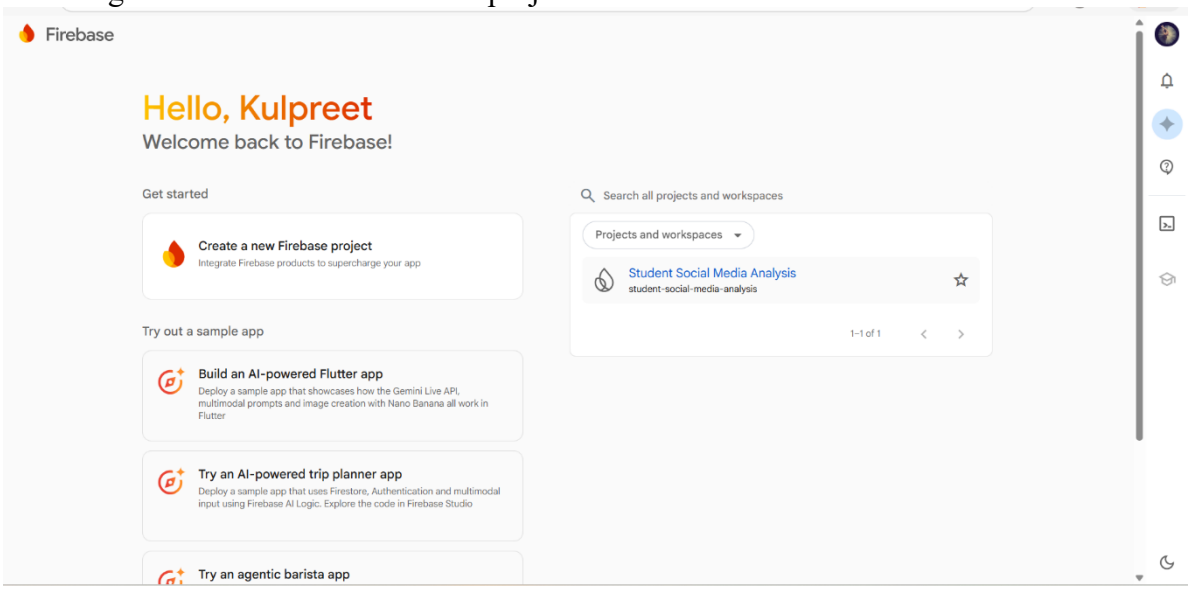


Figure 89: Project hs been created on the firebase console.

- **Step 4: Initialization of Firebase in Project Folder**
  Inside the project directory, Firebase was initialized using:
  firebase init
  During setup:
  - Hosting was selected
  - The public directory was set (e.g., /public or root folder)
  - index.html was defined as the entry point
  - Single Page App option was not configured as we have three html pages as individuals, namely: index.html, contact.html and login.html

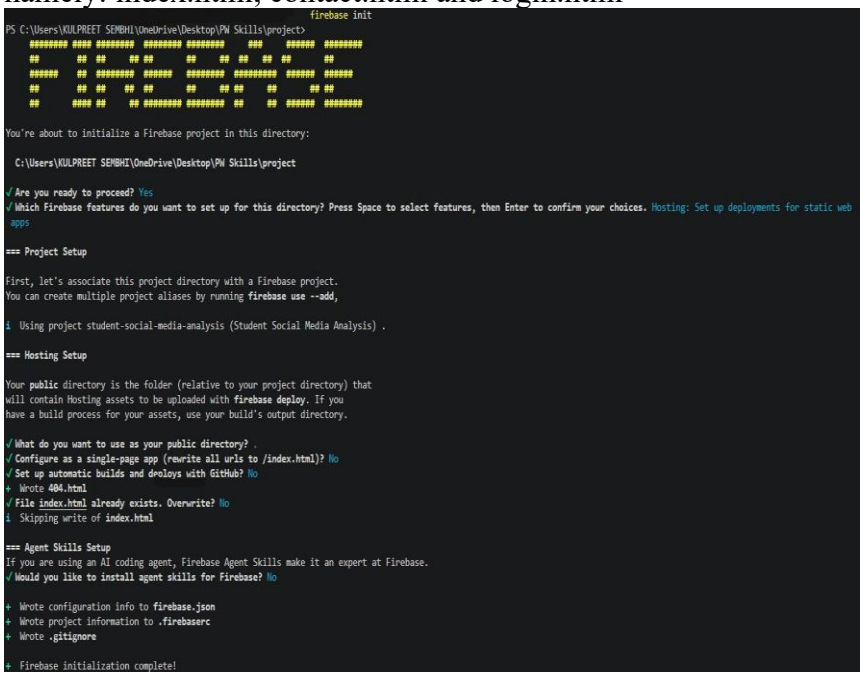


Figure 60: Firebase has been initialised and details for the project are being provided in the terminal.

- **Step 5: Configuration:**
  Firebase created configuration files such as:
  - firebase.json – Contains hosting settings

```
project > firebase.json > ...
1  {
2    "hosting": {
3      "public": ".",
4      "ignore": [
5        "firebase.json",
6        "**/.*",
7        "**/node_modules/**"
8      ]
9    }
10 }
11
```

Figure 61: firebase.json code

  - .firebaserc – Stores project details

```
project > .firebaserc > ...
1  {
2    "projects": {
3      "default": "student-social-media-analysis"
4    }
5  }
6
```

Figure 62: firebaserc code

  These files ensure proper deployment and routing of the website.

- **Step 6: Deployment of Website**
  The website was deployed using the command:
  firebase deploy
  After successful deployment:
  - Firebase generated a live URL
  - The website became accessible globally

```
PS C:\Users\KULPREET SEMBHI\OneDrive\Desktop\PW Skills\project> firebase deploy

=== Deploying to 'student-social-media-analysis'...

i  deploying hosting
i  hosting[student-social-media-analysis]: beginning deploy...
i  hosting[student-social-media-analysis]: found 49 files in .
+  hosting[student-social-media-analysis]: file upload complete
i  hosting[student-social-media-analysis]: finalizing version...
+  hosting[student-social-media-analysis]: version finalized
i  hosting[student-social-media-analysis]: releasing new version...
+  hosting[student-social-media-analysis]: release complete

+  Deploy complete!

Project Console: https://console.firebase.google.com/project/student-social-media-analysis/overview
Hosting URL: https://student-social-media-analysis.web.app
```

Figure 63: firebase deploy- command is used to deploy the website globally.

- **The website has been successfully deployed. The following are its links for accessing them:**
  - **Login | Student Analytics : login page of the web-app**
  - **Student Analytics Portfolio : index page of the web-app**
  - **Contact Us | Student Analytics : contact us page of the web-app**

- **Features of Firebase Hosting Used**
  - Global CDN
    Ensures fast loading by serving content from the nearest server
  - Secure Hosting (HTTPS)
    Automatically provides SSL encryption for safe browsing
  - Instant Deployment
    Changes can be deployed instantly with a single command
  - Version Control & Rollback
    Allows reverting to previous versions if needed
  - Scalability
    Can handle multiple users without performance issues

- **Output and Results**
  - The website was successfully hosted online
  - All sections including dashboards, graphs, and analysis are accessible
  - The website performs efficiently across different devices
  - Users can navigate between sections smoothly

- **Advantages Achieved**
  - Easy sharing via a public URL
  - Professional presentation of project work
  - High performance and reliability
  - Secure and stable hosting environment
  - Minimal cost (free tier used)

- **Limitations**
  - Static hosting (limited backend functionality)
  - Requires internet connection for access
  - Advanced features require Firebase upgrades

- **Conclusion**
  The deployment of the Student Social Media Analytics website using Firebase Hosting was successful and efficient. Firebase provided a simple yet powerful platform to host the project with high performance, security, and scalability. It enhanced the overall accessibility and professionalism of the project, making it suitable for real-world demonstration and evaluation.

Future Scope

The Analysis of Social media Platforms in students project serves as a data-driven system that analyses the relationship between social media usage and key aspects of students' lives, including mental health, sleep patterns, addiction levels, and academic performance. While the current implementation focuses on analysis and visualization, the insights generated by this project have wide-ranging practical applications across educational, psychological, and societal domains.

One of the primary uses of this project is in the field of education. Educational institutions such as schools, colleges, and universities can utilize this system to better understand student behaviour in the digital age. By analysing usage patterns and their correlation with academic performance, educators can identify students who may be at risk of poor academic outcomes due to excessive social media engagement. This enables institutions to design targeted interventions, awareness programs, and counselling sessions to promote healthier study habits and improve overall academic performance.

In addition, the project has significant applications in the domain of mental health awareness and support. The analysis highlights how increased screen time and social media dependency can negatively impact mental well-being. Psychologists, counsellors, and mental health professionals can use these insights to monitor behavioural patterns and identify early signs of stress, anxiety, or digital addiction among students. The project can act as a supportive tool in developing strategies for improving emotional well-being and encouraging balanced digital usage.

Another important application lies in student self-awareness and personal development. By interacting with the dashboards and insights, students can gain a better understanding of their own digital habits. This awareness can help them recognize excessive usage patterns, understand their impact, and make informed decisions to manage their time more effectively. The project can thus serve as a self-assessment tool, encouraging responsible social media usage and promoting a healthier lifestyle.

The project can also be used by parents and guardians to monitor and guide their children's online behaviour. With increasing concerns about screen addiction and its effects, parents can benefit from data-backed insights to understand how social media influences their children's routines, sleep cycles, and academic performance. This can help them establish better guidelines and create a balanced environment for digital engagement at home.

From a research perspective, this project provides a foundation for further studies in the fields of data analytics, behavioural science, and digital sociology. Researchers can use the findings to explore deeper relationships between technology usage and human behaviour. The dataset and analysis can also be extended to different demographics, allowing for comparative studies across regions, age groups, or educational levels.

Moreover, the project has potential applications in policy-making and institutional planning. Educational boards and policymakers can use such analytical insights to design guidelines around digital usage in academic environments. For example, policies related to screen time, digital learning, and student well-being can be better informed using real data and trends identified through this system.

The system can also be utilized in awareness campaigns and workshops. Institutions and organizations can use the visual dashboards and findings to educate students about the risks of excessive social media usage. The graphical representation of data makes it easier to communicate complex patterns in a simple and impactful manner, making it an effective tool for presentations and training sessions.

Furthermore, the project can support the development of digital well-being tools and applications. The insights generated can be integrated into apps that track screen time, provide usage alerts, and suggest healthier habits. This extends the project's utility beyond analysis into practical solutions that directly influence user behaviour.

In conclusion, the future scope of this project lies in its wide applicability across multiple domains. It can be used as an educational aid, a mental health support tool, a self-awareness platform, and a research foundation. By leveraging the insights generated, various stakeholders—including students, educators, parents, and policymakers—can make informed decisions to promote balanced digital usage and improve overall well-being. The project thus holds strong potential to contribute meaningfully to both academic understanding and real-world problem-solving in the context of increasing social media influence.

List of Figures and Tables

Section	Description	Page No.
1.	Form 2: Sample of Google form used for manual data collection	10 - 12
2.	Form 2: Sample of response to the survey by students	12 - 14
3.	Table 1: Dataset consisting of responses of the survey	14
4.	Figure 1: Student Performance Analysis	15
5.	Table 2: Format of dataset taken from Kaggle	16
6.	Table 3: Format of normal responses as a dataset without any changes	17
7.	Table 4: Format of dataset taken from Kaggle	17
8.	Table 5: Changes made to the manual responses of the survey	18
9.	Figure 2: Importing python libraries	19
10.	Figure 3: Reading datasets using pandas.read_csv()	19
11.	Figure 4: Checking columns of both datasets.	19
12.	Figure 5: Checking shapes of both the datasets(rows*columns)	20
13.	Figure 6: Describes the statistics of all the numerical columns of both the datasets using describe() of python.	20
14.	Figure 7: checking for any null values in both datasets.	20
15.	Figure 8: checking datatypes of both data frames	21
16.	Figure 9: Concatenation of both data frames and checking the final data frame's profile	21
17.	Figure 10: Checking for any duplicated values in the final_df	21
18.	Figure 11: dividing the columns into list of categorical and numerical columns using list comprehension	22
19.	Figure 12: Performing categorical feature analysis by checking each one’s values and value counts	22
20.	Figure 13: This figure shows the value counts of “Gender”, “Academic_Level” and “Country”.	22

21.	Figure 14: This figure shows the value counts of “Most_used_platform”, “Affects_academic_Performance”, and “Relationship_status”	23
22.	Figure 15: Shows the number of females and males individually.	23
23.	Figure 16: Counts the number of students at each level of Academics	24
24.	Figure 17: Shows the distribution of countries by using a pie chart	24
25.	Figure 18: Depicts most used platforms using a bar plot	25
26.	Figure 19: Depicts the values counts of effects on academic performance according on students	25
27.	Figure 20: Shows the count plot of relationship status	26
28.	Figure 21: Shows the histogram of “Age” feature, which ranges from 18 to 26	26
29.	Figure 22: Distribution plot of Average Daily Usage hours of students	27
30.	Figure 23: Distribution plot of sleep hours per night of students	27
31.	Figure 24: Box plot depicting mental health score with median value of 6	28
32.	Figure 25: Histogram depicting Conflicts over social media	28
33.	Figure 26: Distribution plot of Addiction score of students	29
34.	Figure 27: Scatter plot depicting relationship between Age on the x axis and sleep hours per nights on the y axis	29
35.	Figure 28: Pairplot depiction relationship between multiple pair of features namely: “Age”,”Addiction score” and Avg_daily_usage_hours”	30
36.	Figure 29: Correlation between multiple features	31
37.	Figure 30: Scatter plot between mental health score and Avg daily usage hours. This graph shows a negative relation between the two features	32
38.	Figure 31: Scatter plot between sleep hours per night and Avg daily usage hours. This graph shows a negative relation between the two features	32
39.	Figure 32: Scatter plot between sleep hours per night and addiction score. This graph shows negative relation between the two features	33
40.	Figure 33: Scatter plot between conflicts over social media and Avg daily usage hours. This graph shows a positive relation between the two features	33
41.	Figure 34: Scatter plot between addiction score and Avg daily usage hours. This graph show positive relation between the two features.	34
42.	Figure 35: Scatter plot between mental health score and sleep hours per nights. This graph shows a positive relation between the two features.	34
43.	Figure 36: Scatter plot between conflicts over social media and sleep hours per night. This graph shows a positive relation between the two features	35
44.	Figure 37:Bar plot depicting relationship between conflicts over social media and mental	35

	health score.	
45.	Figure 38: Bar plot depicting relationship between mental health score and addiction score	36
46.	Figure 39: Barplot depicting relationship between conflicts over social media and addiction score.	36
47.	Figure 40: Simple Linear Model is trained and fitted	38
48.	Figure 41: Model's Coefficient and intercept is calculated	38
49.	Figure 42: Linear regression plot on addiction score	39
50.	Figure 43: R2 Score of the SLR model is checked	39
51.	Figure 44: Multiple linear regression model – 1 on addiction score.	39
52.	Figure 45: MLR model's coefficient and intercept are calculated	40
53.	Figure 46: Model's R2 score is calculated	40
54.	Figure 47: MLR model's Actual values distinguished with the predicted values	41
55.	Figure 48: MLR model on mental health score is trained and fitted. Its coefficients and intercept values are calculated. Further its R2 scoe is being calculated.	42
56.	Figure 49: Actual values of this MLR model is being distinguished to its predicted values	42
57.	Figure 50: independent and dependent variables are divided for developing a classification model.	43
58.	Figure 51: a classification model is being trained, fitted and predicted values on the basis of trained model. Further, its classification metrics has been calculated which generates 97% of accuracy	43
59.	Figure 52: ROC Curve for the classification model	44
60.	Graph 1: KPI Dashboards	45
61.	Graph 2: Usage Analysis Dashboard	46
62.	Graph:3 Advanced insights Dashboard	46
63.	Figure 53: Index.html code	48 - 52
64.	Figure 54: login.html code	52 – 53
65.	Figure 55: contact.html code	53 – 54
66.	Figures 56: style.css code	54 – 60
67.	Webpage 1: Admin login page	61
68.	Webpage 2: About page	62

69.	Webpage 3: KPI page	63
70.	Webpage 4: Basic Insights (Pagination – 1,2,3,4)	64 – 65
71.	Webpage 5: Advanced Insights (Pagination – 1,2,3,4,5,6)	66 - 68
72.	Webpage 6: Findings page	68
73.	Webpage 7: Contact Us page	69
74.	Figure 57: Showcasing VS-Code terminal for installing firebase tools for deploying the project	70
75.	Figure 58: Login to firebase from your terminal for providing connectivity to your code	70
76.	Figure 59: Project has been created on the firebase console	71
77.	Figure 60: Firebase has been initialized and details for the project are being provided in the terminal	71
78.	Figure 61: firebase.json code	72
79.	Figure 62: firebaserc code	72
80.	Figure 63: firebase deploy- command is used to deploy the website globally	72